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- J3. **[Under Review]** J. Portella and A. Goel, “A Constraint-Lifting Framework for Safe and Stable Nonlinear Control,” *IEEE Transactions on Automatic Control*.
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- C16. **[Submitted]** J. Portella and Ankit Goel, “Stability Preserving Safe Control of a Bicopter,” *American Control Conference 2026*
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ABSTRACT

Title of dissertation: Adaptive and Safe Control of Nonlinear Systems with Multiplicative Uncertainty and State Constraints

Jhon Manuel Portella Delgado

Dissertation directed by: Professor Ankit Goel
Department of Mechanical Engineering

The demand for reliable and safe control has increased significantly across engineering domains such as robotics, aerospace, nuclear systems, and emerging quantum technologies. Despite notable advances, control techniques with rigorous performance guarantees remain limited, leaving many nonlinear systems beyond the reach of existing methods. This thesis addresses two related classes of nonlinear control problems motivated by these limitations.

The first problem concerns the control of mechanical systems with uncertain inertial or actuation properties, where unknown parameters enter the dynamics multiplicatively with the control input, rendering classical adaptive backstepping impractical. The second problem focuses on designing controllers that stabilize the dynamics while enforcing state constraints without relying on online optimization, as required in safety-critical applications on resource-constrained platforms.

The first part of the dissertation develops an adaptive control framework for pure-feedback systems with multiplicative uncertainties. Rather than inverting parameter estimates and risking singularities, the proposed method directly estimates

inverse parameters, preventing unbounded internal signals. The resulting controller preserves the recursive backstepping structure and ensures robustness to uncertainty. A Lyapunov-based analysis establishes boundedness of all closed-loop signals and asymptotic convergence of the tracking error. A dynamic extension is introduced to address systems whose final state equation is non-affine in the control input.

The second part addresses safe nonlinear control for systems subject to explicit state constraints. A constraint-lifting framework maps the constrained state space into an unconstrained domain using smooth, invertible sigmoid transformations. To avoid ill-conditioning near constraint boundaries, a Lyapunov function referred to as the sigmoid integral function is introduced. The resulting controller guarantees closed-loop stability and constraint satisfaction without barrier terms or online solvers.

Together, these contributions advance nonlinear control in complementary directions: adaptive design for uncertain pure-feedback systems and safe control for constrained nonlinear dynamics. Their connection reveals that constraint-lifting transformations often induce pure-feedback structures, suggesting opportunities for integration. By avoiding parameter inversion, enforcing constraints through smooth transformations, and providing rigorous stability guarantees, this dissertation presents practical and theoretically grounded tools for controlling uncertain and safety-critical nonlinear systems.

Adaptive and Safe Control of Nonlinear Systems with Multiplicative
Uncertainty and State Constraints

by

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Dedication

I dedicate this dissertation first and foremost to God, who is the source of all wisdom, strength, and inspiration. He is the one who has placed the ideas, the curiosity, and the courage in my heart that have led to the development of the methods presented in this work. Truly, I can do everything with Him and nothing without Him. Every step of this journey has been guided by His hand, and every accomplishment is a reflection of His grace and love.

To my beloved wife, Josselin Gamboa, your love has been my anchor, my inspiration, and my joy. You have walked beside me through every challenge, celebrating triumphs and providing strength in moments of doubt. Your faith, patience, and unwavering belief in me have made this journey possible. This work stands as a testament to your devotion, your heart, and the incredible light you bring into my life. You are not only my partner but the reason I dare to dream higher every day.

To my parents, Manuel Portella and Marlene Delgado, your sacrifices, love, and teachings have shaped the person I am today. From the richness of our Peruvian roots, you instilled in me resilience, perseverance, and the courage to pursue my dreams. Your guidance and support have been a foundation that allowed me to reach for the impossible and remain grounded in humility and gratitude.

This dissertation is the culmination of faith, love, and dedication. To God, my wife Josselin, and my parents Manuel, and Marlene: thank you for being the heart, the guide, and the inspiration behind every word, every idea, and every effort contained within these pages.

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Goel's supervision. Beyond being a colleague, Juan has been a generous mentor and a true friend, always ready to offer guidance and perspective in both research and life.

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And to my wife, Josselin Gamboa, nothing I write here can fully express the depth of my gratitude and love. You have been the steady feedback signal in every moment of instability, the Lyapunov function that guaranteed my emotional boundedness, and the adaptive law that helped me converge when uncertainties arose. Your patience, understanding, and unwavering faith gave meaning to every equation and purpose to every late night. Thank you for walking beside me not only in this academic journey but also in the larger system we are learning to control together, life itself. This achievement is as much yours as it is mine.

Finally, to God, the ultimate Designer, the perfect Controller. Every inspiration, every convergence proof, and every moment of clarity in this dissertation comes from Him. The journey of my Ph.D. has been a control problem in disguise: uncertain, nonlinear, and yet always stabilizable under divine supervision. To capture this truth, I close these acknowledgments with a few theorems that express, in the language of mathematics, my faith and gratitude.

Theorem 0.0.1 (Existence of a Divine Lyapunov Function) *Let $x(t)$ denote*

the trajectory of my life and $V(x)$ a Lyapunov candidate representing peace and purpose. Then, there exists $V : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that $\dot{V}(x) = -\|G(x)\|^2$, where $G(x)$ represents God's grace acting as a stabilizing feedback term. Hence, $V(x)$ is non-increasing and all trajectories remain bounded under the control of God.

Proof: For every disturbance $d(t)$ in life, exists grace $G(x)$ such that $G(x) = -Kx + d(t)$, with $K > 0$ chosen by divine design. By faith, $\dot{V}(x) = x^T(-Kx + d(t)) \leq -\lambda_{\min}(K)\|x\|^2 + \|x\|\|d(t)\|$. Since God's grace always dominates disturbance, $\dot{V}(x) < 0$ for all $x \neq 0$. Therefore, $x(t) \rightarrow 0$, meaning the soul remains stable. ■

Theorem 0.0.2 (Adaptive Faith Law) Let $\theta(t) \in \mathbb{R}$ denote the true understanding and $\hat{\theta}(t) \in \mathbb{R}$ the human estimate of it. Let $e(t) \in \mathbb{R}$ represent the deviation from the desired path (God's path), and let $G \in \mathbb{R}$ be a known constant gain.

Consider the adaptive update law $\dot{\hat{\theta}}(t) = \gamma e(t) G$, where $\gamma > 0$ is a learning rate.

Then, under this update law, the estimation error $\tilde{\theta}(t) := \hat{\theta}(t) - \theta(t)$ converges asymptotically to zero as $t \rightarrow \infty$, i.e., the human estimate $\hat{\theta}(t)$ approaches the true understanding $\theta(t)$.

Proof: Define the Lyapunov candidate function $V(t) \triangleq \frac{1}{2}e(t)^2 + \frac{1}{2\gamma}\tilde{\theta}(t)^2 \geq 0$.

Its time derivative along the system trajectories is $\dot{V}(t) = e\dot{e} + \frac{1}{\gamma}\tilde{\theta}\dot{\tilde{\theta}}$.

Assuming the system satisfies $\dot{e}(t) = -G\tilde{\theta}(t)$, and using the adaptive law $\dot{\hat{\theta}}(t) = \dot{\hat{\theta}} - \dot{\theta} = \gamma Ge(t)$, we get $\dot{V}(t) = e(-G\tilde{\theta}) + \frac{1}{\gamma}\tilde{\theta}(\gamma Ge) = 0$.

Since $\dot{V} \leq 0$ and $V \geq 0$, we can invoke *LaSalle's Invariance Principle*. The largest invariant set contained in $\{ (e, \tilde{\theta}) \mid \dot{V} = 0 \}$ is $\{ (e, \tilde{\theta}) \mid e = 0, \tilde{\theta} = 0 \}$.

Therefore, all trajectories asymptotically converge to the origin: $e(t) \rightarrow 0, \tilde{\theta}(t) \rightarrow 0$ as $t \rightarrow \infty$.

Hence, the human estimate $\hat{\theta}(t)$ aligns perfectly with true understanding $\theta(t)$.

■

Theorem 0.0.3 (Global Stability Under Divine Feedback) *Consider the system of life $\dot{x} = f(x) + G(x)u$, where $x \in \mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is locally Lipschitz, and $G(x) \in \mathbb{R}^{n \times n}$ is invertible for all x .*

If the system is controlled by $u(x) = -G^{-1}(x)(f(x) + K(x - x^))$, where $G \in \mathbb{R}^{n \times n}$ is positive definite, $x^* \in \mathbb{R}^n$, is the equilibrium point established by God's will, and $K > 0$. Then, the closed-loop system is globally asymptotically stable at x^* i.e., all trajectories satisfy $x(t) \rightarrow x^*$ as $t \rightarrow \infty$.*

Proof: Define the Lyapunov candidate function $V(x) \triangleq \frac{1}{2}(x - x^*)^T(x - x^*) \geq 0$.

Its derivative along closed-loop trajectories is

$$\begin{aligned} \dot{V}(x) &= (x - x^*)^T \dot{x} \\ &= (x - x^*)^T [f(x) + G(x)u(x)] \\ &= (x - x^*)^T [f(x) + G(x)(-G^{-1}(x)(f(x) + K(x - x^*)))] \\ &= -K(x - x^*)^T(x - x^*). \end{aligned}$$

Since $\dot{V}(x) < 0$, except at the origin ($x - x^* = 0$). By Luapunov Stability, $x(t) \rightarrow x^*$ as $t \rightarrow \infty$, proving *global asymptotic stability* of x^* . ■

To God be all the glory, for He not only stabilized the nonlinear dynamics of my

mind but also ensured convergence to a meaningful equilibrium.

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Chapter 1

Introduction

1.1 Overview of Pure-feedback Nonlinear Systems and Control Problems

A broad and important class of systems that arises in robotics, aerospace, and mechanical systems takes the form

$$\dot{x}_1 = f_1(x_1) + g_1(x_1, x_2), \quad (1.1)$$

$$\dot{x}_2 = f_2(x_1, x_2) + g_2(x_1, x_2, u), \quad (1.2)$$

where x_1, x_2 are states and u is the input. The functions f_1, f_2, g_1 , and g_2 are appropriately defined vector functions. This form is referred to as the *pure-feedback form*. Pure-feedback systems are characterized by a control input that influences the first state only indirectly through a cascade of nonlinear, nonaffine interactions, with each subsequent state appearing nonaffinely. In contrast, strict-feedback systems admit an affine structure in the subsequent states, allowing the control input to enter each layer of the state recursion in an affine manner. This complex interaction, where the subsequent states appear nonaffinely, makes the design of stabilizing controllers more challenging.

A broad variety of engineering systems are in fact pure-feedback systems. For example, coupled translation and rotational dynamics of multi-rigid-body systems,

which are widespread in robotics, aerospace, and mechanical engineering domains, can be written in the pure-feedback form [1]. Chemical Process Systems [2], Economic and Socioeconomic Systems [3], and Electromechanical systems [4] can also be written in pure-feedback form after appropriate manipulation. On the other hand, certain systems may be transformed into the pure-feedback form as a consequence of required coordinate or structural transformations. In fact, the two main problems investigated in this thesis naturally yield system representations in pure-feedback form after the necessary transformations.

We focus on two related control problems arising in the context of pure-feedback systems, each addressing key challenges in extending nonlinear control methods to uncertain and constrained settings. The first problem focuses on the design of an adaptive controller for pure-feedback systems with multiplicative uncertainties, which frequently occur in mechanical and aerospace applications due to unknown actuation gains, aerodynamic coefficients, or inertial properties. The second problem focuses on designing a safety-enforcing nonlinear controller for systems subject to state constraints, which naturally arise in applications where physical limits or safety requirements restrict the allowable state trajectories.

1.1.1 Classical Methods

Classical nonlinear control methods primarily focus on strict-feedback systems, in which each subsequent state enters the dynamics affinely. This structural simplicity makes strict-feedback systems more amenable to analysis and control, and

has motivated the development of a wide range of nonlinear control strategies for them.

Feedback linearization is one such technique. It transforms a nonlinear system into an equivalent linear, triangular form, assuming that the zero-dynamics are stable. This linearization is achieved through an appropriate combination of coordinate transformation and state feedback. Consider the system

$$\dot{x} = f(x) + g(x)u, \quad (1.3)$$

$$y = h(x), \quad (1.4)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the input, and f, g and h are appropriately defined functions. Feedback linearization focuses on finding In this technique, a diffeomorphism $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ and a control law $u = \alpha(x) + \beta(x)v$ is designed such that the resulting dynamics reduces to

$$\dot{\eta} = f(\eta, \xi), \quad (1.5)$$

$$\dot{\xi} = A\xi + Bv, \quad (1.6)$$

$$y = C\xi \quad (1.7)$$

where $f(\eta, \xi)$ is nonlinear internal dynamics excluded from the linear strict-feedback structure of ξ dynamics, the dynamics of η is called the zero dynamics when $\xi = 0$, and (A, B) is controllable by construction [5]. Then, any linear technique can be used to design a control law for v to obtain a desired y . The technique requires the system to be regular, that is $\beta(x)$ must not have singularities by construction [6], as well as stable zero dynamics [7]. Unlike Jacobian local linearization, feedback linearization

cancels nonlinearities through feedback and provides global results under the stated assumptions [8].

The backstepping framework is also widely used for controller design in strict-feedback systems. It is a recursive nonlinear control method that utilizes the triangular structure of strict-feedback systems [2]. In the backstepping framework, a control law is recursively designed by introducing a virtual control for each subsystem. Consider the strict-feedback system

$$\dot{x}_1 = f_1(x_1) + g_1(x_1)x_2, \quad (1.8)$$

$$\dot{x}_2 = f_2(x_1, x_2) + g_2(x_1, x_2)u, \quad (1.9)$$

where x_1 and x_2 are the states, and u is the control input, and f_1, f_2, g_1, g_2 are appropriately defined functions.

The backstepping procedure begins by designing a controller for x_1 by assuming that x_2 is the control signal. For example, using the Lyapunov function

$$V_1 \triangleq \frac{1}{2}x_1^T x_1, \quad (1.10)$$

and assuming that $x_2 = x_{2d}$, where

$$x_{2d} \triangleq -g_1(x_1)^{-1}f_1(x_1) - k_1 x_1, \quad (1.11)$$

where $k_1 > 0$, the x_1 dynamics is stabilized. However, x_2 cannot be arbitrarily chosen. Therefore, the backstepping procedure continues to design a control law such that x_2 follows x_{2d} . Note that stabilization of x_1 requires the inversion of $g_1(x_1)$.

To track x_{2d} , define the error

$$e_2 \triangleq x_2 - x_{2d}, \quad (1.12)$$

and the Lyapunov function,

$$V_2 \triangleq V_1 + \frac{1}{2} e_2^T e_2. \quad (1.13)$$

By choosing

$$u = g_2(x_1, x_2)^{-1} (-f_2(x_1, x_2) + \dot{e}_2 - k_2 e_2), \quad (1.14)$$

where $k_2 > 0$ and following the standard Lyapunov design procedure, $\dot{V}_2 < 0$, which ensure that $e_2 \rightarrow 0$ and thus x_2 tracks x_{2d} . Note that stabilization of e_2 requires the inversion of $g_2(x_1, x_2)$. The requirement to invert the functions g_1 and g_2 limits the application of the backstepping technique to a broader class of nonlinear systems.

1.2 Motivation for Current Thesis

The overarching goal of this thesis is to develop systematic control methodologies for nonlinear systems that cannot be adequately addressed using classical techniques described above. In particular, the first problem is motivated by the need to design a controller precisely in the cases where g_1 and g_2 might need to be inverted. This is precisely the case where the maps g_1 and g_2 are unknown or uncertain, which frequently occur in mechanical and aerospace applications due to unknown actuation gains, aerodynamic coefficients, or inertial properties. In such cases, classical backstepping is not directly applicable because the control law involves the inversion

of terms that may be singular [2], [5], [9]–[13]. This scenario frequently occurs in practical systems in aerospace and robotics. For instance, unmanned aerial vehicles, robotic manipulators, and underactuated vehicles often include parameters such as inertia, masses, or actuator gains that are unknown and appear multiplicatively in the control input. Consequently, there is a pressing need for a control methodology capable of stabilizing such systems while avoiding inversion of uncertain parameters. The approach developed in this thesis introduces direct estimates of the inverse parameters within the backstepping framework. This strategy eliminates the need to invert uncertain terms, preserves numerical robustness, and maintains the recursive structure inherent to backstepping. An illustration of the main idea proposed in the first part of this thesis is presented in Figure 1.1.

An alternative approach to solve this problem is to ensure that x_1 and x_2 remain *constrained* within a *safe set* such that g_1 and g_2 are well-behaved inside the safe set, thereby extending the application of the classical controller design. This problem naturally includes the problem of designing controllers that guarantee forward invariance of prescribed sets [14]. Classical control methods typically emphasize stabilization or tracking objectives and do not provide systematic mechanisms for constraint satisfaction. Existing approaches, including reference governors, saturation schemes, and barrier Lyapunov functions, can mitigate constraint violations, but they often suffer from singularities, approximation errors, or computational overhead due to online optimization [15]–[17]. Control barrier functions (CBFs) provide a systematic way to enforce forward invariance of admissible state sets [18]–[20], yet their reliance on constrained optimization at each control step can limit appli-

cability in high-speed or resource-constrained systems [21]–[24]. To address these limitations, this thesis introduces the *constraint-lifting framework*, a novel methodology that transforms constrained control problems into equivalent unconstrained problems, as shown in Figure 1.2, using smooth, sigmoid-based mappings [25], [26]. This transformation allows conventional control design tools to be applied while maintaining well-conditioned behavior across the entire state space, avoiding singularities or unbounded growth near constraint boundaries. Additionally, a new Lyapunov function, termed the sigmoid integral function, is proposed to rigorously guarantee numerical robustness and closed-loop stability [27], [28]. Unlike barrier or predictive control approaches, the resulting control law is explicit and analytic, eliminating the need for online optimization and enabling real-time implementation on complex nonlinear systems, such as the attitude dynamics of aerial vehicles and constrained bicopters.

1.3 Summary of Contributions

In this section, we summarize the major results and contributions of this thesis.

1. **Inversion-free estimation.** We propose an inversion-free estimation step in the context of adaptive control of systems where the unknown parameters multiply the input maps. This technique prevents division by zero in the control laws and thus ensures global asymptotic stability.
2. **Dynamic extension of multicopter dynamics.** We propose a dynamic extension of the multicopter dynamics that yields a triangular pure-feedback

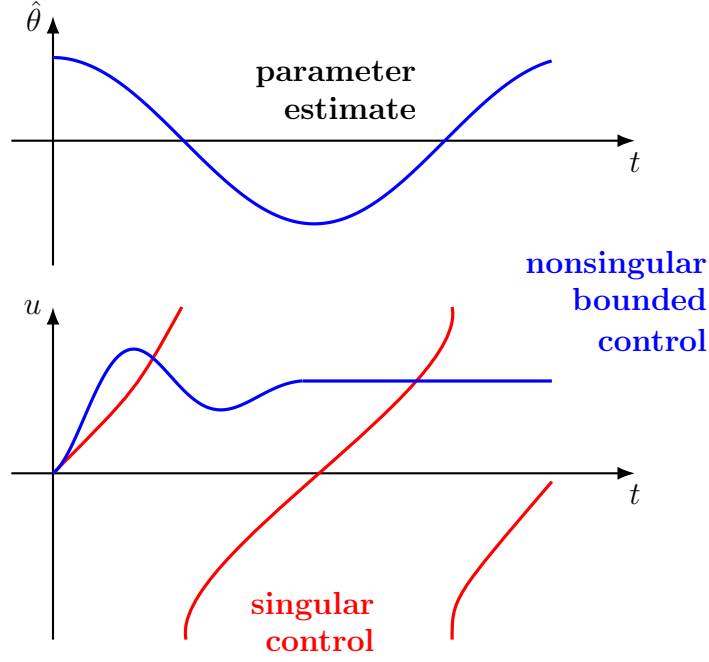


Figure 1.1: Inversion Free Parameter Estimate Adaptive Control.

form, thereby enabling the application of the classical backstepping framework.

3. **Constraint lifting functions.** We propose constraint lifting functions, based on the classical smooth sigmoid functions, to convert a constrained optimization problem into an unconstrained optimization problem. Note that this transformation usually produces a transformed system in pure-feedback form.
4. **Sigmoid integral function.** We propose a new Lyapunov function, which we refer to as the sigmoid integral function, to construct stabilizing control laws for globally non-Lipschitzian dynamics. The proposed function exhibits quadratic behavior near the origin and linear behavior away from the origin, thereby mitigating numerical instability issues associated with the classical choice of quadratic Lyapunov functions.

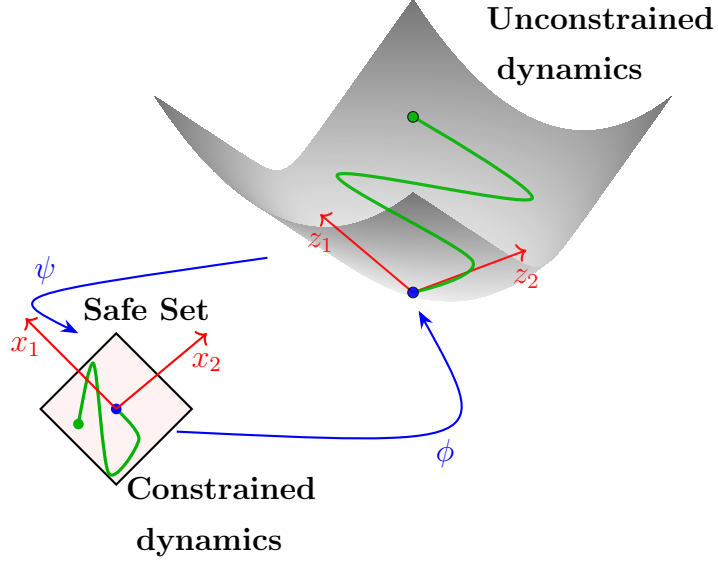


Figure 1.2: Enforcing forward invariance via the constraint lifting framework.

5. **State-constraint-enforcing control.** We use constraint lifting functions and sigmoid integral functions to develop a control framework that guarantees stability and forward invariance of a user-defined safety set. The synthesis is entirely Lyapunov-based and requires no online optimization, and the associated stability analysis is presented rigorously.
6. **Prevention of gimbal lock in attitude dynamics.** We apply the constraint-lifting framework to synthesize a safe controller for the attitude dynamics, preventing gimbal lock when using Euler angles for parameterization. In this implementation, both the angles and the angular velocities are constrained.
7. **Safe control of Bicopter.** We apply the constraint-lifting framework to enforce position and velocity constraints in a bicopter system.

1.4 Organization

This thesis is organized into two major parts, each addressing one of the central research problems introduced in Chapter 1. Chapter 2 focuses on the development of an inversion-free adaptive control framework for pure-feedback systems subject to multiplicative uncertainties. The chapter begins with an introduction and formal problem statement, followed by the derivation of the adaptive controller and a rigorous stability analysis. To illustrate the method’s applicability, the chapter presents examples for both strict-feedback and pure-feedback systems, and applies the proposed controller to a bicopter with unknown mass and inertia. This application includes a detailed step-by-step controller design, stability analysis, and numerical simulations demonstrating waypoint-following and elliptical-trajectory tracking. The chapter concludes with a synthesis of findings and insights gained from the adaptive control formulation.

Chapter 3 addresses the second major contribution of the thesis: a constraint-lifting framework for safe nonlinear control in systems subject to state constraints. The chapter begins by establishing the problem formulation and notation, then introduces the class of sigmoid and sigmoid-integral functions that form the basis of the constraint-lifting transformation. The proposed method is developed through a state transformation, controller synthesis, and Lyapunov-based stability analysis, supported by theoretical justification for the constraint-lifting function as a valid Lyapunov candidate. The framework is then applied to two safety-critical problems, safe attitude control and safe bicopter control, each accompanied by a complete

controller derivation, stability proofs, and numerical simulations that validate the approach. The thesis concludes in Chapter 4 with a summary of the contributions and a discussion of future research directions.

Chapter 2

Inversion-free Adaptive Control of Pure Feedback Systems

In practical control applications, nonlinear systems frequently exhibit pure-feedback structures, where each state depends nonlinearly on subsequent states and the control input appears only in the final equation. Such configurations arise in robotic manipulators, aerospace vehicles, and electromechanical systems, where the dynamic coupling among subsystems is significant and often uncertain. The design of controllers for these systems becomes particularly challenging when the system dynamics involve parametric uncertainties or unmodeled nonlinearities, which prevent the direct application of classical feedback linearization techniques. To address these challenges, adaptive estimation can be systematically integrated with recursive backstepping to compensate for unknown parameters and nonlinear interdependencies among the states. This combination enables the construction of control laws that achieve asymptotic tracking without requiring exact model knowledge or linear parameterization of the nonlinearities. Therefore, incorporating adaptive mechanisms within the backstepping framework is essential to ensure stability and robust performance of nonlinear pure-feedback systems under model uncertainties and complex interconnections.

In this chapter, a new adaptive control framework is developed to address the stabilization of nonlinear pure-feedback systems with unknown multiplicative

parameters, that is, parameters that affect the input maps. A rigorous Lyapunov-based stability analysis is provided to establish the boundedness of all closed-loop signals and the asymptotic convergence of the tracking error. The design principles are systematically presented to illustrate how adaptive estimation and recursive backstepping are integrated to handle the nonlinear interdependencies among the system states. Finally, We discuss how our solution can be applied to the Bicopter control problem with unknown multiplicative uncertainties.

2.1 Introduction

Complex dynamic systems, such as aircraft, multicopters, and multi-body robotic systems, are governed by nonlinear dynamics, in which the unknown or uncertain parameters often appear as multiplicative terms in the input map. For example, in aerial vehicles, the inertial properties, including the mass and the moment of inertia, are often uncertain and appear as multiplicative terms in the input map. Moreover, such system parameters can change over time, thus necessitating adaptive controllers to maintain stability guarantees.

Adaptive backstepping is the most widely used technique to design stabilizing controllers for nonlinear systems. The classical backstepping framework is motivated by systems represented in the strict feedback form [5]. In this framework, each state is recursively stabilized by the succeeding state, where each succeeding state serves as a virtual control signal. The virtual control is designed to stabilize each state, usually by choosing a Lyapunov candidate function and a virtual control law to yield

a stabilizing controller. The process repeats until the real control appears in the state equation, at which point the actual stabilizing controller is finally designed. By judicious aggregation of Lyapunov candidate functions, the closed loop's stability is guaranteed. The control laws are chosen to cancel the nonlinearities in the dynamics of each state to obtain the desired transient performance. However, the nonlinearities may not be cancelable if they are unknown or constitute parametric uncertainties.

Adaptive extensions of the classical backstepping framework have been developed with stability guarantees for the cases where the uncertainty appears in the dynamics map [2, p 157]. An adaptive backstepping-based technique is presented in [12] for nonlinear systems in strict feedback form with linearly parameterized uncertainty in the dynamics of the last state. A fixed-time adaptive controller to deal with unknown system disturbances is presented in [29]; however, the adaptation is focused on the disturbance rather than the uncertainties in the system's dynamics.

In the adaptive-backstepping framework, if the uncertainty appears in the input map of the virtual or the actual control, the general approach of designing stabilizing controllers may not be applicable since the input map with the parameter estimate may become singular for some values of the parameter estimates. Approaches using tuning functions [30], [31] and Lyapunov barrier functions [32] have been explored to address this problem; however, they do not guarantee stability. Nussbaum functions have recently been explored to deal with uncertain parameters in nonlinear systems with unknown control coefficients [9]–[11], [33]–[39]. In this approach an unbounded function is introduced to compensate for uncertainties,

ensuring stability despite parameter variations. A key limitation of this approach, however, is the difficulty of constructing Nussbaum functions and the fact that the resulting internal signals in the closed loop may be unbounded. Furthermore, their application in multivariable adaptive control is limited due to the need for multiple Nussbaum functions, whose combined effects can counteract each other [39]. Adaptive extension of the classical backstepping framework for systems with uncertainty in the input maps thus remains a challenging problem.

The problem of synthesizing a stabilizing controller is further aggravated if the dynamics of each state are in the pure feedback form [2], [40]. Several techniques have been investigated to design stabilizing controllers for nonlinear systems in pure feedback form [41]. A dynamic extension is explored in [42]. An observer-based approach is used to design an output feedback controller for pure feedback systems in [43]. However, these techniques apply to the cases where the dynamics, including state dynamics and the input map, are completely known.

This chapter focuses on developing a stabilizing controller for a class of nonlinear systems with dynamics in the pure-feedback form, where unknown parameters appear in the input map. This generalized problem is motivated by the problem of designing an adaptive controller for a multi-copter with unknown inertial properties. A backstepping controller with a finite-time parameter estimator was designed in [13] for this problem. However, the proposed controller required inverting estimates of mass and inertia, thus necessitating careful tuning to avoid zero crossings of the estimates. The inversion of the parameter estimates, therefore, weakened the stability property of the closed-loop system.

Unlike the current approaches, this chapter proposes a technique to deal with the multiplicative unknown parameters by introducing estimates of the inverse of the unknown parameters instead of inverting the estimates of the unknown parameters. By judiciously choosing the Lyapunov candidate functions, the resulting feedback controller does not contain the inverse of any internal signal, thus guaranteeing global asymptotic stability. The intermediate Lyapunov candidate functions, however, do require knowledge of the sign of the unknown parameters, which is usually known in practice. For example, inertial properties such as mass and moments of inertia are positive numbers. The contributions of this chapter are thus

1. the development of an adaptive backstepping control method that addresses uncertainties in the input map for pure-feedback systems without imposing saturation constraints on the parameter update law gains or requiring parameter estimate inversion;
2. a stability analysis guaranteeing global stability of the closed-loop system; and
3. numerical validation of the proposed controller.

2.2 Problem formulation

Consider the pure feedback system with linear affine uncertain parameters

$$\dot{x}_1 = f_1(x_1) + g_1(x_1, x_2)\theta_1, \quad (2.1)$$

$$\dot{x}_2 = f_2(x_1, x_2) + g_2(x_1, x_2, u)\theta_2, \quad (2.2)$$

where $x_1, x_2 \in \mathbb{R}^n$ are the states, $u \in \mathbb{R}^n$ is the input, and $\theta_1, \theta_2 \in \mathbb{R}$ are unknown parameters. The functions $f_1 : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $f_2 : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, $g_1 : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, and $g_2 : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ are assumed to be continuously differentiable of sufficient order.

System (2.1), (2.2) is assumed to satisfy the following conditions.

Assumption 2.2.1 *The sign of the parameters θ_1 and θ_2 is known.*

Assumption 2.2.2 *The functions f_1, f_2, g_1 , and g_2 satisfy $f_1(0) = f_2(0, 0) = g_1(0, 0) = g_2(0, 0, 0) = 0$.*

Assumption 2.2.3 *$g_1(0, x_2) = 0$ if and only if $x_2 = 0$ and $g_2(0, 0, u) = 0$ if and only if $u = 0$.*

Assumption 2.2.4 *There exists a set $\mathcal{D} \subseteq \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n$ such that*

$$\frac{\partial g_1(x_1, x_2)}{\partial x_2} \frac{\partial g_2(x_1, x_2, u)}{\partial u} \in \mathbb{R}^{n \times n} \quad (2.3)$$

is nonsingular and $(0, 0, 0) \in \mathcal{D}$.

Remark 2.2.1 *Assumption 2.2.2 implies that $x_1 = 0, x_2 = 0, u = 0$ is an equilibrium point. Note that this assumption is not restrictive since any nonzero equilibrium point can be mapped to a zero equilibrium point with a linear transformation.*

Remark 2.2.2 *If g_1 and g_2 are affine functions of x_2 and u , respectively, then Assumption 2.2.3 is trivially satisfied.*

Remark 2.2.3 *Assumption 2.2.4, which requires that $\frac{\partial g_1(x_1, x_2)}{\partial x_2}$ and $\frac{\partial g_2(x_1, x_2, u)}{\partial u}$ be nonsingular, ensures the existence of the proposed control law proposed in Section 2.3.*

Remark 2.2.4 Assumptions 2.2.3 and 2.2.4 ensure that both states are controllable.

The objective is to construct a control law that stabilizes the system (2.35), (2.36), without the knowledge of parameters θ_1 and θ_2 .

2.3 Controller Synthesis

This section presents the control law and parameter update rules designed to stabilize the nonlinear system, and formulates the necessary functions for analyzing the stability of the closed-loop system. Figure 2.1 shows the block diagram illustrating the adaptive controller implementation.

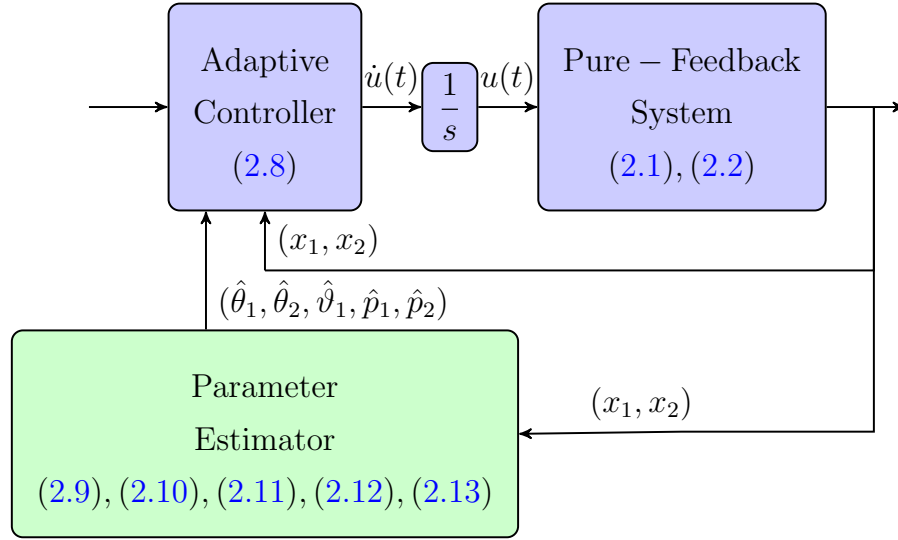


Figure 2.1: Block diagram illustrating the adaptive controller implementation.

To emphasize the generalizability of the proposed controller, we make the following Assumption.

Assumption 2.3.1 There exists a positive definite function $V: \mathbb{R}^n \rightarrow [0, \infty)$, where

$V_x(0) = 0$ and a function $\kappa: \mathbb{R}^n \rightarrow \mathbb{R}^n$, where $\kappa(0) = 0$, such that for all $x \neq 0 \in \mathbb{R}^n$,

$$V_x(x)\kappa(x) > 0. \quad (2.4)$$

Note that $V_x(x) \triangleq \frac{\partial V(x)}{\partial x}$.

Note that the Assumption 2.3.1 is satisfied by the typical choice of the Lyapunov candidate function $V(x) = x^2$ and the control law $\kappa(x) = x_1$. The proposed generalization, however, allows the choice of functions V and κ that may yield stronger stability properties such as finite-time and prescribed-time stability.

Proposition 2.3.1 *Consider the system (2.35), (2.36). Let $\gamma_1, \gamma_2, \alpha_1, \alpha_2, \alpha_3, k_1, k_2, k_3 > 0$. Consider functions V and κ satisfying Assumption 2.3.1. Define $p_1 \triangleq \theta_1^{-1}$, $p_2 \triangleq \theta_2^{-1}$. Consider the function*

$$\begin{aligned} \mathcal{V} \triangleq & V(x_1) + \frac{1}{2}\gamma_1^{-1}|\theta_1|(\hat{p}_1 - p_1)^2 + \frac{1}{2}e_2^T e_2 \\ & + \frac{1}{2}\gamma_2^{-1}|\theta_2|(\hat{p}_2 - p_2)^2 + \frac{1}{2}\alpha_1^{-1}(\theta_1 - \hat{\theta}_1)^2 + \frac{1}{2}e_3^T e_3 \\ & + \frac{1}{2}\alpha_2^{-1}(\theta_1 - \hat{\vartheta}_1)^2 + \frac{1}{2}\alpha_3^{-1}(\theta_2 - \hat{\theta}_2)^2, \end{aligned} \quad (2.5)$$

where $\hat{\theta}_1$ and $\hat{\vartheta}_1$ are estimates of θ_1 , $\hat{p}_1$ is an estimate of p_1 , $\hat{\theta}_2$ is an estimate of θ_2 , $\hat{p}_2$ is an estimate of p_2 , and

$$e_2 \triangleq g_1(x_1, x_2) + \hat{p}_1 f_1(x_1) + k_1 \text{sign}(\theta_1)\kappa(x_1) \in \mathbb{R}^n, \quad (2.6)$$

$$\begin{aligned} e_3 \triangleq & \frac{\partial e_2}{\partial x_2} g_2(x_1, x_2, u) + \hat{p}_2 \left[\left(\frac{\partial V}{\partial x_1} + \frac{\partial e_2}{\partial x_1} g_1(x_1, x_2) \right) \hat{\theta}_1 \right. \\ & \left. + \frac{\partial e_2}{\partial x_1} f_1(x_1) + \frac{\partial e_2}{\partial x_2} f_2(x_1, x_2) + \frac{\partial e_2}{\partial \hat{p}_1} \dot{\hat{p}}_1 \right] + k_2 \text{sign}(\theta_2)e_2 \in \mathbb{R}^n. \end{aligned} \quad (2.7)$$

Consider the control law

$$\begin{aligned} \dot{u} = & - \left[\frac{\partial e_3}{\partial u} \right]^{-1} \left[e_2 \hat{\theta}_2 + \frac{\partial e_3}{\partial x_1} \left(f_1(x_1) + g_1(x_1, x_2) \hat{v}_1 \right) \right. \\ & \left. + \frac{\partial e_3}{\partial x_2} \left(f_2(x_1, x_2) + g_2(x_1, x_2, u) \hat{\theta}_2 \right) + \frac{\partial e_3}{\partial \hat{\theta}_1} \dot{\hat{\theta}}_1 + \frac{\partial e_3}{\partial \hat{p}_1} \dot{\hat{p}}_1 + \frac{\partial e_3}{\partial \hat{p}_2} \dot{\hat{p}}_2 + k_3 e_3 \right], \end{aligned} \quad (2.8)$$

and the parameter update laws

$$\dot{\hat{p}}_1 = \gamma_1 \text{sign}(\theta_1) \frac{\partial V}{\partial x_1} f_1(x_1), \quad (2.9)$$

$$\begin{aligned} \dot{\hat{p}}_2 = & \gamma_2 \text{sign}(\theta_2) \left(e_2^T \left[\left(\frac{\partial V}{\partial x_1} \right)^T + \frac{\partial e_2}{\partial x_1} g_1(x_1, x_2) \right] \hat{\theta}_1 \right. \\ & \left. + \frac{\partial e_2}{\partial x_1} f_1(x_1) + \frac{\partial e_2}{\partial x_2} f_2(x_1, x_2) + \frac{\partial e_2}{\partial \hat{p}_1} \dot{\hat{p}}_1 \right), \end{aligned} \quad (2.10)$$

$$\dot{\hat{\theta}}_1 = \alpha_1 \left[e_2^T \left(\left(\frac{\partial V}{\partial x_1} \right)^T + \frac{\partial e_2}{\partial x_1} g_1(x_1, x_2) \right) \right], \quad (2.11)$$

$$\dot{\hat{v}}_1 = \alpha_2 e_3^T \left[\frac{\partial e_3}{\partial x_1} g_1(x_1, x_2) \right], \quad (2.12)$$

$$\dot{\hat{\theta}}_2 = \alpha_3 e_3^T [e_2 + g_2(x_1, x_2, u)]. \quad (2.13)$$

Then,

$$\dot{V} = -k_1 |\theta_1| \frac{\partial V}{\partial x_1} \kappa(x_1) - k_2 |\theta_2| e_2^T e_2 - k_3 e_3^T e_3. \quad (2.14)$$

Proof: Consider the function

$$V_1(x_1, \hat{p}_1) \triangleq V(x_1) + \frac{1}{2} \gamma_1^{-1} |\theta_1| (\hat{p}_1 - p_1)^2.$$

Note that

$$\dot{V}_1 = \frac{\partial V}{\partial x_1} [f_1(x_1) + g_1(x_1, x_2) \theta_1] + \gamma_1^{-1} |\theta_1| (\hat{p}_1 - p_1) \dot{\hat{p}}_1.$$

Substituting $\dot{\hat{p}}_1$ from the parameter update law (2.9) and substituting e_2 from (2.6) in the equation above yields

$$\dot{V}_1 = -w + e_2^T \frac{\partial V}{\partial x_1} \theta_1,$$

where $w \triangleq k_1 |\theta_1| V_{x_1}(x_1) \kappa(x_1)$. Note that it follows from Assumption 2.3.1 that, for all $x_1 \neq 0 \in \mathbb{R}^n$, $w > 0$.

Next, consider the function

$$V_2(x_1, x_2, \hat{p}_1, \hat{\theta}_1, \hat{p}_2) \triangleq V_1 + \frac{1}{2} e_2^T e_2 + \frac{1}{2} \gamma_2^{-1} |\theta_2| (\hat{p}_2 - p_2)^2 + \frac{1}{2} \alpha_1^{-1} (\theta_1 - \hat{\theta}_1)^2.$$

Note that

$$\begin{aligned} \dot{V}_2 = & -w + e_2^T \left[\frac{\partial V}{\partial x_1} \theta_1 + \frac{\partial e_2}{\partial x_1} (f_1(x_1) + g_1(x_1, x_2) \theta_1) \right. \\ & \left. + \frac{\partial e_2}{\partial x_2} (f_2(x_1, x_2) + g_2(x_1, x_2, u) \theta_2) + \frac{\partial e_2}{\partial \hat{p}_1} \dot{\hat{p}}_1 \right] \\ & + \gamma_2^{-1} |\theta_2| (\hat{p}_2 - p_2) \dot{\hat{p}}_2 - \alpha_1^{-1} (\theta_1 - \hat{\theta}_1) \dot{\hat{\theta}}_1. \end{aligned}$$

Substituting $\dot{\hat{p}}_2$ and $\dot{\hat{\theta}}_1$ from the parameter update laws (2.10) and (2.11) and substituting e_3 from (2.7) in the equation above yields

$$\dot{V}_2 = -w - k_2 |\theta_2| e_2^T e_2 + e_2^T e_3 \theta_2.$$

Finally, consider (2.5), which can be written as

$$\mathcal{V}(x_1, x_2, u, \hat{p}_1, \hat{\theta}_1, \hat{v}_1, \hat{\theta}_2, \hat{p}_2) = V_2 + \frac{1}{2} e_3^T e_3 + \frac{1}{2} \alpha_2^{-1} (\theta_1 - \hat{v}_1)^2 + \frac{1}{2} \alpha_3^{-1} (\theta_2 - \hat{\theta}_2)^2.$$

Note that

$$\begin{aligned}\dot{\mathcal{V}} = & -w - k_2|\theta_2|\text{sign}^2(\theta_2)e_2^T e_2 + e_3^T \left[e_2\theta_2 + \frac{\partial e_3}{\partial x_1} (f_1(x_1) + g_1(x_1, x_2)\theta_1) \right. \\ & \left. + \frac{\partial e_3}{\partial x_2} (f_2(x_1, x_2) + g_2(x_1, x_2, u)\theta_2) + \frac{\partial e_3}{\partial \hat{\theta}_1} \dot{\hat{\theta}}_1 + \frac{\partial e_3}{\partial \hat{p}_1} \dot{\hat{p}}_1 + \frac{\partial e_3}{\partial \hat{p}_2} \dot{\hat{p}}_2 + \frac{\partial e_3}{\partial u} \dot{u} \right] \\ & - \alpha_2^{-1} (\theta_1 - \hat{\vartheta}_1) \dot{\hat{\vartheta}}_1 - \alpha_3^{-1} (\theta_3 - \hat{\theta}_2) \dot{\hat{\theta}}_2.\end{aligned}$$

Substituting $\dot{\hat{\vartheta}}_1$ and $\dot{\hat{\theta}}_2$ from the parameter update laws (2.12) and (2.13) and the control law from (2.8) in the equation above yields (2.14). $\blacksquare$

Remark 2.3.1 *The control law (2.8) generates the signal $\dot{u}$. The control signal u is obtained by integrating $\dot{u}$ given by the control law (2.8).*

Remark 2.3.2 *The control law (2.8) requires $\frac{\partial e_3}{\partial u} = \frac{\partial g_1(x_1, x_2)}{\partial x_2} \frac{\partial g_2(x_1, x_2, u)}{\partial u}$ to be nonsingular, which is guaranteed by Assumption 2.2.4. Note that $\frac{\partial g_1(x_1, x_2)}{\partial x_2} \frac{\partial g_2(x_1, x_2, u)}{\partial u}$ is a product of two matrices.*

Remark 2.3.3 *Proposition 2.3.1 assumes that the dimensions of x_1 , x_2 , and u are the same. However, $\frac{\partial e_3}{\partial u}$ may be nonsingular in some cases where x_1 , x_2 , and u have different dimensions. The control law (2.8) thus can be extended to such nonlinear systems.*

Remark 2.3.4 *The map κ does not need to be continuous.*

2.4 Stability Analysis

This section presents the stability analysis of the closed-loop system (2.1), (2.2), (2.8) and the parameter update laws (2.9), (2.10), (2.11), (2.12), and (2.13).

In particular, Propositions 2.4.1, 2.4.2, 2.4.3, and 2.4.4 define sets used in the stability proof and establish their properties. Finally, Barbashin-Krasovskii-LaSalle's invariance principle, which originally appeared as Theorem 1 and 2 in [44], is used to prove the asymptotic stability of $x_1 = x_2 = u = 0$ and the boundedness of the parameter estimates.

Proposition 2.4.1 *Consider the differential equations (2.1), (2.2), (2.8), (2.9), (2.10), (2.11), (2.12), (2.13) and Assumptions 2.2.2, 2.3.1 Then, $x_1 = x_2 = u = 0$ and $\hat{p}_1, \hat{\theta}_1, \hat{\vartheta}_1, \hat{\theta}_2, \hat{p}_2 \in \mathbb{R}$ is an equilibrium point.*

Proof: First, note that if $x_1 = x_2 = u = 0$, then it follows from (2.1) and (2.2) that $\dot{x}_1 = \dot{x}_2 = 0$. Additionally, for any $\hat{p}_1, \hat{\theta}_1, \hat{\vartheta}_1, \hat{\theta}_2, \hat{p}_2 \in \mathbb{R}$, then it follows from (2.9) that $\dot{\hat{p}}_1 = 0$. Next, it follows from (2.6) that $e_2 = 0$, and thus it follows from (2.10) that $\dot{\hat{p}}_2 = 0$ and it follows from (2.11) that $\dot{\hat{\theta}}_1 = 0$. Next, it follows from (2.7) that $e_3 = 0$, and thus it follows from (2.12) that $\dot{\hat{\vartheta}}_1 = 0$ and it follows from (2.13) that $\dot{\hat{\theta}}_2 = 0$. Finally, it follows from Assumption 2.2.2, the fact that $e_2 = e_3 = 0$, and $\dot{\hat{\theta}}_1 = \dot{\hat{p}}_1 = \dot{\hat{p}}_2 = 0$, that $\dot{u} = 0$, which completes the proof. ■

Proposition 2.4.2 *Let $\mathcal{D}$ be a set that satisfies Assumption 2.2.4. Let $\ell > 0$. Define*

$$\Omega \triangleq \left\{ x_1, x_2, u \in \mathcal{D} \subseteq \mathbb{R}^n, \hat{p}_1, \hat{\theta}_1, \hat{\vartheta}_1, \hat{\theta}_2, \hat{p}_2 \in \mathbb{R} : \mathcal{V} \leq \ell \right\}. \quad (2.15)$$

Then, Ω is bounded and a positively invariant set.

Proof: Since $\mathcal{V}$ is radially unbounded and Ω is defined by $V(x) \leq \ell$, it follows from Theorem 2 in [44] that Ω is bounded. Since $\dot{\mathcal{V}} \leq 0$ and $\mathcal{V} > 0$, Ω is a positively invariant set. ■

Proposition 2.4.3 Define $E \subset \Omega$ such that $x_1 = 0, x_2 = 0, u = 0$. Then, E is the set of all points in Ω where $\dot{\mathcal{V}} = 0$.

Proof: It follows from (2.14) that $\dot{\mathcal{V}} = 0$ if and only if $x_1 = 0, x_2 = 0, u = 0$. ■

Proposition 2.4.4 E is the largest positively invariant set in E .

Proof: It follows from Proposition 2.4.1 that $x_1 = 0, x_2 = 0, u = 0$, and $\hat{p}_1, \hat{\theta}_1, \hat{\vartheta}_1, \hat{\theta}_2, \hat{p}_2 \in \mathbb{R}$ is an equilibrium point, thus E is the largest positively invariant set in E . ■

Theorem 2.4.1 Consider the system (2.1), (2.2), (2.8), where $\gamma_1, \gamma_2, \alpha_1, \alpha_2, \alpha_3, k_1, k_2, k_3 > 0$ and the functions V and κ satisfy the assumptions in Assumption 2.3.1. Then,

$$\lim_{t \rightarrow \infty} x_1(t) = 0, \quad (2.16)$$

$$\lim_{t \rightarrow \infty} x_2(t) = 0, \quad (2.17)$$

$$\lim_{t \rightarrow \infty} u(t) = 0, \quad (2.18)$$

and the estimates $\hat{p}_1(t), \hat{\theta}_1(t), \hat{\vartheta}_1(t), \hat{\theta}_2(t), \hat{p}_2(t)$ remain bounded.

Proof: Consider the set Ω defined in Proposition 2.4.2 and the set E defined in Proposition 2.4.3. It follows from Proposition 2.4.4 that E is the largest positively invariant set in E . It follows from Theorem 2 in [44] that any solution starting in Ω approaches E as $t \rightarrow \infty$, which implies that (2.16), (2.17), (2.18) and the estimates $\hat{p}_1(t), \hat{\theta}_1(t), \hat{\vartheta}_1(t), \hat{\theta}_2(t), \hat{p}_2(t)$ remain bounded. ■

Remark 2.4.1 It can not be concluded that $\hat{\theta}_1$ and $\hat{\vartheta}_1$ converge to θ_1 and $\hat{\theta}_2$ converges to θ_2 . However, they remain bounded.

2.5 Examples

The examples in this section demonstrate the application of the proposed control design methodology to the stabilization of nonlinear systems. Specifically, the technique is used to design controllers for both strict-feedback and pure-feedback nonlinear systems.

We first consider a strict-feedback nonlinear system and design a stabilizing controller based on the procedure developed in this work, as well as the Nussbaum technique, which is reviewed in Appendix [A.3](#). Note that the Nussbaum technique applies only to the strict-feedback systems. Finally, we consider a pure-feedback nonlinear system, previously studied in [\[40\]](#), [\[45\]](#), [\[46\]](#), and design an adaptive controller to stabilize it.

2.5.1 Strict-feedback System

In this example, we consider the problem of designing an adaptive controller for a strict feedback system and compare its performance with a similarly tuned Nussbaum controller. In particular, we consider the nonlinear system

$$\dot{x}_1 = x_1^2 + x_2, \tag{2.19}$$

$$\dot{x}_2 = x_1 x_2 + \theta u, \tag{2.20}$$

where $x_1, x_2, u \in \mathbb{R}$. Let $\theta = 5$, which is assumed to be unknown. The objective is to design a control law that stabilizes the origin without using the knowledge of the parameter θ .

Note that the nonlinear system (2.19)-(2.20) is in strict-feedback form. Furthermore, since

$$\frac{\partial g_1}{\partial x_2} = 1, \quad \frac{\partial g_2}{\partial u} = \theta, \quad (2.21)$$

are nonzero for all $x_1, x_2, u \in \mathbb{R}$, Assumption 2.2.4 is satisfied.

To design the controller, let $V(x_1) = \frac{1}{2}x_1^2$ and $\kappa(x_1) = x_1$ in (2.6), (2.7). The controller is then given by (2.8), that is,

$$\dot{u} = - \left(e_2 \hat{\theta} + \frac{\partial e_3}{\partial x_1} (x_1^2 + x_2) + \frac{\partial e_3}{\partial x_2} (x_1 x_2 + \hat{\theta} u) + \frac{\partial e_3}{\partial \hat{p}} \dot{\hat{p}} + k_3 e_3 \right), \quad (2.22)$$

and the parameter estimators are

$$\dot{\hat{p}} = \gamma \text{sgn}(\theta) e_2 \left(x_1 + (2x_1 + k_1)(x_1^2 + x_2) + x_1 x_2 \right), \quad (2.23)$$

$$\dot{\hat{\theta}} = \alpha e_3 (e_2 + u). \quad (2.24)$$

The explicit expressions of the variables in this example are given in Appendix A. Let the controller gains $k_1 = k_2 = k_3 = 1$ and the parameter estimator gains $\alpha = \gamma = 1$. In the estimators, we set $\hat{p}(0) = \hat{\theta}(0) = 0$.

The initial state is assumed to be $x_1(0) = 1.5, x_2(0) = 0$, and $u(0) = 0$.

Figure 2.2 shows the closed-loop response with both the proposed controller and the Nussbaum controller (briefly described in Appendix C). To ensure fairness in performance comparison, all the controller gains in the Nussbaum controller are also set to 1. Note that the parameter estimates $\hat{p}$ and $\hat{\theta}$ cross zero multiple times, however, since the controller does not require inversion of the parameter estimates, the resulting closed-loop performance is not affected and closed-loop stability is guaranteed. Furthermore, note that, compared to the Nussbaum controller, the

control law (2.22), with similar gains, yields better transients with a smaller control effort.

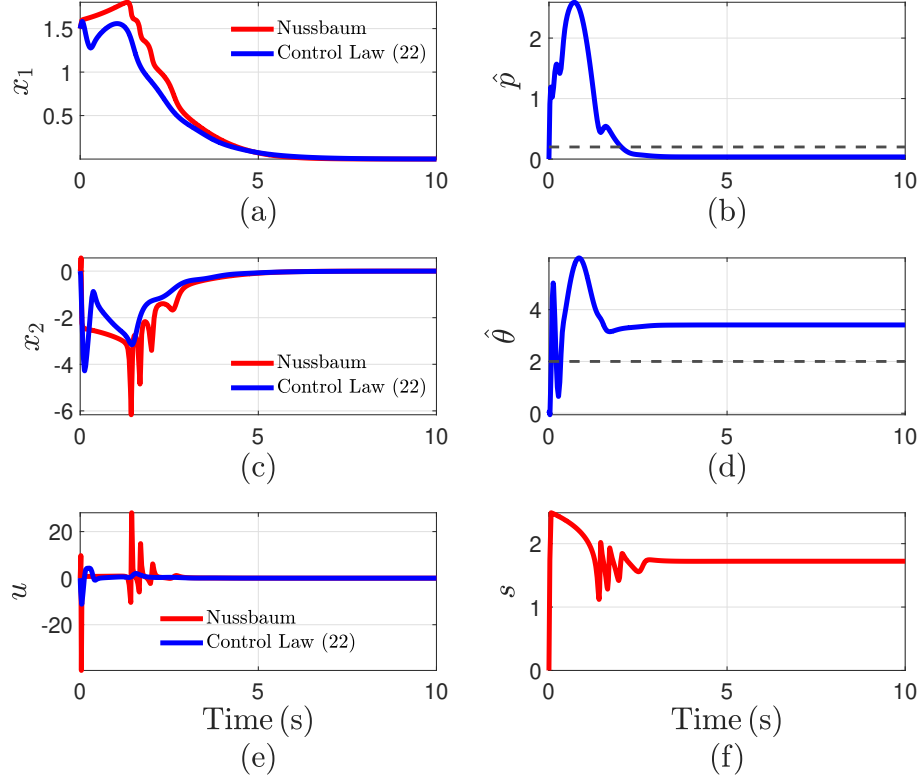


Figure 2.2: Closed-loop response of the system (2.19), (2.20) with control law (2.22) (blue) and the Nussbaum controller (red). (a) and (c) show the states x_1 and x_2 , respectively, and (e) shows both the control signal u obtained with control law (2.22) and the Nussbaum controller. (b) and (d) show the parameter estimates $\hat{p}$ and $\hat{\theta}$. Note that both estimates cross zero multiple times without affecting the closed-loop stability. Finally, (e) shows the internal variable s of the Nussbaum controller.

2.5.2 Pure-feedback system

In this example, we consider the problem of designing an adaptive controller for a pure feedback system considered in [45], [46], and [40]. In particular, we

consider the nonlinear system

$$\dot{x}_1 = x_1 + \left(x_2 + \frac{x_2^3}{5}\right)\theta_1, \quad (2.25)$$

$$\dot{x}_2 = x_1x_2 + \left(u + \frac{u^3}{7}\right)\theta_2, \quad (2.26)$$

where $x_1, x_2, u \in \mathbb{R}$. Let $\theta_1 = -3$ and $\theta_2 = 8$. The parameters θ_1, θ_2 are assumed to be unknown. The objective is to design a control law that stabilizes the origin without using the knowledge of the parameters θ_1 and θ_2 .

Note that the nonlinear system (2.25)-(2.26) is in pure feedback form. Furthermore, since

$$\frac{\partial g_1}{\partial x_2} = 1 + \frac{3x_2^2}{5}, \quad \frac{\partial g_2}{\partial u} = 1 + \frac{3u^2}{7} \quad (2.27)$$

are nonzero for all $x_1, x_2, u \in \mathbb{R}$, Assumption 2.2.4 is satisfied.

To design the stabilizing controller, let $V(x_1) = \frac{1}{2}x_1^2$ and $\kappa(x_1) = x_1$. The controller is then given by (2.8), that is,

$$\dot{u} = -\left(\frac{\partial e_3}{\partial u}\right)^{-1} \left(e_2 \hat{\theta} + \frac{\partial e_3}{\partial x_1} \left(x_1 + x_2 + \frac{x_2^3}{5} \right) + \frac{\partial e_3}{\partial x_2} \left(x_1x_2 + \left(u + \frac{u^3}{7} \right) \hat{\theta} \right) + \frac{\partial e_3}{\partial \hat{p}} + k_3 e_3 \right), \quad (2.28)$$

and the parameter estimators are

$$\dot{\hat{p}}_1 = \gamma_1 \text{sgn}(\theta_1) x_1^2, \quad (2.29)$$

$$\begin{aligned} \dot{\hat{p}}_2 = \gamma_2 \text{sgn}(\theta_2) e_2 & \left((x_1 + \hat{p}_1 + k_1 \text{sgn}(\theta_1)) \hat{\theta}_1 \right. \\ & \left. + (\hat{p}_1 + k_1 \text{sgn}(\theta_1)) x_1 + \left(1 + \frac{3x_2^2}{5} \right) x_1 x_2 + x_1 \dot{\hat{p}}_1 \right), \end{aligned} \quad (2.30)$$

$$\dot{\hat{\theta}}_1 = \alpha_1 e_2 \left(x_1 + (\hat{p}_1 + k_1 \text{sgn}(\theta_1)) \left(x_2 + \frac{x_2^3}{5} \right) \right), \quad (2.31)$$

$$\dot{\hat{\vartheta}}_1 = \alpha_2 e_3 \left(\frac{\partial e_3}{\partial x_1} \left(x_2 + \frac{x_2^3}{5} \right) \right), \quad (2.32)$$

$$\dot{\hat{\theta}}_2 = \alpha_3 e_3 \left(e_2 + u + \frac{u^3}{7} \right), \quad (2.33)$$

The explicit expressions of the variables in this example are given in Appendix A.2. Let the controller gains $k_1 = k_2 = k_3 = 1$ and the parameter estimator gains $\gamma_1 = \gamma_2 = \alpha_1 = \alpha_2 = \alpha_3 = 10$. In the estimators, we set $\hat{p}_1(0) = \hat{p}_2(0) = \hat{\theta}_1(0) = \hat{\vartheta}_1(0) = \hat{\theta}_2(0) = 0$.

The initial state is assumed to be $x_1(0) = 1, x_2(0) = 0.5$, and $u(0) = 0$.

Figure 2.3 shows the states x_1 and x_2 , the control signal u , and the estimates $\hat{p}_1, \hat{p}_2, \hat{\theta}_1, \hat{\theta}_2, \hat{\vartheta}_1$. Note that, as shown in Figure 2.3(c) and 2.3(d), the parameter estimates $\hat{p}_1, \hat{p}_2, \hat{\theta}_1, \hat{\theta}_2, \hat{\vartheta}_1$ cross zero multiple times, however, since the controller does not require inversion of the parameter estimates, the resulting closed-loop performance is not affected and closed-loop stability is guaranteed.

Remark 2.5.1 *In the two examples considered in this section, we emphasize that the transient performance of the closed-loop system can be improved by appropriately tuning the controller and estimator gains, or by choosing alternative functions*

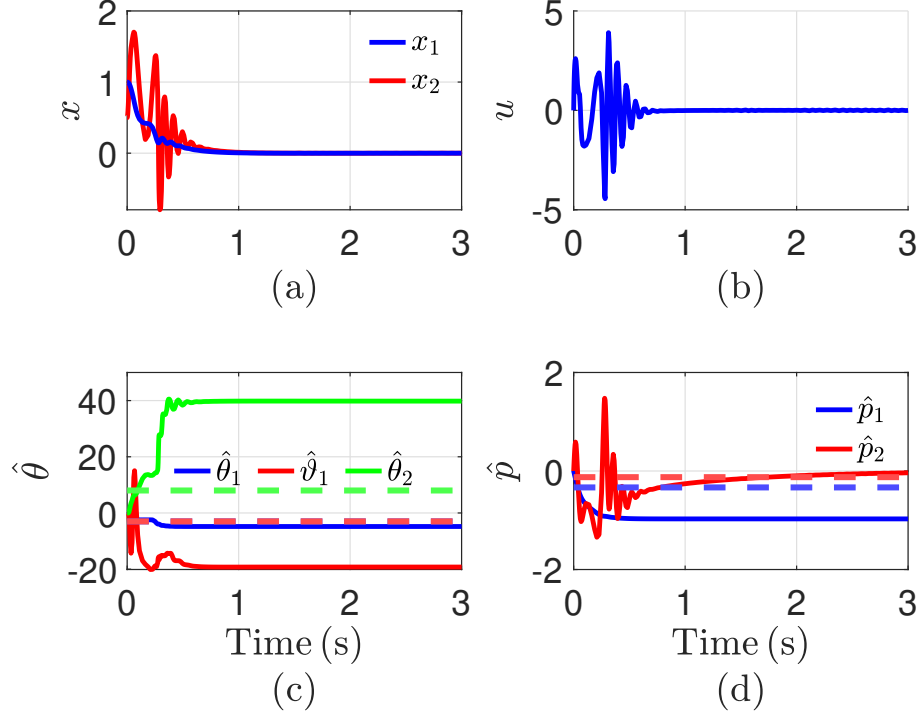


Figure 2.3: Closed-loop response of the system (2.25), (2.26), (2.28). (a) shows the states x_1 and x_2 , (b) shows control signal u , (c) shows the parameter estimates $\hat{\theta}_1$, $\hat{\vartheta}_1$, and $\hat{\theta}_2$, and (d) shows the parameter estimates $\hat{p}_1$ and $\hat{p}_2$. The corresponding true values of the parameters θ_1 and θ_2 are shown in dashed lines.

V and κ to obtain stronger convergence properties. Nevertheless, the origin of the closed-loop system remains asymptotically stable, as the control law does not involve inversion of parameter estimates. This avoids potential issues associated with inverting estimates that may approach or cross zero. In particular, an adaptive controller based on the classical backstepping approach would have failed, as it relies on the inversion of the parameter estimate.

Remark 2.5.2 The adaptive controller presented in this chapter ensures closed-loop stability at the origin, meaning that the system states converge to zero. However, it does not guarantee that the estimator states converge to the true parameters of

the system dynamics. Therefore, the estimator states should not be interpreted as accurate or reliable estimates of the actual system parameters.

2.6 Application to Bicopter control with Unknown Mass and Inertia

This section focuses on the control problem of a Bicopter, where both the mass and inertia are considered unknown for the purpose of controller design. The Bicopter is used as a representative example to illustrate the applicability of the proposed framework to higher-order systems.

Multicopters are extensively used in applications like precision agriculture [47], environmental surveys [48], [49], construction management [50], and load transportation [51]. Their growing popularity drives interest in enhancing their capabilities, but reliable control remains challenging due to nonlinear, time-varying dynamics and uncertain environments. Model-based control techniques for multicopters [52]–[54] depend on accurate model parameters, making them sensitive to uncertainties [55], [56]. Adaptive methods like model reference adaptive control [57], [58], L1 adaptive control [59], adaptive sliding mode control [60]–[62], and retrospective cost adaptive control [63], [64] address some of these issues, but often lack stability guarantees or require pre-existing stabilizing controllers.

Conventional control architectures separate the dynamics of multicopters into translational and rotational control loops. [65], which simplifies design since each loop is responsible for controlling a lower-order simpler system. This approach relies on the well-known time separation principle, where successive loops are designed to

be faster than the previous ones. Although this approach simplifies control system design, it does not guarantee closed-loop stability.

Adaptive backstepping-based approaches can guarantee the stability of the closed loop in multi-loop systems, but current adaptive backstepping approaches only apply to systems where the unknown parameter appears linearly in the dynamics map [42], [66]–[68]. However, in the case of multicopters, unknown parameters, which are the multicopter’s inertial properties, appear in the input map. Conventional adaptive backstepping methods fail for systems with uncertainties in the input map because the parameter estimates may lead to singular input mappings, rendering the control design invalid. Existing approaches, such as Nussbaum functions [34], [35] and tuning functions [30], [31], can address this issue but do not always guarantee closed-loop stability and may introduce unbounded internal signals.

This section focuses on developing an adaptive control system for a multicopter with guaranteed closed-loop stability. In particular, this section develops an adaptive controller for a bicopter system with unknown mass and inertia. A bicopter is a multicopter constrained to a vertical plane and thus is modeled by a 6th-order nonlinear system instead of a 12th-order nonlinear system. Despite the lower dimension of the state space, the 6th-order bicopter retains the complexities of the nonlinear dynamics of an unconstrained multicopter.

We consider the bicopter system described in Section II of [13]. The objective is to design an adaptive feedback control law that guarantees asymptotic tracking of a desired trajectory. As shown in our previous work in [1], the classical backstepping technique can not be used in a bicopter system due to the non-invertibility of the

input map. In a bicopter system, the input map is a tall matrix, and thus, the non-invertibility is structural rather than due to rank deficiency at a few points in the state space. As shown in [13], a dynamic extension of the equations of motion overcomes the structural non-invertibility of the input map. The derivation of the equations of motion and their dynamic extension is described in more detail in [13] and is briefly reviewed below to establish notation.

We consider the bicopter system shown in Figure 2.4 and described in detail in Section II of [13]. The equations of motion of a bicopter are

$$m\ddot{r}_1 = -F \sin(\theta), \quad m\ddot{r}_2 = F \cos(\theta) - mg, \quad J\ddot{\theta} = M, \quad (2.34)$$

where m and J are the mass and the moment of inertia of the bicopter, respectively, r_1 and r_2 are the horizontal and vertical positions of the center of mass, θ is the roll angle, and $F \triangleq f_1 + f_2$ and $M \triangleq (f_2 - f_1)\ell$ are the total force and the total moment applied to the bicopter, respectively, where ℓ is the length of the bicopter arm.

As shown in our previous work in [1], the backstepping framework can not be used with the bicopter system due to the non-invertibility of the input map. In a bicopter system, the input map is a tall matrix, and thus, the non-invertibility is structural rather than due to rank deficiency at a few points in the state space. As shown in [13], a dynamic extension of the equations of motion overcomes the structural non-invertibility of the input map. The derivation of the equations of motion and their dynamic extension is described in more detail in [1]. The dynamically

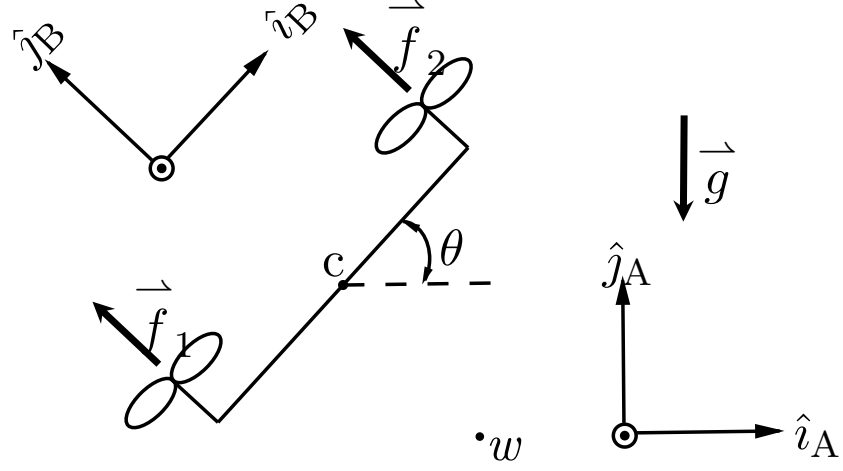


Figure 2.4: Bicopter configuration considered in this paper. The bicopter is constrained to the $\hat{i}_A - \hat{j}_A$ plane and rotates about the $\hat{k}_A$ (out of the page) axis of the inertial frame F_A . F_B is fixed to the bicopter such that the forces $\vec{f}_1$ and $\vec{f}_2$ are along $\hat{j}_B$. Note that $\vec{r}_{c/w} = r_1 \hat{i}_A + r_2 \hat{j}_A$.

extended equations of motion are

$$\dot{x}_1 = x_2, \quad (2.35)$$

$$\dot{x}_2 = f_2 + g_2(x_3), \quad (2.36)$$

$$\dot{x}_3 = x_4, \quad (2.37)$$

$$\dot{x}_4 = g_4 u, \quad (2.38)$$

where

$$x_1 \triangleq \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}, x_2 \triangleq \begin{bmatrix} \dot{r}_1 \\ \dot{r}_2 \end{bmatrix}, x_3 \triangleq \begin{bmatrix} \theta \\ F \end{bmatrix}, x_4 \triangleq \begin{bmatrix} \dot{\theta} \\ \dot{F} \end{bmatrix}, \quad (2.39)$$

$$u \triangleq \begin{bmatrix} \ddot{F} \\ M \end{bmatrix}, f_2 \triangleq \begin{bmatrix} 0 \\ -g \end{bmatrix}, \quad (2.40)$$

and the functions

$$g_2(x_3) \triangleq m^{-1} \begin{bmatrix} -S_{x_{31}} \\ C_{x_{31}} \end{bmatrix} x_{32}, \quad g_4 \triangleq \begin{bmatrix} 0 & J^{-1} \\ 1 & 0 \end{bmatrix}. \quad (2.41)$$

Note that the system (2.35)-(2.38) is in pure feedback form since x_3 appears non-affinely in (2.36). Finally, since g_4 is a constant nonsingular matrix, the backstepping framework can be used with the extended equations.

2.6.1 Controller Synthesis

In this section, we develop an adaptive controller based on the backstepping framework to stabilize the bicopter system. Unlike the controller designed in [13], the proposed controller does not require inversion of the parameter estimates.

In this work, to simplify the controller construction, we assume that the objective for the bicopter is to reach a constant waypoint; that is, the desired value of state x_1 is a constant. However, the following procedure can be easily extended to cases where the desired state trajectory consists of sufficiently smooth functions of time, as described in [69], by including the derivatives in the control law and the parameter update laws.

Let $\xi_1 \triangleq \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T \in \mathbb{R}^2$ denote the desired state and define the tracking error $e_1 \triangleq x_1 - \xi_1$.

2.6.1.1 e_1 stabilization

Consider the function

$$V_1 \triangleq \frac{1}{2} e_1^T e_1. \quad (2.42)$$

Differentiating (2.42) and using (2.35) yields $\dot{V}_1 = e_1^T \dot{x}_1 = e_1^T x_2$. Note that if $x_2 = -k_1 e_1$, where $k_1 > 0$, then $\dot{V}_1 < 0$. However, x_2 is not the control signal and thus cannot be chosen arbitrarily. Instead, following the backstepping process, we design a control law that yields the desired x_2 response, as follows.

Define

$$e_2 \triangleq x_2 - \xi_2, \quad \xi_2 \triangleq -k_1 e_1, \quad (2.43)$$

and $k_1 > 0$. Then, differentiating (2.42) yields

$$\dot{V}_1 = -k_1 e_1^T e_1 + e_2^T e_1. \quad (2.44)$$

2.6.1.2 e_2 stabilization

Define $p_1 \triangleq \theta_1^{-1}$ and consider the function

$$V_2 \triangleq V_1 + \frac{1}{2} e_2^T e_2 + \frac{1}{2} \gamma_1^{-1} |\theta_1| (\hat{p}_1 - p_1)^2, \quad (2.45)$$

where $\gamma_1 > 0$ and $\hat{p}_1$ is an estimate of p_1 . Differentiating (2.45) and using (2.44) yields $\dot{V}_2 = -k_1 e_1^T e_1 + e_2^T (e_1 + f_2 + g_2(x_3)\theta_1 + k_1 x_2) + \gamma_1^{-1} |\theta_1| (\hat{p}_1 - p_1) \dot{\hat{p}}_1$. However, $g_2(x_3)$ is not the control signal and thus can not be chosen arbitrarily. Instead, following the backstepping process, we design a control law that yields the desired $g_2(x_3)$ response, as follows. Define

$$e_3 \triangleq g_2(x_3) - \xi_3, \quad (2.46)$$

where

$$\xi_3 \triangleq -\hat{p}_1 (e_1 + f_2 + k_1 x_2) - k_2 \sigma(\theta_1) e_2, \quad (2.47)$$

$\sigma(\theta_1)$ is the sign of θ_1 , and $k_2 > 0$. Consider the parameter update law

$$\dot{\hat{p}}_1 = \gamma_1 \sigma(\theta_1) e_2^T (e_1 + f_2 + k_1 x_2). \quad (2.48)$$

Then, differentiating (2.45) yields

$$\dot{V}_2 = -k_1 e_1^T e_1 - k_2 |\theta_1| e_2^T e_2 + e_3^T e_2 \theta_1. \quad (2.49)$$

2.6.1.3 e_3 stabilization

Consider the function

$$V_3 \triangleq V_2 + \frac{1}{2} e_3^T e_3. \quad (2.50)$$

Differentiating (2.50) and using (2.49) yields $\dot{V}_2 = -k_1 e_1^T e_1 - k_2 |\theta_1| e_2^T e_2 + e_3^T (\theta_1 e_2 + \frac{\partial g(x_3)}{\partial x_3} x_4 + \dot{\hat{p}}_1 (e_1 + f_2 + k_1 x_2) + \hat{p}_1 (x_2 + k_1 f_2) + k_2 \sigma(\theta_1) (f_2 + k_1 x_2) + (k_1 \hat{p}_1 + k_2 \sigma(\theta_1)) g_2(x_3) \theta_1)$.

However, x_4 is not the control signal and thus can not be chosen arbitrarily. Instead, following the backstepping process, we design a control law that yields the desired x_4 response, as follows. Define

$$e_4 \triangleq x_4 - \xi_4, \quad (2.51)$$

where

$$\begin{aligned} \xi_4 \triangleq \mathcal{G}_2^{-1} & \left[\dot{\hat{p}}_1 (e_1 + f_2 + k_1 x_2) + \hat{p}_1 (x_2 + k_1 f_2) + k_2 \sigma(\theta_1) (f_2 + k_1 x_2) \right. \\ & \left. + (e_2 + (k_1 \hat{p}_1 + k_2 \sigma(\theta_1)) g_2(x_3)) \hat{\theta}_1 + k_3 e_3 \right], \end{aligned} \quad (2.52)$$

$k_3 > 0$ and

$$\mathcal{G}_2 \triangleq \frac{\partial g_2}{\partial x_3} = \begin{bmatrix} -C_{x_{3,1}}x_{3,2} & -S_{x_{3,1}} \\ -S_{x_{3,1}}x_{3,2} & C_{x_{3,1}} \end{bmatrix}. \quad (2.53)$$

Then, differentiating (2.50) yields

$$\begin{aligned} \dot{V}_3 = & -k_1 e_1^T e_1 - k_2 |\theta_1| e_2^T e_2 - k_3 e_3^T e_3 + e_4^T \mathcal{G}_2^T e_3 \\ & + (\theta_1 - \hat{\theta}_1) e_3^T (e_2 + (k_1 \hat{p}_1 + k_2 \sigma(\theta_1)) g_2(x_3)). \end{aligned} \quad (2.54)$$

2.6.1.4 e_4 stabilization

Define $p_2 \triangleq \theta_2^{-1}$ and consider the function

$$V_4 \triangleq V_3 + \frac{1}{2} e_4^T e_4 + \frac{1}{2} \gamma_2^{-1} |\theta_2| (\hat{p}_2 - p_2)^2 + \frac{1}{2} \alpha_1^{-1} (\theta_1 - \hat{\theta}_1)^2 + \frac{1}{2} \alpha_2^{-1} (\theta_1 - \hat{\vartheta}_1)^2, \quad (2.55)$$

where $\gamma_2 > 0$, $\alpha_1 > 0$, $\alpha_2 > 0$, $\hat{p}_2$ is an estimate of p_2 , and $\hat{\theta}_1$ and $\hat{\vartheta}_1$ are estimates of θ_1 . Differentiating (2.55) yields

$$\begin{aligned} \dot{V}_4 = & -k_1 e_1^T e_1 - k_2 |\theta_1| e_2^T e_2 - k_3 e_3^T e_3 + (\theta_1 - \hat{\theta}_1) e_3^T (e_2 + (k_1 \hat{p}_1 + k_2 \sigma(\theta_1)) g_2(x_3)) - \\ & (\theta_1 - \hat{\theta}_1) e_3^T \alpha_1^{-1} \dot{\hat{\theta}}_1 + e_4^T (g_4(\theta) u + \Sigma + \Psi \theta_1) + \gamma_2^{-1} |\theta_2| (\hat{p}_2 - p_2) \dot{\hat{p}}_2 - \alpha_2^{-1} (\theta_1 - \hat{\vartheta}_1) \dot{\hat{\vartheta}}_1, \end{aligned} \quad (2.56)$$

where

$$\begin{aligned}
\Sigma \triangleq & \mathbf{d}_t[\mathcal{G}_2^{-1}] \left[\dot{\hat{p}}_1 (e_1 + k_1 x_2 + f_2) + \hat{p}_1 (x_2 + k_1 f_2) + k_2 \sigma(\theta_1) (f_2 + k_1 x_2) \right. \\
& \left. + \hat{\theta}_1 (e_2 + k_1 \hat{p}_1 g_2(x_3) + k_2 \sigma(\theta_1) g_2(x_3)) + k_3 e_3 \right] \\
& + \mathcal{G}_2^{-1} \left[\gamma_1 \sigma(\theta_1) \left[(f_2^T + k_1 x_2^T) (e_1 + k_1 x_2 + f_2) + e_2^T (x_2 + k_1 f_2) \right] (e_1 + k_1 x_2 + f_2) \right. \\
& + 2 \dot{\hat{p}}_1 (x_2 + k_1 f_2) + 2 \hat{p}_1 f_2 + k_2 \sigma(\theta_1) k_1 f_2 \\
& + \dot{\hat{\theta}}_1 (e_2 + k_1 \hat{p}_1 g_2(x_3) + k_2 \sigma(\theta_1) g_2(x_3)) + \hat{\theta}_1 (f_2 + k_1 x_2 + k_1 \hat{p}_1 g_2(x_3) \\
& + k_1 \hat{p}_1 \mathcal{G}_2 x_4 + k_2 \sigma(\theta_1) \mathcal{G}_2 x_4) + k_3 (\mathcal{G}_2 x_4 + \dot{\hat{p}}_1 (e_1 + k_1 x_2 + f_2) + \hat{p}_1 (x_2 + k_1 f_2) \\
& \left. \left. + k_2 \sigma(\theta_1) (f_2 + k_1 x_2) \right) \right] \in \mathbb{R}^2, \tag{2.57}
\end{aligned}$$

$$\begin{aligned}
\Psi \triangleq & \mathcal{G}_2^{-1} \left[\gamma_1 \sigma(\theta_1) (g_2(x_3)^T \Phi) \Phi + \dot{\hat{p}}_1 k_1 g_2(x_3) + \hat{p}_1 g_2(x_3) + k_1 k_2 \sigma(\theta_1) g_2(x_3) + \hat{\theta}_1 g_2(x_3) \right. \\
& \left. + k_1 k_3 \hat{p}_1 g_2(x_3) + k_2 k_3 \sigma(\theta_1) g_2(x_3) \right] \in \mathbb{R}^2, \tag{2.58}
\end{aligned}$$

$$\Phi \triangleq e_1 + k_1 x_2 + f_2 \in \mathbb{R}^2. \tag{2.59}$$

Note that

$$\mathcal{G}_2^{-1} = \begin{bmatrix} -\frac{C_{x_{3,1}}}{x_{3,2}} & -\frac{S_{x_{3,1}}}{x_{3,2}} \\ -S_{x_{3,1}} & C_{x_{3,1}} \end{bmatrix}, \tag{2.60}$$

$$\mathbf{d}_t[\mathcal{G}_2^{-1}] = \begin{bmatrix} \frac{S_{x_{3,1}} x_{4,1}}{x_{3,2}} + \frac{C_{x_{3,1}} x_{4,2}}{x_{3,2}^2} & \frac{S_{x_{3,1}} x_{4,2}}{x_{3,2}^2} - \frac{C_{x_{3,1}} x_{4,1}}{x_{3,2}} \\ -C_{x_{3,1}} x_{4,1} & -S_{x_{3,1}} x_{4,1} \end{bmatrix}. \tag{2.61}$$

Finally, consider the control law

$$u = - \left[\begin{bmatrix} 0 & 1 \\ \hat{p}_2 & 0 \end{bmatrix} (\Sigma + \Psi \hat{\vartheta}_1) + k_4 \begin{bmatrix} 0 & 1 \\ \sigma(\theta_2) & 0 \end{bmatrix} e_4 \right], \quad (2.62)$$

where $k_4 > 0$ and the parameters update laws

$$\dot{\hat{p}}_2 = \gamma_2 \sigma(\theta_2) \begin{bmatrix} 1 & 0 \end{bmatrix} e_4 \begin{bmatrix} 1 & 0 \end{bmatrix} (\Sigma + \Psi \hat{\vartheta}_1), \quad (2.63)$$

$$\dot{\hat{\theta}}_1 = \alpha_1 e_3^T (e_2 + k_1 \hat{p}_1 g_2(x_3) + k_2 \sigma(\theta_1) g_2(x_3)), \quad (2.64)$$

$$\dot{\hat{\vartheta}}_1 = \alpha_2 \Sigma^T e_4. \quad (2.65)$$

Substituting the control law and the parameter update laws above in (2.56) yields

$$\dot{V}_4 = -k_1 e_1^T e_1 - k_2 |\theta_1| e_2^T e_2 - k_3 e_3^T e_3 - k_4 e_4^T \begin{bmatrix} |\theta_2| & 0 \\ 0 & 1 \end{bmatrix} e_4 \leq 0. \quad (2.66)$$

Note that the control (2.62) depends on Σ and Ψ , which in turn, require the inverse of $\mathcal{G}_2$, given by (2.60). The matrix $\mathcal{G}_2$, given by (2.53) is nonsingular if and only if $x_{3,2} = F \neq 0$. Although the control (2.62) does not explicitly guarantee that $F \neq 0$ for all $t \geq 0$, the net force required near any desired equilibrium is not zero, and thus the net force is not likely to converge to zero in well-posed problems. In practice, to deal with zero-crossings of the net force, projecting the net force F to $-\epsilon$ and ϵ , where $\epsilon > 0$ is a small number, if $F \in (-\epsilon, \epsilon)$, ensures well-posed computation of (2.61).

Remark 2.6.1 *Although the controller is formulated with $\ddot{F}$ and M as inputs, in the actual implementation only the force F and the moment M are applied to the system. The derivatives of F , including $\ddot{F}$, are employed solely in the computation*

of u , whose integration yields the applied value of F . Therefore, the implementation effectively relies only on F and M .

2.6.2 Stability Analysis

The following theorem uses the Barbashin-Krasovskii-LaSalle invariance principle, which originally appeared as Theorem 1 and 2 in [44], to establish the asymptotic stability of the closed-loop system (2.35), (2.36), (2.37), (2.38), (2.62). In particular, the Barbashin-Krasovskii-LaSalle's invariance principle is used to prove the asymptotic stability of $x_1 = \xi_1 = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T$, $x_2 = x_4 = u = 0$, $x_3 = \begin{bmatrix} 0 & p_1 g \end{bmatrix}^T = \begin{bmatrix} 0 & mg \end{bmatrix}^T$, $\hat{p}_1 = p_1$, $\hat{\theta}_1 = \theta_1$ and the boundedness of the remaining parameter estimates.

Next, Propositions 2.6.1, 2.6.2, 2.6.3, and 2.6.4 define sets used in the stability proof and establish their properties.

Proposition 2.6.1 *Consider the differential equations (2.35), (2.36), (2.37), (2.38), (2.48), (2.62), (2.63), (2.64), (2.65). Then, $x_1 = \xi_1 = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T$, $x_2 = x_4 = u = 0$, $x_3 = \begin{bmatrix} 0 & p_1 g \end{bmatrix}^T = \begin{bmatrix} 0 & mg \end{bmatrix}^T$, $\hat{p}_1 = p_1$, $\hat{\theta}_1 = \theta_1$ and any $\hat{p}_2, \hat{\vartheta}_1 \in \mathbb{R}$ is an equilibrium point.*

Proof: First, note that if $x_1 = \xi_1 = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T$, $x_2 = x_4 = u = 0$, $x_3 = \begin{bmatrix} 0 & p_1 g \end{bmatrix}^T = \begin{bmatrix} 0 & mg \end{bmatrix}^T$, then it follows from (2.35), (2.36), (2.37), and (2.38) that $\dot{x}_1 = \dot{x}_2 = \dot{x}_3 = \dot{x}_4 = 0$. Additionally, for any $\hat{p}_1, \hat{p}_2, \hat{\vartheta}_1 \in \mathbb{R}$, it follows from (2.61) that $d_t \text{inv} \mathcal{G}_2 = 0$, and thus it follows from (2.57) that $\Sigma = 0$. Then, it follows from (2.65) that $\dot{\hat{\vartheta}}_1 = 0$. Next, it follows from (2.43) that $e_2 = 0$, and thus it follows from

(2.48) that $\dot{\hat{p}}_1 = 0$. Next, it follows from (2.47) that $\xi_3 = -\hat{p}_1 f_2$. Thus, it follows from (2.46) that $e_3 = 0$, which along with (2.64) implies that $\dot{\hat{\theta}}_1 = 0$. Finally, it follows from (2.52) that $\xi_4 = 0$. Thus, it follows from (2.51) that $e_4 = 0$, which along with (2.63) implies that $\dot{\hat{p}}_2 = 0$. ■

Proposition 2.6.2 *Let $\ell > 0$. Define*

$$\Omega \triangleq \{x_1, x_2, x_3, x_4, u \in \mathbb{R}^2, \hat{p}_1, \hat{p}_2, \hat{\theta}_1, \hat{\vartheta}_1 \in \mathbb{R} : V_4 \leq \ell\}. \quad (2.67)$$

Then, Ω is bounded and a positively invariant set.

Proof: Since V_4 is radially unbounded and Ω is defined by $V_4 \leq \ell$, it follows from Theorem 2 in [44] that Ω is bounded. Since $\dot{V}_4 \leq 0$ and $V_4 > 0$, Ω is a positively invariant set. ■

Proposition 2.6.3 *Define $E \subset \Omega$ such that $x_1 = \xi_1 = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T$, $x_2 = x_4 = u = 0$, $x_3 = \begin{bmatrix} 0 & p_1 g \end{bmatrix}^T = \begin{bmatrix} 0 & m g \end{bmatrix}^T$, $\hat{p}_1 = p_1$, $\hat{\theta}_1 = \theta_1$. Then, E is the set of all points in Ω where $\dot{V}_4 = 0$.*

Proof: It follows from (2.56), that $\dot{V}_4 = 0$ if and only if $e_1 = e_2 = e_3 = e_4 = 0$, which necessarily requires $x_1 = \xi_1 = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T$, $x_2 = x_4 = u = 0$, $x_3 = \begin{bmatrix} 0 & p_1 g \end{bmatrix}^T = \begin{bmatrix} 0 & m g \end{bmatrix}^T$, $\hat{p}_1 = p_1$, $\hat{\theta}_1 = \theta_1$. ■

Proposition 2.6.4 *E is the largest positively invariant set in E .*

Proof: It follows from Proposition 2.6.1 that $x_1 = \xi_1 = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T$, $x_2 = x_4 = u = 0$, $x_3 = \begin{bmatrix} 0 & p_1 g \end{bmatrix}^T = \begin{bmatrix} 0 & m g \end{bmatrix}^T$, $\hat{p}_1 = p_1$, $\hat{\theta}_1 = \theta_1$ and any $\hat{p}_2, \hat{\vartheta}_1$ is an equilibrium point. Thus, E is the largest positively invariant set in E . ■

Theorem 2.6.1 Consider the system (2.35), (2.36), (2.37), (2.38), the controller (2.62), and the parameter update laws (2.48), (2.63), (2.64), (2.65), where $\gamma_1, \gamma_2, \alpha_1, \alpha_2, \alpha_3, k_1, k_2, k_3, k_4 > 0$. Then,

$$\lim_{t \rightarrow \infty} x_1(t) = \xi_1 = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T, \quad (2.68)$$

$$\lim_{t \rightarrow \infty} x_2(t) = 0, \quad (2.69)$$

$$\lim_{t \rightarrow \infty} x_3(t) = \begin{bmatrix} 0 & p_1 g \end{bmatrix}^T = \begin{bmatrix} 0 & mg \end{bmatrix}^T, \quad (2.70)$$

$$\lim_{t \rightarrow \infty} x_4(t) = 0, \quad (2.71)$$

$$\lim_{t \rightarrow \infty} u(t) = 0, \quad (2.72)$$

$$\lim_{t \rightarrow \infty} \hat{\theta}_1(t) = \theta_1, \quad (2.73)$$

$$\lim_{t \rightarrow \infty} \hat{p}_1(t) = p_1, \quad (2.74)$$

Proof: Consider the set Ω defined in Proposition 2.6.2 and the set E defined in Proposition 2.6.3. It follows from Proposition 2.6.4 that E is the largest positively invariant set in Ω . It follows from Theorem 2 in [44] that any solution starting in Ω converge to E as $t \rightarrow \infty$, which implies (2.68), (2.69), (2.70), (2.71), (2.72), (2.73), (2.74), and the estimates $\hat{p}_2, \hat{\vartheta}_1$ remain bounded. ■

Remark 2.6.2 It can not be concluded that $\hat{p}_2$ and $\hat{\vartheta}_1$ converge to p_2 and θ_1 . However, they remain bounded.

2.6.3 Numerical Simulation

In this section, we implement the adaptive backstepping controller developed earlier to address the trajectory-tracking problem. Specifically, the adaptive con-

troller is used to follow waypoints defined by a second-order Hilbert curve and an elliptical trajectory. Note that the adaptive controller is given by (2.62) with the parameter update laws (2.48), (2.63), (2.64), and (2.65). The control architecture is shown in Figure 2.5, and the estimation architecture is shown in Figure 2.6.

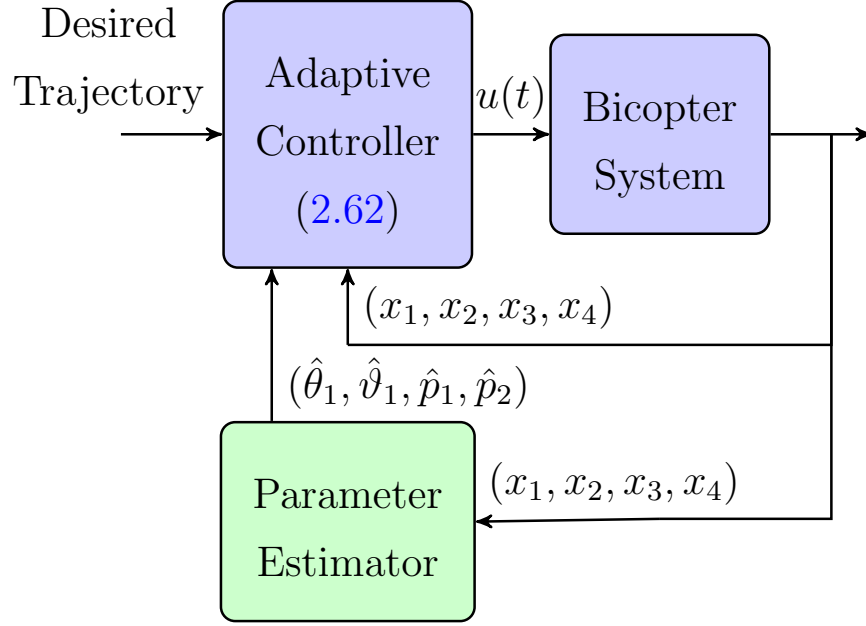


Figure 2.5: Inversion-free adaptive control architecture.

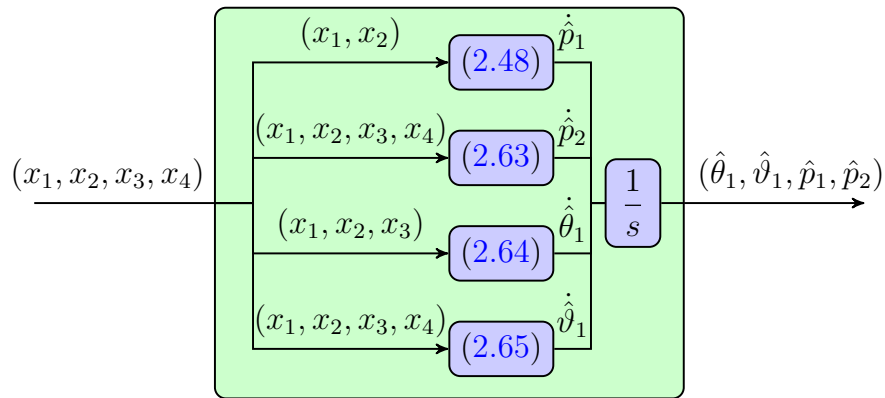


Figure 2.6: Parameter estimation architecture.

To simulate the bicopter, we assume that the mass of the bicopter m is 1 kg and its moment of inertia J is 0.2 kgm². In the controller, we set $k_1 = 1$, $k_2 = 5$,

$k_3 = 10$, and $k_4 = 20$. In the parameter update laws presented in Figure 2.6, we set the estimator gains $\gamma_1 = \gamma_2 = \alpha_1 = \alpha_2 = 0.1$, and the initial estimates $\hat{\theta}_1(0) = \hat{v}_1(0) = \hat{p}_1(0) = \hat{p}_2(0) = 0$.

2.6.3.1 Waypoint following

The bicopter is commanded to follow a waypoints defined by a second-order Hilbert curve. In particular, the trajectory is constructed using the algorithm described in Appendix A of [70] with a maximum velocity $v_{\max} = 1$ m/s and a maximum acceleration $a_{\max} = 1$ m/s². Figure 2.7 shows the trajectory-tracking response of the bicopter, where the desired trajectory is shown in black dashes, and the output trajectory response is shown in blue. Figure 2.8 shows the position r_1 and r_2 response, the roll angle ϕ response of the bicopter and the control u generated by the adaptive backstepping controller (2.62).

Figure 2.9 shows the absolute values of estimation errors $\hat{\theta}_1 - \theta_1$, $\hat{p}_1 - p_1$, $\hat{v}_1 - \theta_1$, and $\hat{p}_2 - p_2$ on a logarithmic scale. As shown in Theorem 2.6.1, for constant desired state, the estimation errors $\hat{\theta}_1 - \theta_1$ and $\hat{p}_1 - p_1$ converge asymptotically towards zero, and the estimation errors $\hat{v}_1 - \theta_1$ and $\hat{p}_2 - p_2$ remain bounded. Note that although the estimation errors $\hat{v}_1 - \theta_1$ also appear to converge asymptotically towards zero, the paper's main result does not guarantee it nor requires it.

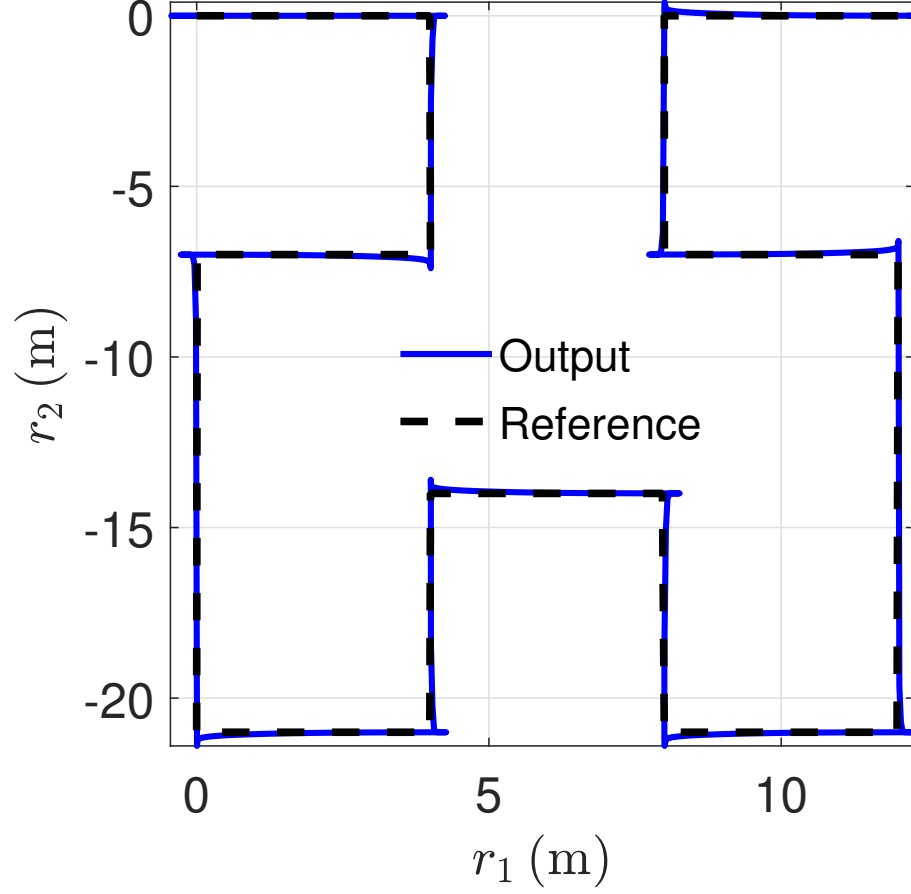


Figure 2.7: **Waypoint following.** Tracking response of the bicopter with adaptive backstepping controller. Note that the output trajectory is in solid blue, and the desired trajectory is in dashed black.

2.6.3.2 Elliptical Trajectory

Next, the bicopter is commanded to follow an elliptical trajectory given by

$$r_{d1}(t) = 5 \cos(\psi) - 5 \cos(\psi) \cos(\omega t) - 3 \sin(\psi) \sin(\omega t), \quad (2.75)$$

$$r_{d2}(t) = 5 \sin(\psi) - 5 \sin(\psi) \cos(\omega t) + 3 \cos(\psi) \sin(\omega t), \quad (2.76)$$

where $\psi = 45 \text{ deg}$ and $\omega = 0.1 \text{ rad} \cdot \text{s}^{-1}$. Figure 2.10 shows the trajectory-tracking response of the bicopter, where the desired trajectory is shown in black dashes, and the output trajectory response is shown in blue. Figure 2.11 shows the position

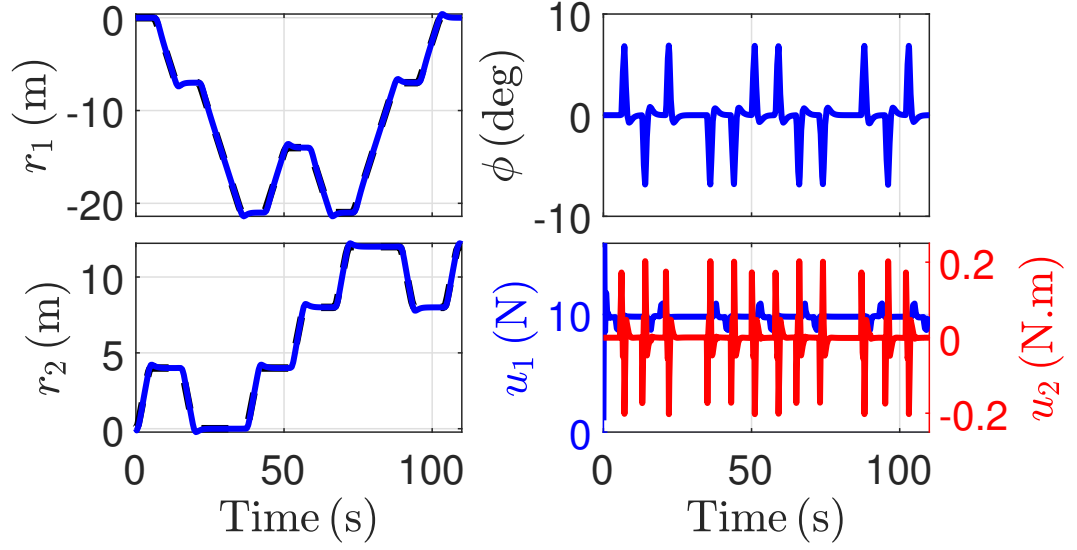


Figure 2.8: **Waypoint following.** Positions (r_1, r_2) and roll angle ϕ response of the bicopter with adaptive backstepping controller. Note that the references r_{d1}, r_{d2} are shown in black dashed lines

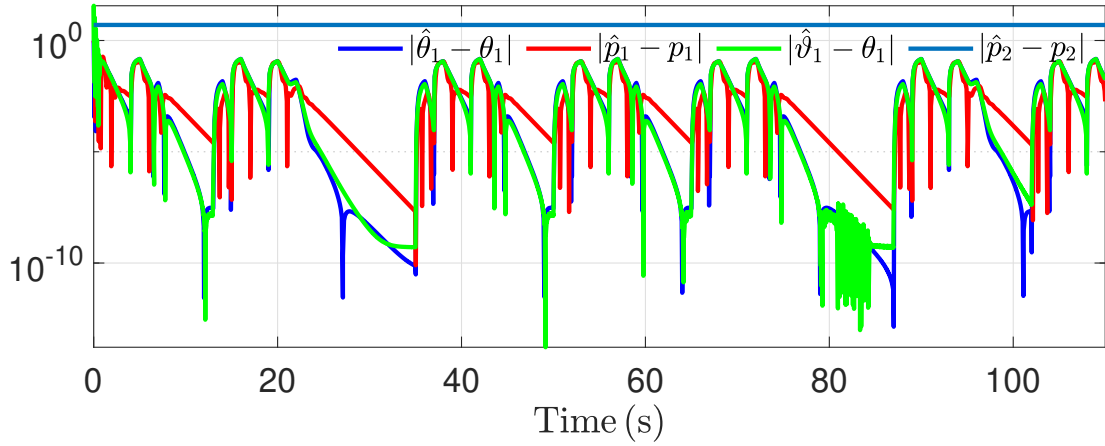


Figure 2.9: **Waypoint following.** Estimation errors obtained with parameter update laws (2.64), (2.65), (2.48), and (2.63).

r_1, r_2 response, the roll angle ϕ response of the bicopter and the control u generated by the adaptive backstepping controller (2.62).

Figure 2.12 shows the estimation errors $\hat{\theta}_1 - \theta_1, \hat{v}_1 - \theta_1, \hat{p}_1 - p_1, \hat{p}_2 - p_2$ on a logarithmic scale. Note that all parameter estimates remain bounded. In the adaptive

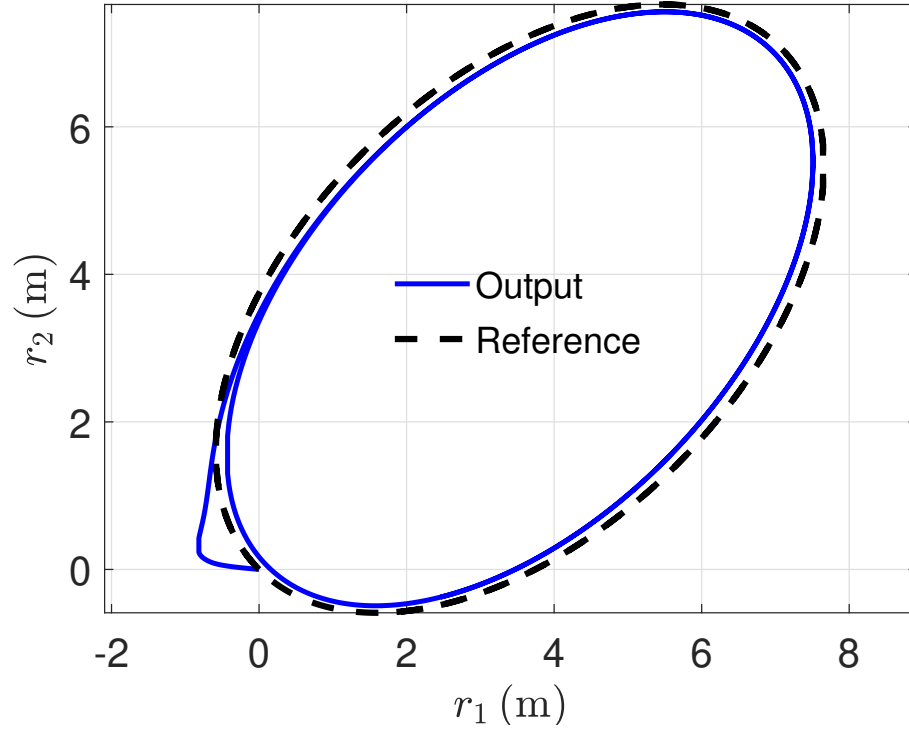


Figure 2.10: **Elliptical trajectory.** Tracking response of the bicopter with the adaptive backstepping controller. Note that the output trajectory is shown in solid blue, and the reference trajectory is in dashed black.

controller design, since the parameter update laws are chosen to cancel undesirable factors and not to estimate the parameters, the estimates do not necessarily need to converge ¹.

¹ The simulation files are in <https://github.com/estimation-control-learning-laboratory/inverse-free-adaptive-control.git>

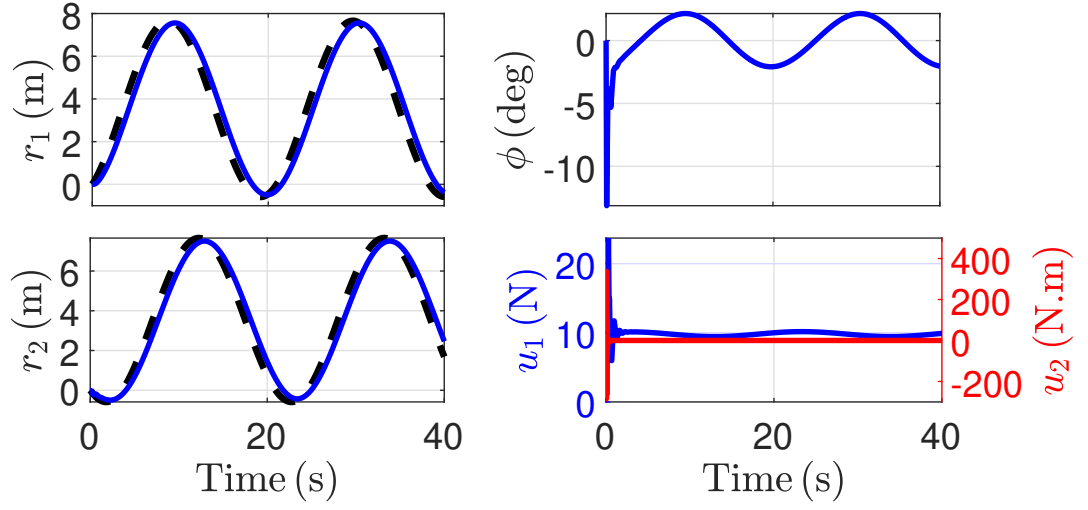


Figure 2.11: **Elliptical trajectory**. Position (r_1, r_2) , roll angle ϕ response and the input signal applied to the bicopter obtained with adaptive backstepping controller (2.62). Note that the references r_{d1}, r_{d2} are shown in black dashed lines

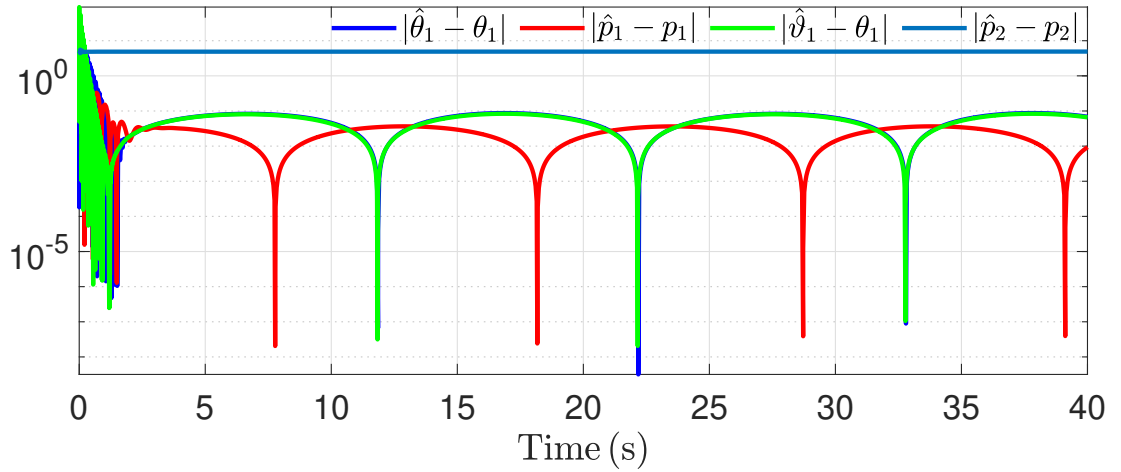


Figure 2.12: **Elliptical trajectory**. Estimation errors obtained with parameter update laws (2.64), (2.65), (2.48), and (2.63).

2.7 Conclusions

This chapter presented an adaptive control technique that eliminates the need to invert parameter estimates and guarantees closed-loop stability for a class of nonlinear systems with uncertain input maps. The technique, based on the backstepping procedure, applies to pure feedback systems with linearly parameterized uncertainty in the input maps. A rigorous theoretical analysis is presented to prove the asymptotic stability of the proposed approach. Numerical simulations demonstrate the effectiveness of the proposed technique for second-order nonlinear systems in strict and pure feedback form, and for a Bicopter with unknown mass and inertia.

Chapter 3

A Constraint-Lifting Framework for Safe and Stable Nonlinear Control

In practical applications, systems are inherently subject to *state constraints*. For instance, it may be necessary to constrain the position of a body to prevent collisions, limit its velocity to reduce momentum or ensure safety, and bound its acceleration to enhance passenger comfort or mitigate mechanical stress. When such constraints are not explicitly incorporated into the controller design, undesirable phenomena may arise. A notable example is *integrator windup*, which occurs in controllers containing integral action, commonly used to eliminate steady-state errors. When the actuator reaches its saturation limit, the integrator continues to accumulate error as if the actuator were still responding linearly. This accumulation leads to a significant mismatch between the commanded and actual control inputs. As a result, once the actuator returns to its normal operating range, the excessively large integral term drives the system beyond its intended behavior, causing overshoot, oscillations, or even instability. Therefore, explicitly accounting for state constraints during the controller synthesis is essential to ensure that the integrator remains bounded and that the overall closed-loop performance remains safe under practical operating conditions.

In this chapter, we develop a new control framework that addresses the prob-

lem of enforcing constraint satisfaction while guaranteeing the stability of the closed loop. This chapter also presents a rigorous stability analysis of the resulting closed-loop system. Furthermore, the motivation underlying the key components of the framework is discussed in detail. Finally, we demonstrate the applicability of the proposed approach to two representative examples: the attitude tracking problem with constrained Euler angles and the Bicopter control problem with constrained positions and velocities.

3.1 Introduction

The problem of enforcing state and output constraints in nonlinear systems is an emerging theme in control theory, with significant relevance in the safety-critical applications in aerospace and robotics. Classical control theory primarily addresses the stabilization and command-following problems. State-of-the-art approaches to handle state constraints often use techniques, such as saturation functions [71] and reference governors [72]–[75]. However, designing a controller that enforces state constraints while guaranteeing stability remains an open challenge. This chapter thus addresses the *state-constrained control problem* by developing a control framework that ensures user-specified state constraints are satisfied while maintaining closed-loop stability.

While optimal control frameworks can be formulated to handle constraints, the resulting constrained optimization problems are typically complex and rarely admit analytical solutions. Model predictive control (MPC), a subclass of optimal control

techniques, provides a systematic framework for handling constraints by embedding them into an optimization problem [76], [77]. However, MPC has several well-known limitations. First, solving the full nonlinear optimization problem is computationally demanding, and in many cases, infeasible, thereby limiting its applicability in real-time embedded implementations [27]. To circumvent this problem, linearization techniques are often employed to reduce the optimization problem to a quadratic program [28]; however, this approach inevitably introduces modeling errors due to approximations. Furthermore, because constraints are enforced over a prediction horizon, modeling errors arising from linearizations and/or approximations can lead to constraint violations. Finally, due to the computational burden associated with solving the optimization problem at each sampling instant, performance guarantees cannot be maintained in fast or highly dynamic applications.

Incorporating constraints into Lyapunov-based formulations represents an alternative approach to handling constraints. Nevertheless, the presence of constraints usually introduces singularities in the resulting control synthesis. Barrier Lyapunov functions (BLFs), a class of these methods, provide a framework to handle constraints by using Lyapunov candidate functions that grow unbounded near constraint boundaries [15]–[17]. Although BLFs can theoretically prevent constraint violations, they typically introduce singular terms into the resulting controller [78], [79]. Mitigating these singularities requires careful gain tuning, and stability guarantees often rely on conservative assumptions [80]. Furthermore, as shown in [31], [81], [82], the BLF framework enforces constraints on the error states rather than directly on the system states, thereby limiting its application.

Recently, control barrier functions (CBFs) have emerged as an alternative tool for enforcing safety constraints [21]–[24], [83]. CBFs [18]–[20] are based on Nagumo’s invariance principle [14], [84], which introduced the notion of *lower integrals* to guarantee positive forward invariance of a prescribed set. In particular, Nagumo’s invariance principle yields a *safety inequality* that ensures the system’s state remains within the prescribed safe set if the inequality is satisfied. Note that the CBFs themselves do not generate a control signal; instead, they modify a control signal to enforce safety. In practice, the CBFs are implemented as discrete-time safety filters that, at each step, solve a constrained optimization problem to find a signal as close as possible to the nominal control signal while satisfying the safety inequality [85]. Since the CBFs are necessarily implemented using sample data, this complicates the closed-loop stability analysis. Ad-hoc techniques such as fixed margins can be used to mitigate inter-sample violations, but they generally degrade performance. Control Lyapunov functions (CLFs), similar to CBFs, have been proposed to guarantee simultaneous safety and stability, but these approaches ultimately rely on solving a constrained optimization problem to compute the control signal. Furthermore, the construction of the forward invariant set becomes nontrivial in the case where the relative degree of the CBF is greater than one [86].

An alternative approach, which also motivates the constraint-lifting framework developed in this chapter, uses state transformations to embed constraints into the system dynamics [21], [25], [26]. In particular, continuous functions that diverge to infinity at the constraint boundaries are used to transform the system dynamics. However, these nonlinear mappings may induce numerical instability

Method	Central Idea	Limitations	Contrast with Proposed Method
MPC	Embeds constraints in an online optimization framework.	High computational cost; requires solving a nonlinear or quadratic program at each step; prone to inter-sample violations.	Provides an explicit control law without online optimization, avoiding inter-sample constraint issues.
BLF	Uses Lyapunov functions that diverge near constraint boundaries.	Contains singular terms causing numerical ill-conditioning; constraints are often applied to error states.	Employs smooth sigmoid-integral Lyapunov functions, thus improving numerical conditioning.
CBF	Uses Nagumo's principle to ensure forward invariance.	Requires solving QPs at every instant; discrete-time filters complicate stability.	Yields an analytic control law with continuous enforcement and guaranteed stability.
State Transformation	Maps constrained space to unconstrained via a nonlinear diffeomorphism.	Mappings diverge near boundaries, causing numerical instability and compounded ill-conditioning.	Uses smooth sigmoid-based maps, ensuring bounded gradients and numerical robustness.

Table 3.1: Comparison of existing constraint-enforcing control techniques with the proposed method.

near the constraint boundaries. Furthermore, constraints that are applied to the entire state result in the composition of several such nonlinear functions, leading to

even more pronounced numerical instabilities. Ad hoc techniques, such as switching controllers, have been investigated to address numerical instability [87].

This chapter introduces the constraint-lifting framework for designing a control law that simultaneously enforces constraints and guarantees closed-loop stability for a class of nonlinear systems. Similar to state-transformation techniques, the proposed framework converts a state-constrained control problem into an equivalent unconstrained one, as shown in Figure 1.2, by *lifting* the constrained state space into an unconstrained state space. In contrast to existing approaches, however, this work introduces a novel Lyapunov function that mitigates the numerical instabilities commonly observed in prior methods. Specifically, a controller is synthesized for the unconstrained system using the classical backstepping approach with the proposed Lyapunov function, called the *sigmoid integral function*. This function exhibits quadratic behavior near the origin and linear growth away from it, a key property that eliminates the exponentially growing terms responsible for numerical ill-conditioning near constraint boundaries in conventional Lyapunov-based control designs. Unlike MPC and CBF-based approaches, the proposed method yields an explicit control law rather than requiring the solution of an optimization problem at each sampling instant, thereby eliminating inter-sample constraint violations. Furthermore, unlike BLF- and state-transformation-based techniques, the proposed method avoids the use of singular terms, thereby preventing numerical ill-conditioning. The comparison of existing constraint-enforcing control techniques with the proposed approach is summarized in Table 3.1.

The main contributions of this chapter are as follows. First, a constraint-

lifting, sigmoid-based diffeomorphic map is introduced to reformulate the constrained control problem as an equivalent unconstrained one. Second, a Lyapunov function, called the sigmoid integral function, is proposed, which exhibits quadratic behavior near the origin and linear behavior away from it, thereby eliminating exponentially growing terms introduced by the state transformation. Third, a rigorous stability analysis of the resulting closed-loop system is provided. Finally, the effectiveness of the proposed method is demonstrated through the constrained attitude control problem.

3.2 Problem Formulation

This section establishes the notation, formulates the control problem, and introduces the sigmoid-function-based mappings used in the controller synthesis.

3.2.1 Notation

Let $q \in \mathbb{R}^n$, define

$$D(q) \triangleq \text{diag}(q_1, q_2, \dots, q_n) \in \mathbb{R}^{n \times n}. \quad (3.1)$$

If, for $i = 1, \dots, n$, $q_i \neq 0$, then define

$$D_I(q) \triangleq \text{diag}(q_1^{-1}, \dots, q_n^{-1}) \in \mathbb{R}^{n \times n}. \quad (3.2)$$

Note that $D_I(q)^{-1} = D(q)$. The set of all $q \in \mathbb{R}^n$ such that each $|q_i| < 1$ is denoted by $\mathcal{C}$,

$$\mathcal{C} \triangleq \{x \in \mathbb{R}^n : \|x\|_\infty < 1\}. \quad (3.3)$$

Note that $\mathcal{C}$ is a unit norm ball induced by the infinity norm. The set of n -dimensional real vectors with strictly positive components is denoted by $\mathbb{R}_+^n$, that is,

$$\mathbb{R}_+^n \triangleq \{x \in \mathbb{R}^n : x_i > 0, \forall i = 1, \dots, n\}. \quad (3.4)$$

3.2.2 System Description

Consider the system

$$\dot{x}_1 = g_1(x_1)x_2, \quad (3.5)$$

$$\dot{x}_2 = f_2(x_1, x_2) + g_2(x_1, x_2)u, \quad (3.6)$$

where $x_1, x_2 \in \mathbb{R}^n$ denote the states, and $u \in \mathbb{R}^n$ is the control input. Let $g_1 : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$, $f_2 : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, and $g_2 : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ be continuously differentiable functions of sufficient order.

The objective is to design a control law that asymptotically stabilizes a desired state x_{1d} and ensures that the system states remain bounded within user-specified limits. Specifically, the states (x_1, x_2) are required to remain inside a safe set $\mathcal{S}$ defined as

$$\mathcal{S} \triangleq \{(x_1, x_2) \in \mathbb{R}^n \times \mathbb{R}^n : |x_1| < \bar{x}_1, |x_2| < \bar{x}_2\}, \quad (3.7)$$

where $\bar{x}_1, \bar{x}_2 \in \mathbb{R}_+^n$ denote the user-specified upper bounds on x_1 and x_2 , respectively.

In other words, the objective is to design a control law that renders $(x_{1d}, 0)$ is asymptotically stable and guarantees the forward invariance of $\mathcal{S}$.

The following assumptions are required for controller synthesis.

Assumption 3.2.1 *The function $f_2(x_1, x_2) = 0$ if and only if $x_2 = 0$.*

Assumption 3.2.2 *Matrices $g_1(x_1)$ and $g_2(x_1, x_2)$ are nonsingular for $(x_1, x_2) \in \mathcal{S}$.*

3.3 Sigmoid and Associated Functions

This section provides a brief review of the *sigmoid functions* used to transform the state x for the construction of a control law that ensures asymptotic stability of the desired state x_{1d} and guarantees forward invariance of $\mathcal{S}$. In addition, these sigmoid functions are used to construct the *constraint-lifting function* and to define a class of Lyapunov functions, called *sigmoid integral functions*, that will later facilitate the controller synthesis.

3.3.1 Sigmoid Functions

We revisit the definition and key properties of *sigmoid functions*, presented in [88], including the fact that a strictly increasing sigmoid function has an inverse.

Definition 3.3.1 (Simple sigmoids) *A function $\sigma: \mathbb{R} \rightarrow (-1, 1)$ is a simple sigmoid if it satisfies the following conditions.*

1. σ is a smooth function, that is, $\sigma(x)$ is C^∞ .
2. σ is an odd function, that is, $\sigma(-x) = -\sigma(x)$.
3. σ has $y = \pm 1$ as horizontal asymptotes, that is, $\lim_{x \rightarrow \infty} \sigma(x) = 1$.
4. $\sigma(x)/x$ is a completely convex function in $(0, 1)$.

The following result, reproduced from [88], establishes that if the simple sigmoid is strictly increasing, then its inverse exists.

Proposition 3.3.1 *Let $y = \sigma(x)$ be a strictly increasing simple sigmoid (that is, $\forall x \in \mathbb{R}, \sigma'(x) > 0$.) Then,*

1. $\eta \triangleq \sigma^{-1} : (-1, 1) \rightarrow \mathbb{R}$ exists.
2. $\eta(y)$ is a strictly increasing function, analytic in the interval $(-1, 1)$.
3. $\eta'(y) = 1/\sigma'(\eta(y))$, where η' and σ' are the first derivatives of η and σ , respectively.
4. $\eta(y)/y$ is absolutely monotone in $(0, 1)$.

3.4 Constraint-Lifting Function

Using the strictly increasing sigmoid functions presented above, we define the *constraint-lifting function*, which is used to convert a constrained control problem into an equivalent unconstrained problem. In particular, the constraint-lifting function $\phi: \mathcal{C} \rightarrow \mathbb{R}^n$ is constructed using the inverse of a strictly increasing sigmoid function, thereby providing a diffeomorphic transformation from the constrained state x to the unconstrained state z , which will be defined in Section 3.6. Similarly, the mapping $\psi: \mathbb{R}^n \rightarrow \mathcal{C}$, used to recover x from z , is constructed using the strictly increasing sigmoid function.

In particular, the constraint-lifting function is

$$\phi(x) \triangleq \begin{bmatrix} \eta_1(x_1) & \eta_2(x_2) & \dots & \eta_n(x_n) \end{bmatrix}^T \in \mathbb{R}^n, \quad (3.8)$$

where each η_i is the inverse of a strictly increasing sigmoid function. Note that the constraint-lifting function maps an element from $\mathcal{C}$ to $\mathbb{R}^n$. Similarly, the inverse of the constraint-lifting function is

$$\psi(z) \triangleq \begin{bmatrix} \sigma_1(z_1) & \sigma_2(z_2) & \dots & \sigma_n(z_n) \end{bmatrix}^T \in \mathbb{R}^n, \quad (3.9)$$

where each σ_i is the inverse of η_i . Table 3.2 lists several strictly increasing sigmoid functions and their inverses in the second and the first column, respectively, and Figure 3.1 shows these functions and their inverses.

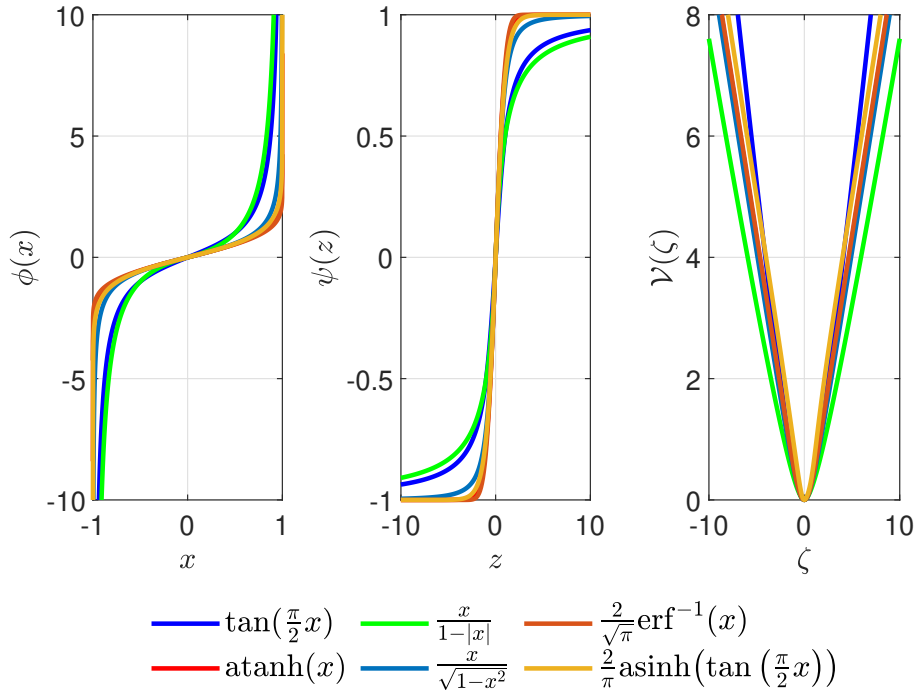


Figure 3.1: Various sigmoid functions, their inverses, and the corresponding sigmoid integral function. Note that the legend includes only the $\phi(x)$ functions, while the associated $\psi(z)$ and $\nu(\zeta)$ functions are shown using the same color scheme to maintain visual consistency.

Remark 3.4.1 *For the controller synthesis in Section 3.6 and the example in Section 3.8, the same constraint-lifting function is used for both state variables to simplify the presentation. However, the components of the constraint-lifting function,*

$\phi(x) = \eta(x)$	$\psi(z) = \sigma(z)$	$\mathcal{V}(\zeta)$
$\tan\left(\frac{\pi}{2}x\right)$	$\frac{2}{\pi}\text{atan}(z)$	$\zeta \text{atan}(\zeta) - \frac{1}{2} \log(1 + \zeta^2)$
$\text{atanh}(x)$	$\tanh(z)$	$\log(\cosh(\zeta))$
$\frac{x}{1 - x }$	$\frac{z}{1 + z }$	$ \zeta - \log(1 + \zeta)$
$\frac{x}{\sqrt{1 - x^2}}$	$\frac{z}{\sqrt{1 + z^2}}$	$\sqrt{1 + \zeta^2} - 1$
$\frac{2}{\sqrt{\pi}}\text{erf}^{-1}(x)$	$\text{erf}\left(\frac{\sqrt{\pi}}{2}z\right)$	$\zeta \text{erf}\left(\frac{\sqrt{\pi}}{2}\zeta\right) + \frac{2}{\pi} \left(e^{-\frac{\pi}{4}\zeta^2} - 1 \right)$
$\frac{2}{\pi}\text{asinh}\left(\tan\left(\frac{\pi}{2}x\right)\right)$	$\frac{2}{\pi}\text{atan}\left(\sinh\left(\frac{\pi}{2}z\right)\right)$	$\frac{2}{\pi}\zeta \text{atan}\left(\sinh\frac{\pi}{2}\zeta - \int_0^{\frac{\pi}{2}\zeta} r \text{sech}(r) dr\right)$

Table 3.2: Various sigmoid functions, their inverses, and the corresponding sigmoid integral function.

that is, the sigmoid functions σ_i in (3.8) and their corresponding inverses η_i in (3.9) may be selected differently depending on the application.

Remark 3.4.2 Note that the constraint-lifting function is defined using the inverse of the sigmoid function, whereas its own inverse is constructed using the sigmoid function itself. This clarification is emphasized to avoid confusion arising from the use of the term “inverse” in both contexts.

The following property of the constraint-lifting function is used in the stability analysis of the closed-loop system, presented in Section 3.7.

Proposition 3.4.1 *Consider the constraint-lifting function (3.8). Then, for all $x \in \mathcal{C}$, the Jacobian $\partial_x \phi(x)$ is nonsingular.*

Proof: The Jacobian $\partial_x \phi(x)$ is a diagonal matrix whose diagonal entries are strictly positive. Since all diagonal elements are positive, $\partial_x \phi(x)$ is nonsingular. ■

3.5 Sigmoid Integral Function

Using the sigmoid functions presented above, we define the *sigmoid integral function*. The sigmoid integral function will be used to construct candidate Lyapunov functions in Section 3.6.

Definition 3.5.1 *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a simple sigmoid function. The sigmoid integral function $\mathcal{V} : \mathbb{R} \rightarrow [0, \infty)$ is defined as*

$$\mathcal{V}(\zeta) \triangleq \int_0^\zeta \sigma(s) ds. \quad (3.10)$$

The sigmoid integral function defined above has following easily verifiable properties.

1. $\mathcal{V}(0) = 0$, and for all $\zeta \in \mathbb{R} \setminus \{0\}$, $\mathcal{V}(\zeta) > 0$.
2. $\lim_{\zeta \rightarrow \pm\infty} \mathcal{V}(\zeta) = \pm\infty$.
3. $\partial_\zeta \mathcal{V}(\zeta) = \sigma(\zeta)$, and thus the sigmoid integral function is continuous.

Note that the sigmoid integral functions $\mathcal{V}(\zeta)$ exhibit quadratic behavior near $\zeta = 0$ and approach linear behavior for $|\zeta| \gg 0$. Table 3.2 lists the sigmoid integral

functions $\mathcal{V}(\zeta)$ corresponding to each sigmoid function in the third column and Figure 3.1 shows these sigmoid integral functions.

3.5.1 Motivation for the Sigmoid Integral Function as a Lyapunov Candidate

The sigmoid integral functions, introduced in Section 3.5, are selected as the Lyapunov candidate functions for control synthesis because the classical quadratic Lyapunov function leads to a control law that exhibits numerical instability arising from specific ill-conditioned terms. In contrast, employing the sigmoid integral function eliminates these problematic terms, yielding a numerically robust closed-loop system. The following example demonstrates the numerical difficulties associated with the classical quadratic Lyapunov function. Consider the double integrator

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = u, \quad (3.11)$$

where $x_1, x_2, u \in \mathbb{R}$. To simplify presentation, we consider the problem of only constraining x_2 with $\bar{x}_2 = 1$. Note that x_1 is unconstrained. Following the procedure outlined in Section 3.6.1, the transformed system in z coordinates is

$$\dot{z}_1 = \tanh(z_2), \quad \dot{z}_2 = \cosh^2(z_2) u. \quad (3.12)$$

Next, following the procedure outlined in Section 3.6.2 with $\phi(x) = \text{atanh}(x)$ as the constraint lifting function, the controller is

$$u = -k_2 \text{sech}^2(z_2) (\tanh(z_2) + k_1 e_1), \quad (3.13)$$

which implies that

$$\dot{z}_2 = -k_2 (\tanh(z_2) + k_1 e_1). \quad (3.14)$$

If, instead of the sigmoid integral function, the classical quadratic function were used, that is,

$$V = V_1 + \frac{1}{2} e_2^T e_2, \quad (3.15)$$

then $\dot{V} = -k_1 e_1^2 + e_2 (e_1 + k_1 \tanh(z_2) + u)$. Furthermore, the control law

$$u = -(e_1 + k_1 \tanh(z_2) + k_2 e_2), \quad (3.16)$$

where $k_1, k_2 > 0$ ensure that $\dot{V} = -k_1 e_1^2 - k_2 e_2^2 \leq 0$. Moreover, z_2 satisfies

$$\dot{z}_2 = -\cosh^2(z_2) (e_1 + k_1 \tanh(z_2) + k_2 e_2). \quad (3.17)$$

Note that the variables e_1 and e_2 for both formulations are exactly the same and are given by $e_1 = z_1 - z_{1d}$, and $e_2 = \tanh(z_2) + k_1 e_1$. A straightforward Lyapunov analysis shows that the closed-loop dynamics are asymptotically stable in both cases. However, $\dot{z}_2$ in (3.17) includes the term $\cosh(z_2)^2$, while $\dot{z}_2$ in (3.14) does not. As shown in Figure 3.2, the function $\cosh(z_2)^2$ grows exponentially, implying that $\dot{z}_2$ in (3.17) also increases exponentially, which leads to numerical instability.

We emphasize that both controllers theoretically guarantee asymptotic stability, as established by the Lyapunov analysis. However, the corresponding closed-loop dynamics exhibit numerical instability, which imposes practical limitations on implementation. This behavior has been consistently observed in multiple numerical investigations with various choices of constraint lifting functions. The numerical instability arising from the classical quadratic Lyapunov function thus motivates the

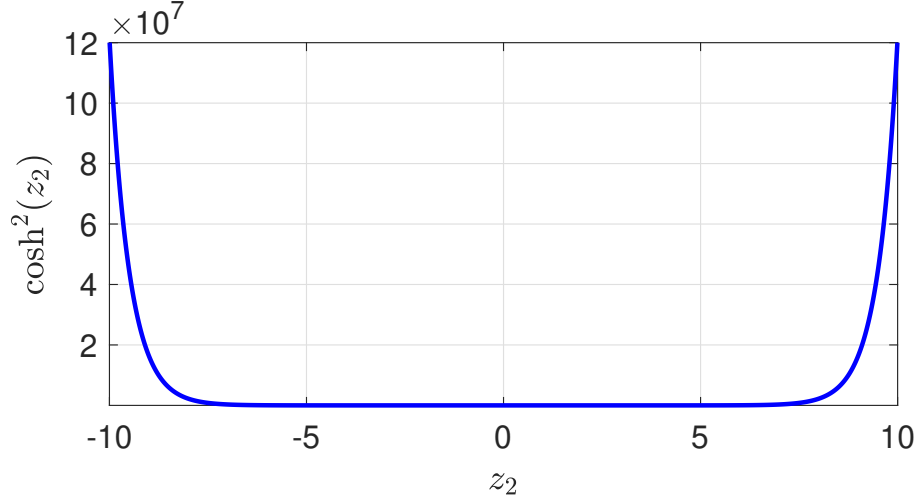


Figure 3.2: The function $\cosh^2(z_2)$. Note the exponential growth in $\cosh^2(z_2)$. For a value of $z_2 \approx 10$, $\cosh^2(z_2) \approx 10^7$.

use of sigmoid integral functions for constructing a numerically robust Lyapunov candidate.

3.6 Controller Synthesis

This section develops a control law that guarantees the forward invariance of the safe set $\mathcal{S}$ for the system described by (3.5)–(3.6) while simultaneously guaranteeing asymptotic stability of the desired state x_{1d} .

3.6.1 State Transformation

The states subject to constraints are *lifted* using the constraint-lifting function to obtain an unconstrained representation in the lifted state space. The key idea of transforming the constrained dynamics into an equivalent unconstrained form is illustrated in Figure 3.3.

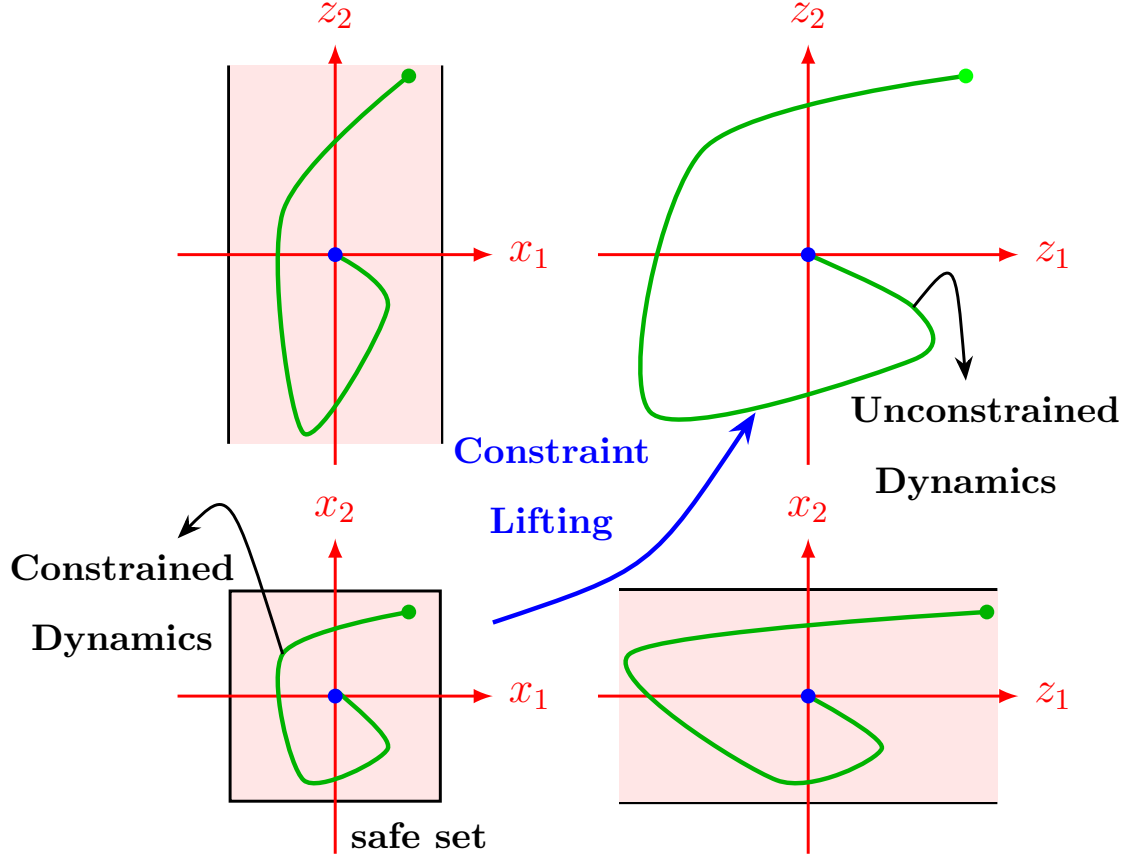


Figure 3.3: Illustration of the constraint-lifting concept, where the constrained dynamics in the original state space (x_1, x_2) are transformed into equivalent unconstrained dynamics in the lifted state space (z_1, z_2) .

In particular, to facilitate controller synthesis, the states (x_1, x_2) are first transformed to intermediate state variables (χ_1, χ_2) such that the bounds on components of x_1 and x_2 are normalized to ± 1 . This transformation ensures that each component of χ_1 and χ_2 lies within $(-1, +1)$ whenever $(x_1, x_2) \in \mathcal{S}$, that is, $\chi_1, \chi_2 \in \mathcal{C}$. Next, (χ_1, χ_2) are transformed to $(z_1, z_2) \in \mathbb{R}^n$ through a constraint-lifting function, which are detailed in Section 3.4. Due to the properties of the constraint-lifting function, while $\chi_1, \chi_2 \in \mathcal{C}$, the transformed variables z_1, z_2 are unconstrained, that is, $z_1, z_2 \in \mathbb{R}^n$. A stabilizing controller is then designed in the z -coordinates for

the transformed x_{1d} . Finally, an additional transformation from (z_1, z_2) to (ζ_1, ζ_2) is introduced to simplify the notation used in mapping z back to x as well as the stability analysis. The state transformations from x to z and the intermediate state variables χ and ζ are shown in Figure 3.4.

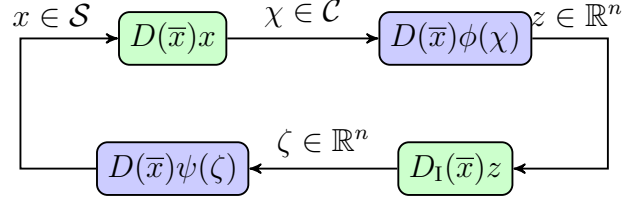


Figure 3.4: State transformations with the constraint-lifting function and its inverse used in the control synthesis.

Define

$$\chi_1 \triangleq D_I(\bar{x}_1)x_1, \quad (3.18)$$

$$\chi_2 \triangleq D_I(\bar{x}_2)x_2. \quad (3.19)$$

Note that the state variables χ_1 and χ_2 are defined to map the constraint boundaries to ± 1 , that is, when the constraints are satisfied, then, $\chi_1, \chi_2 \in \mathcal{C}$.

Next, define

$$z_1 \triangleq D(\bar{x}_1)\phi(\chi_1), \quad (3.20)$$

$$z_2 \triangleq D(\bar{x}_2)\phi(\chi_2), \quad (3.21)$$

where $\phi: \mathcal{C} \rightarrow \mathbb{R}^n$ is a constraint-lifting function as defined in Section 3.4.

Remark 3.6.1 *For simplicity, the same constraint-lifting function ϕ is used for both z_1 and z_2 . This assumption is made without loss of generality, as the subse-*

quent formulation can be easily extended to the case where distinct constraint-lifting functions are used for each state.

Next, define

$$\zeta_1 \triangleq D_1(\bar{x}_1)z_1, \quad (3.22)$$

$$\zeta_2 \triangleq D_1(\bar{x}_2)z_2. \quad (3.23)$$

The state variables ζ_1 and ζ_2 are defined to simplify the controller synthesis, as described below.

Finally, define

$$x_1 = D(\bar{x}_1)\psi(\zeta_1), \quad (3.24)$$

$$x_2 = D(\bar{x}_2)\psi(\zeta_2), \quad (3.25)$$

where $\psi : \mathbb{R}^n \rightarrow \mathcal{S}$ is an inverse of the constraint-lifting function as defined in Section 3.4.

With the state transformations defined above, the dynamics (3.5), (3.6) in terms of the transformed state z is

$$\dot{z}_1 = \mathcal{G}_1(z_1, z_2), \quad (3.26)$$

$$\dot{z}_2 = \mathcal{F}_2(z_1, z_2) + \mathcal{G}_2(z_1, z_2)u, \quad (3.27)$$

where

$$\mathcal{G}_1(z_1, z_2) \triangleq \Phi(\zeta_1)\psi(\zeta_2), \quad (3.28)$$

$$\mathcal{F}_2(z_1, z_2) \triangleq [\partial_{\chi_2}\phi(\psi(\zeta_2))] f_2(D(\bar{x}_1)\psi(\zeta_1), D(\bar{x}_2)\psi(\zeta_2)), \quad (3.29)$$

$$\mathcal{G}_2(z_1, z_2) \triangleq [\partial_{\chi_2}\phi(\psi(\zeta_2))] g_2(D(\bar{x}_1)\psi(\zeta_1), D(\bar{x}_2)\psi(\zeta_2)), \quad (3.30)$$

and

$$\Phi(\zeta_1) \triangleq [\partial_{\chi_1} \phi(\psi(\zeta_1))] g_1(D(\bar{x}_1)\psi(\zeta_1))D(\bar{x}_2). \quad (3.31)$$

Note that ζ_1 and ζ_2 are given by (3.22), (3.23). Algebraic details of the derivation of dynamics in z coordinates are shown in Appendix B.1.

The following propositions establish key properties of the functions $\mathcal{G}_1$, $\mathcal{F}_2$, and $\mathcal{G}_2$, which are subsequently used in the controller design and stability analysis.

Proposition 3.6.1 *Consider $\mathcal{G}_1(z_1, z_2) \in \mathbb{R}^n$ given by (3.28). Then, $\mathcal{G}_1(z_1, 0) = 0$.*

Proof: Since $\psi(0) = 0$ by definition, it follows from (3.28) that $\mathcal{G}_1(z_1, 0) = \Phi(D_1(\bar{x}_1)z_1)\psi(0) = 0$. ■

Proposition 3.6.2 *Consider $\mathcal{F}_2(z_1, z_2) \in \mathbb{R}^n$ given by (3.29). Then, $\mathcal{F}_2(z_1, 0) = 0$.*

Proof: Since $\psi(0) = 0$ by definition, it follows from Assumption 3.2.1 that $f_2(D(\bar{x}_1)\psi(\zeta_1), 0) = 0$, which in turn implies $\mathcal{F}_2(z_1, 0) = 0$. ■

Proposition 3.6.3 *Consider $\mathcal{G}_2(z_1, z_2) \in \mathbb{R}^{n \times n}$ given by (3.30). Then, $\mathcal{G}_2(z_1, z_2)$ is nonsingular.*

Proof: It follows from Proposition 3.4.1 that $\partial_{\chi_2} \phi(\psi(\zeta_2)) = \partial_{\chi_2} \phi(\chi_2)$ is a diagonal matrix with strictly positive elements on the diagonal. Next, it follows from Assumption 3.2.2 that $g_2(D(\bar{x}_2)\psi(\zeta_1), D(\bar{x}_2)\psi(\zeta_2))$ is nonsingular. Since the product of two nonsingular matrices is a nonsingular matrix, it follows that $\mathcal{G}_2(z_1, z_2)$ is nonsingular. ■

Proposition 3.6.4 Consider $\Phi(\zeta_1) \in \mathbb{R}^{n \times n}$ given by (3.31). Then, $\Phi(\zeta_1)$ is nonsingular.

Proof: It follows from Proposition 3.4.1 that $\partial_{\chi_1} \phi(\psi(\zeta_1)) = \partial_{\chi_1} \phi(\chi_1)$ is a diagonal matrix with strictly positive elements on the diagonal. Next, it follows from Assumption 3.2.2 that $g_1(D(\bar{x}_2)\psi(\zeta_1), \cdot)$ is nonsingular. Furthermore $D(\bar{x}_2)$ is positive definite. Since the product of nonsingular matrices is a nonsingular matrix, it follows that $\Phi(\zeta_1)$ is nonsingular. $\blacksquare$

3.6.2 Controller Synthesis

Let $x_{1d} \in \mathcal{S}$ be the desired value of x_1 . Let $\chi_{1d} \in \mathcal{C}$ and z_{1d} be the corresponding desired state values, that is,

$$\chi_{1d} \triangleq D_1(\bar{x}_1)x_{1d}, \quad (3.32)$$

$$z_{1d} \triangleq D(\bar{x}_1)\phi(\chi_{1d}). \quad (3.33)$$

Define the tracking error

$$e_1 \triangleq z_1 - z_{1d}. \quad (3.34)$$

3.6.2.1 e_1 stabilization

Consider the function

$$V_1 \triangleq \frac{1}{2}e_1^T e_1. \quad (3.35)$$

Differentiating (3.35) and using (3.26) yields

$$\dot{V}_1 = e_1^T \mathcal{G}_1(z_1, z_2). \quad (3.36)$$

If z_2 is chosen such that $\mathcal{G}_1(z_1, z_2) = -k_1 e_1$, where $k_1 > 0$, then $\dot{V}_1 < 0$. However, since z_2 is not the control input, $\mathcal{G}_1(z_1, z_2)$ cannot be arbitrarily chosen. Instead, we define

$$e_2 \triangleq \mathcal{G}_1(z_1, z_2) - (-k_1 e_1) = \Phi(\zeta_1)\psi(\zeta_2) + k_1 e_1. \quad (3.37)$$

and design the control law to ensure that e_2 converges to 0, thus implying that $\mathcal{G}_1(z_1, z_2)$ converges to $-k_1 e_1$. Substituting $\mathcal{G}_1(z_1, z_2)$ from (3.37) into (3.36) yields

$$\dot{V}_1 = -k_1 e_1^T e_1 + e_2^T e_1. \quad (3.38)$$

3.6.2.2 e_2 stabilization

Next, consider the function

$$V \triangleq V_1 + \sum_{i=1}^n \mathcal{V}(\zeta_{2i}), \quad (3.39)$$

where $\mathcal{V}$ is a sigmoid integral function given by (3.10). As shown in Appendix B.2, note that

$$d_t \sum_{i=1}^n \mathcal{V}(\zeta_{2i}) = \psi(\zeta_2)^T D_1(\bar{x}_2) \dot{z}_2. \quad (3.40)$$

Thus,

$$\dot{V} = -k_1 e_1^T e_1 + e_2^T e_1 + \psi(\zeta_2)^T D_1(\bar{x}_2) \left(\mathcal{F}_2(z_1, z_2) + \mathcal{G}_2(z_1, z_2) u \right). \quad (3.41)$$

Substituting $\psi(\zeta_2)$ from (3.37) in the equation above yields

$$\dot{V} = -k_1 e_1^2 + e_2^T e_1 + (e_2^T - k_1 e_1^T) \Phi^{-T}(\zeta_1) D_1(\bar{x}_2) \left(\mathcal{F}_2(z_1, z_2) + \mathcal{G}_2(z_1, z_2) u \right). \quad (3.42)$$

Finally, choosing

$$u = -\mathcal{G}_2(z_1, z_2)^{-1} \left(\mathcal{F}_2(z_1, z_2) + k_2 D(\bar{x}_2) \Phi^T(\zeta_1) e_2 \right) \quad (3.43)$$

yields

$$\begin{aligned} \dot{V} &= -k_1 e_1^T e_1 + (1 + k_1 k_2) e_2^T e_1 - k_2 e_2^T e_2 \\ &= -k_1 e_1^T e_1 + (1 + k_1 k_2) e_2^T e_1 - k_2 e_2^T e_2 + 2\sqrt{k_1 k_2} e_1^T e_2 - 2\sqrt{k_1 k_2} e_1^T e_2 \\ &= -\left(\sqrt{k_1} e_1 - \sqrt{k_2} e_2 \right)^T \left(\sqrt{k_1} e_1 - \sqrt{k_2} e_2 \right) + (1 + k_1 k_2 - 2\sqrt{k_1 k_2}) e_2^T e_1. \end{aligned} \quad (3.44)$$

Note that If $k_2 = k_1^{-1}$, then,

$$\dot{V} = -\left(\sqrt{k_1} e_1 - \sqrt{k_2} e_2 \right)^T \left(\sqrt{k_1} e_1 - \sqrt{k_2} e_2 \right). \quad (3.45)$$

Furthermore, if $k_1 e_1 = e_2$, then $\dot{V} = 0$.

3.7 Stability Analysis

The stability analysis of the closed-loop dynamics with the proposed control law uses the Barbashin-Krasovskii-LaSalle invariance principle, which originally appeared as Theorems 1 and 2 in [44]. In particular, the Barbashin-Krasovskii-LaSalle's invariance principle is used to prove the asymptotic stability of the equilibrium point.

Propositions 3.7.1, 3.7.2, 3.7.3, and 3.7.4 introduced below define the sets used in the stability analysis and establish their properties.

Proposition 3.7.1 [*Equilibrium solution.*] Consider the system (3.26), (3.27).

Let $z_{1d} \in \mathbb{R}^n$. Then, the solution

$$(z_1, z_2, u) = (z_{1d}, 0, 0) \quad (3.46)$$

is an equilibrium point.

Proof: If $z_2 = 0$, then it follows from Proposition 3.6.1 that $\mathcal{G}_1(z_1, z_2) = 0$, which implies that $\dot{z}_1 = 0$. Furthermore, it follows from Proposition 3.6.2 that $\mathcal{F}_2(z_1, z_2) = 0$. Moreover, if $u = 0$, then $\dot{z}_2 = 0$, implying that $(z_1, z_2, u) \equiv (z_{1d}, 0, 0)$ is an equilibrium point. $\blacksquare$

Proposition 3.7.2 Consider the system (3.26), (3.27). Consider the Lyapunov candidate function (3.39). Let $k_2 = k_1^{-1}$. Let $\ell > 0$. Define the sublevel set

$$\Omega \triangleq \{(z_1, z_2, u) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n : V \leq \ell\}. \quad (3.47)$$

Then, Ω is bounded and a positively invariant set.

Proof: Since V is radially unbounded and Ω is defined by $V \leq \ell$, it follows from Theorem 2 in [44] that Ω is bounded. Since $\dot{V} \leq 0$ and $V > 0$, V is non-increasing along the trajectory of the system. Therefore, any trajectory starting in Ω cannot leave Ω , thus implying that Ω is a positively invariant set. $\blacksquare$

Proposition 3.7.3 Consider the system (3.26), (3.27). Consider the function (3.39) and the positively invariant set Ω , given in (3.47). Let $k_2 = k_1^{-1}$. Define $E \subset \Omega$ such that $z_2 = 0$. Then, E is the set of all the points in Ω such that $\dot{V} = 0$.

Proof: Note that it follows from (3.23) that $\zeta_2 = 0$ if and only if $z_2 = 0$. Next, $\zeta_2 = 0$ implies that $\psi(\zeta_2) = 0$, which further implies that $\Phi(\zeta_1)\psi(\zeta_2) = 0$. Next, it

follows from (3.37) that $k_1 e_1 = e_2$. Finally, since $k_2 = k_1^{-1}$, it follows from (3.44) that $\dot{V} = 0$ if and only if $k_1 e_1 = e_2$. $\blacksquare$

Proposition 3.7.4 *Consider the system (3.26), (3.27). Consider the function (3.39). Let $k_2 = k_1^{-1}$. Let $M \subset E$ such that $u = 0$. Then, M is the largest positively invariant set in E .*

Proof: It follows from Proposition 3.7.1 that the point $z_1 = z_{1d}$, $z_2 = u = 0$ is an equilibrium of the system. Hence, the set of all equilibrium points constitutes a positively invariant subset of E . Next, consider the case where $u \neq 0$ and $z_2 = 0$. From (3.27) and the fact that $\mathcal{G}_2(z_1, z_2)$ is nonsingular for all $z_1, z_2 \in \mathbb{R}^n$, it follows that $\dot{z}_2 \neq 0$. Consequently, the trajectory leaves the set E . Therefore, $M \subset E$ is the largest positively invariant subset of E . $\blacksquare$

Theorem 3.7.1 *Consider the system (3.5), (3.6) and the system (3.26), (3.27). Assume that the initial conditions of (3.5), (3.6) are inside the safe set, that is, $(x_1(0), x_2(0)) \in \mathcal{S}$, and let $|x_{1d}| < \bar{x}_1$. Consider the control law (3.43), with $k_2 = k_1^{-1}$. Then,*

$$\lim_{t \rightarrow \infty} z_1(t) = z_{1d}, \quad (3.48)$$

$$\lim_{t \rightarrow \infty} z_2(t) = 0, \quad (3.49)$$

$$\lim_{t \rightarrow \infty} u(t) = 0, \quad (3.50)$$

$$\lim_{t \rightarrow \infty} x_1(t) = x_{1d}, \quad (3.51)$$

$$\forall t \geq 0, (x_1(t), x_2(t)) \in \mathcal{S}. \quad (3.52)$$

Proof: Consider the set Ω defined in Proposition 3.7.2 and the set E defined in Proposition 3.7.3. It follows from Proposition 3.7.4 that M is the largest positively invariant set in E . It follows from Theorem 2 in [44] that any solution starting in Ω converge to M as $t \rightarrow \infty$, which implies (3.48)–(3.50). Next, it trivially follows from (3.33) and (3.48) that $\lim_{t \rightarrow \infty} x_1(t) = x_{1d}$. Finally, since the map ψ is a bijection, it follows that $\forall t \geq 0, (x_1(t), x_2(t)) \in \mathcal{S}$. ■

3.8 Application to Safe Attitude Control

This section considers the constrained attitude control problem. Although Euler angles offer an intuitive and compact representation of attitude dynamics and are often preferred over quaternions and direction cosine matrices for visualization purposes, they suffer from the well-known *gimbal lock* phenomenon [89]. Accordingly, the objective of this example is to design a control law that enables tracking of an arbitrary attitude command within the prescribed safe set, while ensuring the body's attitude remains strictly away from gimbal lock configuration.

Consider a rigid body $\mathcal{B}$ with a body fixed-frame F_D . Let F_A be an inertial frame. Assume that F_D is related to F_A by a 3-2-1 Euler angle sequence, that is,

$$F_A \xrightarrow{\psi_3} F_B \xrightarrow{\theta_2} F_C \xrightarrow{\phi_1} F_D, \quad (3.53)$$

where F_B is obtained by rotating F_A about its third axis, F_C is obtained by rotating F_B about its second axis, and F_D is obtained by rotating F_C about its first axis. Note that each rotation is an Euler rotation and the corresponding angle is the Euler angle.

Let $q \triangleq \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^T \in \mathbb{R}^3$ denote the vector of Euler angles ϕ, θ , and ψ . Then, the orientation of $\mathcal{B}$ parameterized by q satisfies the Poisson's equation

$$\dot{q} = S(q)^{-1}\omega, \quad (3.54)$$

where $\omega \in \mathbb{R}^3$ is the angular velocity vector of F_D relative to F_A measured in the body-fixed frame F_D and

$$S(q) \triangleq \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & (\sin \phi) \cos \theta \\ 0 & -\sin \phi & (\cos \phi) \cos \theta \end{bmatrix}. \quad (3.55)$$

Note that, if $\theta \neq \pm\pi/2$, then $S(q)$ is nonsingular. Otherwise, the inverse of $S(q)$, is given by

$$S(q)^{-1} = \begin{bmatrix} 1 & \sin(\phi) \tan(\theta) & \cos(\phi) \tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) \sec(\theta) & \cos(\phi) \sec(\theta) \end{bmatrix}. \quad (3.56)$$

Note that the gimbal lock arises from the singularity of the matrix $S(q)$ at $\theta = \pm\pi/2$. As θ approaches $\pm\pi/2$, certain trigonometric terms in $S(q)$ become unbounded, which can result in ill-conditioned computations and undesirable controller behavior. The control objective is thus to enforce the constraint $\theta \in [-\bar{\theta}, \bar{\theta}]$, where $\bar{\theta} < \pi/2$.

Next, the angular velocity ω satisfies the Euler's equation

$$J\dot{\omega} + \omega \times J\omega = \tau, \quad (3.57)$$

where $J \in \mathbb{R}^{3 \times 3}$ is the moment of inertia matrix of $\mathcal{B}$ in F_D and $\tau \in \mathbb{R}^3$ is the external torque applied to $\mathcal{B}$ as measured in F_D .

The attitude dynamics (3.54), (3.57) is in the form similar to (3.5), (3.6), where $x_1 = q$, $x_2 = \omega$,

$$g_1(x_1) = S(x_1)^{-1}, \quad (3.58)$$

$$f_2(x_1, x_2) = J^{-1}x_2 \times Jx_2, \quad (3.59)$$

$$g_2(x_1, x_2) = J^{-1}. \quad (3.60)$$

Note that Assumptions 3.2.1 and 3.2.2 are satisfied.

In this numerical example, the safe set is defined by $\bar{q} \triangleq \begin{bmatrix} \pi/3 & \pi/3 & \pi/3 \end{bmatrix}^T$, and $\bar{\omega} \triangleq \begin{bmatrix} 0.03 & 0.02 & 0.01 \end{bmatrix}^T$. These constraints imply that each of the Euler angles should remain smaller than $\pi/3$ radians in magnitude and the components of the angular velocity vector should not exceed 0.03 rad/s, 0.02 rad/s, and 0.01 rad/s, respectively in magnitude.

To simulate the attitude dynamics, the body-fixed frame F_D is assumed to coincide with the principal-axis frame. Consequently, the inertia matrix is diagonal, that is, $J = \text{diag}(J_1, J_2, J_3)$, where we set $J_1 = 1$, $J_2 = 2$, and $J_3 = 3$ in this example. The initial conditions to simulate (3.54) and (3.57) and the attitude commands in $\mathcal{S}$ are randomly generated from a uniform distribution over $\mathcal{S}$. In particular, we randomly generate ten initial conditions and attitude commands.

The attitude controller is designed following the procedure outlined in Section 3.6, with the control law given by (3.43). In particular, we use the $\phi(x) = \text{atanh}(x)$ as the constraint-lifting function. Consequently, $\psi(z)$ and $\mathcal{V}(\zeta)$ are given by the second line in Table 3.2. In this numerical example, the control gain is set to $k_1 = 0.1$.

Figure 3.5 shows the closed-loop Euler angle responses for ten randomly generated initial conditions and desired references. The pink-shaded region indicates the safe set for the Euler angles. For each trajectory, the initial attitude is marked by a triangle, and a square of the same color shows the corresponding command.

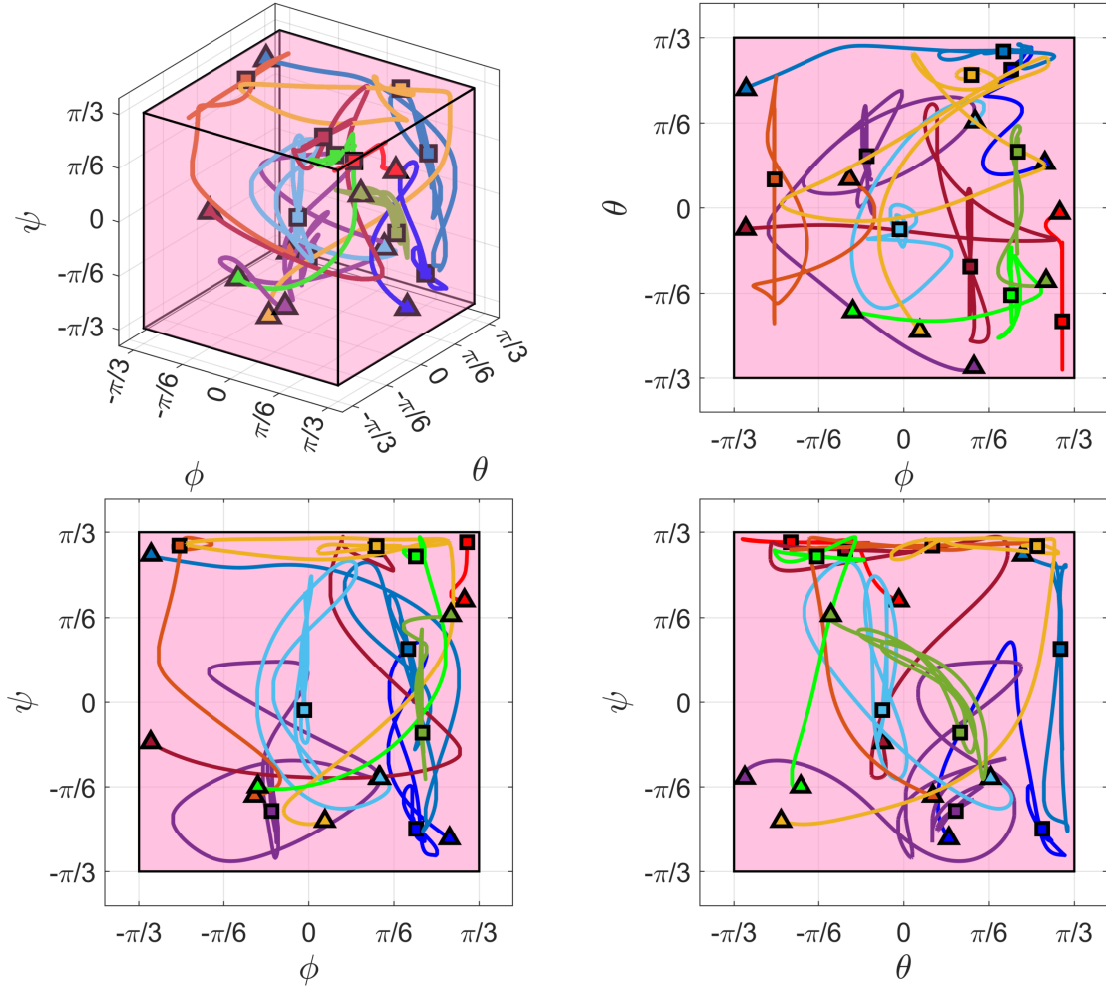


Figure 3.5: Closed-loop angular response of (3.54), (3.57), (3.43) randomly initialized and commanded. The desired angle values are marked with squares, the randomly generated initial conditions are marked with a triangle, and the safe set is shown in pink. Note that the angular response is always contained within the desired safe set $\mathcal{S}$.

Figure 3.6 shows the corresponding closed-loop angular velocity responses.

The yellow-shaded region indicates the safe set for the angular velocities. In each trajectory, the initial angular velocity is marked by a triangle. Since the control objective is attitude stabilization, the angular velocity converges to zero in all trajectories.

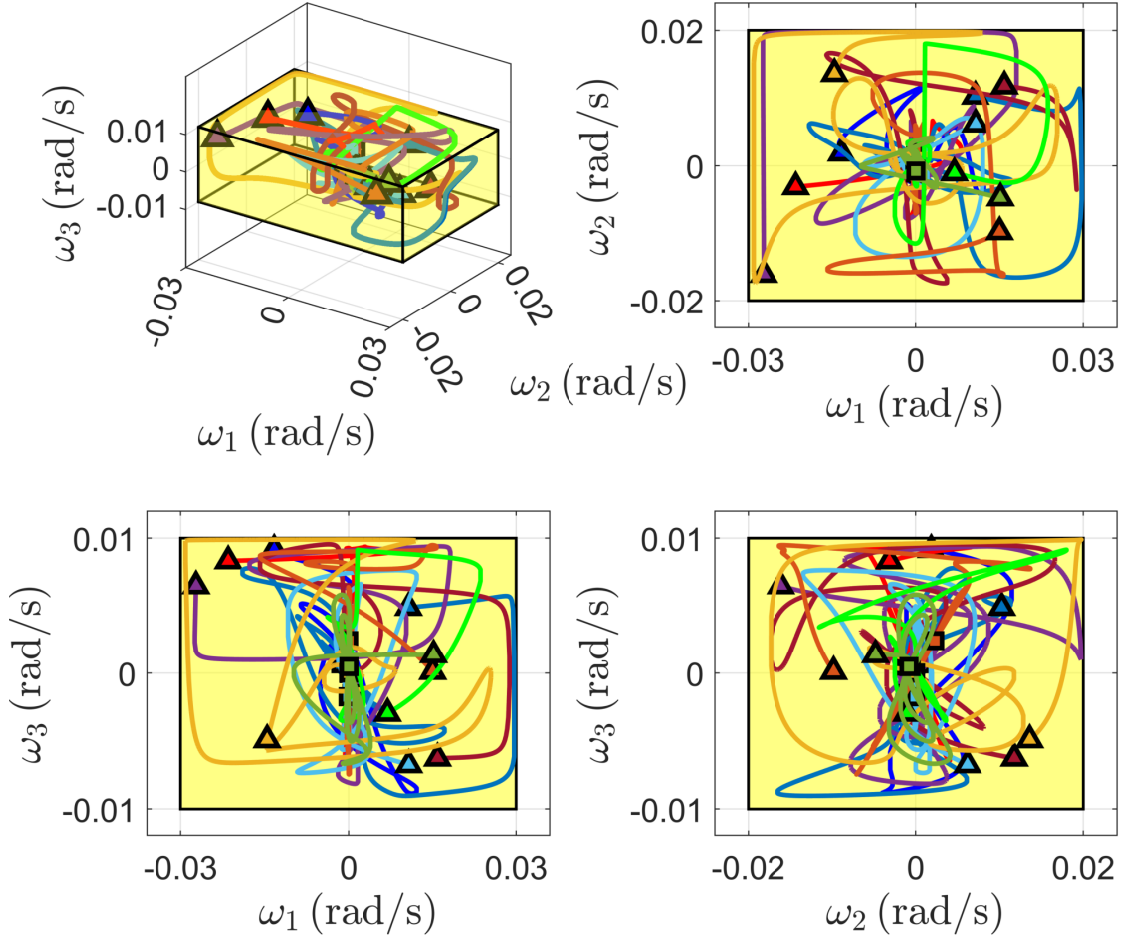


Figure 3.6: Closed-loop angular velocity response of (3.54), (3.57), (3.43) randomly initialized and commanded. The desired angular velocity is marked with a square at the origin, the randomly generated initial conditions are marked with a triangle, and the safe set is shown in yellow. Note that the angular velocity response is always contained within the desired safe set $\mathcal{S}$.

Finally, Figure 3.7 shows the time responses of the closed-loop Euler angles, an-

gular velocities, and control torques for the ten trajectories. The first column shows the Euler angle responses, the second column shows the angular velocity responses, and the third column shows the control torque signals. The pink- and yellow-shaded regions in the first and second columns, respectively, denote the corresponding safe sets.

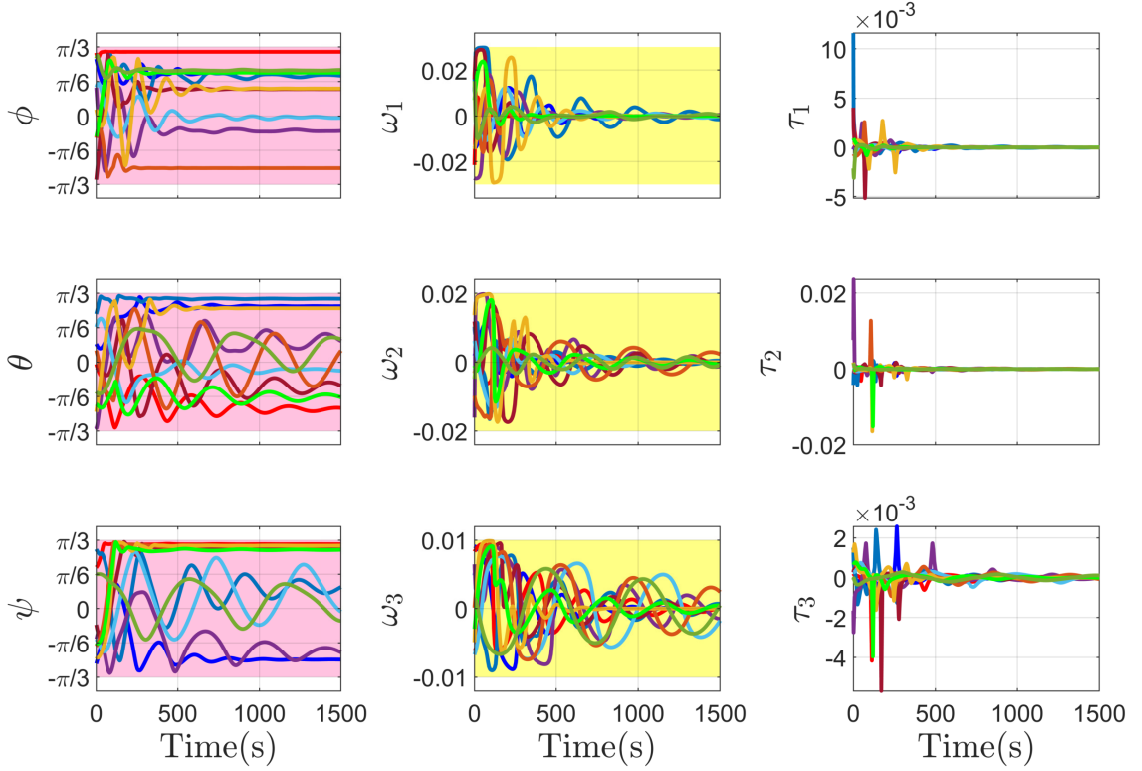


Figure 3.7: Closed-loop responses of the attitude dynamics: angles (ϕ, θ, ψ) and angular velocities $(\omega_1, \omega_2, \omega_3)$ over time for randomly initialized and commanded trajectories, and control inputs (τ_1, τ_2, τ_3) generated by the control law (3.43). Note that the angle and angular responses are always contained within the desired safe set $\mathcal{S}$.

3.9 Application to Safe Bicopter Control

This section addresses the constrained control problem of a Bicopter, serving as a representative example to demonstrate that the proposed framework can be extended to higher-order systems such as multicopters. We revisit the Bicopter example introduced in Section 2.6. Unlike the previous case, where both the mass and inertia were assumed to be unknown during controller synthesis, we now consider a scenario in which these parameters are known. This modification allows us to focus on a different class of challenges, those associated with enforcing safety constraints on the system states. Specifically, both the position and velocity variables are required to remain within prescribed bounds that ensure safe operation.

The inclusion of such constraints reflects practical considerations in aerial robotics, where physical limits on position, velocity, and actuator capabilities must be respected to prevent structural damage or loss of stability. Although the Bicopter serves as a simplified test platform, this scenario captures essential features of constraint handling in multicopters and other aerial vehicles with similar nonlinear and underactuated dynamics. The example therefore provides a useful setting to illustrate how the proposed framework can maintain closed-loop stability while systematically enforcing safety requirements on both positions and velocities.

The objective is to stabilize the bicopter system (2.35)-(2.38) and follow the commanded trajectory, while ensuring that $|x_1| < \bar{x}_1$ and $|x_2| < \bar{x}_2$ for all time, where $\bar{x}_1, \bar{x}_2 \in \mathbb{R}_+^2$ are the user-defined position and velocity bounds.

In set notation, the objective can be formulated as the design of a control law

that ensures the command is followed and the position x_1 and velocity x_2 of the bicopter satisfy $(x_1, x_2) \in \mathcal{S}$, where the safety set $\mathcal{S}$ is

$$\mathcal{S} = \{(p, v) : p, v \in \mathbb{R}^2, |p_1| < \bar{x}_{11}, |p_2| < \bar{x}_{12}, |v_1| < \bar{x}_{21}, |v_2| < \bar{x}_{22}\}. \quad (3.61)$$

The bicopter controller is designed following the procedure outlined in Section 3.6, with the control law given by (3.43). In particular, we use the $\phi(x) = \text{atanh}(x)$ as the constraint-lifting function. Consequently, $\psi(z)$ and $\mathcal{V}(\zeta)$ are given by the second line in Table 3.2. Therefore, define the z dynamics in the bicopter as follows,

$$z_1 \triangleq D(\bar{x}_1) \text{atanh}(\chi_1), \quad (3.62)$$

$$z_2 \triangleq D(\bar{x}_2) \text{atanh}(\chi_2), \quad (3.63)$$

$$z_3 \triangleq x_3, \quad (3.64)$$

$$z_4 \triangleq x_4, \quad (3.65)$$

It thus follows from (3.62)-(3.65) that

$$x_1 = D(\bar{x}_1) \tanh(\zeta_1), \quad (3.66)$$

$$x_2 = D(\bar{x}_2) \tanh(\zeta_2) \quad (3.67)$$

$$x_3 = z_3 \quad (3.68)$$

$$x_4 = z_4. \quad (3.69)$$

Note that the constraints are satisfied for all values of $z_1, z_2 \in \mathbb{R}^2$.

The dynamics of the bicopter in terms of the transformed state z is

$$\dot{z}_1 = \mathcal{F}_1(z_1, z_2) = D(\text{Ch}_{\zeta_1})^2 D(\bar{x}_2) \tanh(\zeta_2) \quad (3.70)$$

$$\dot{z}_2 = D(\text{Ch}_{\zeta_2})^2 (f_2 + g_2(z_3)) \quad (3.71)$$

$$\dot{z}_3 = z_4 \quad (3.72)$$

$$\dot{z}_4 = g_4 u, \quad (3.73)$$

where, for $q \in \mathbb{R}^n$,

$$\text{Ch}_q \triangleq \cosh \left(\begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}^T \right) \in \mathbb{R}^n \quad (3.74)$$

3.9.1 Controller Synthesis

This section presents the controller synthesis for the Bicopter problem. The synthesis procedure is described in detail to demonstrate how the proposed framework can be systematically applied to higher-order nonlinear systems such as the Bicopter. By explicitly outlining each step of the design process, this section highlights the flexibility and scalability of the proposed framework. Let $x_{d1} = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T \in \mathbb{R}^2$ be the desired value of x_1 , χ_{d1} be the corresponding desired value of χ_1 , and $z_{d1} \triangleq D(\bar{x}_1) \text{atanh}(\chi_{d1}) \in \mathbb{R}^2$ be the corresponding desired value of z_1 . Define the tracking error $e_1 \triangleq z_1 - z_{d1}$.

3.9.1.1 e_1 stabilization

Consider the function

$$V_1 \triangleq \frac{1}{2} e_1^T e_1. \quad (3.75)$$

Differentiating (3.75) and using (3.70) yields

$$\dot{V}_1 = e_1^T \dot{z}_1 = e_1^T \mathcal{F}_1(z_1, z_2) \quad (3.76)$$

Note that if

$$\mathcal{F}_2(z_1, z_2) = -k_1 e_1, \quad (3.77)$$

where $k_1 > 0$, then $\dot{V}_1 < 0$. However, $\mathcal{F}_1(z_1, z_2)$ cannot be chosen arbitrarily. Instead, we continue the design process as shown below to formulate a control law that ensures the desired $\mathcal{F}_1(z_2, z_2)$ response.

Define

$$e_2 \triangleq \mathcal{F}_1(z_1, z_2) + k_1 e_1, \quad (3.78)$$

where $k_1 > 0$. Then, differentiating (3.75) yields

$$\dot{V}_1 = -k_1 e_1^T e_1 + e_2^T e_1. \quad (3.79)$$

3.9.1.2 e_2 stabilization

Consider the function

$$V_2 \triangleq V_1 + \log(\cosh(\zeta_{21})) + \log(\cosh(\zeta_{22})). \quad (3.80)$$

Differentiating (3.80) and using (3.27), (3.37), and (3.79) yields

$$\begin{aligned} \dot{V}_2 &= -k_1 e_1^T e_1 + e_2^T e_1 + \tanh(\zeta_2)^T D_1(\bar{x}_2) \dot{z}_2 \\ &= -k_1 e_1^T e_1 + e_2^T e_1 + (e_2 - k_1 e_1)^T Q(f_2 + g_2(z_3)), \end{aligned} \quad (3.81)$$

where

$$Q \triangleq D(\text{Ch}_{\zeta_2})^2 D_I(\text{Ch}_{\zeta_1})^2 D_I(\bar{x}_2)^2 \in \mathbb{R}^{2 \times 2}, \quad (3.82)$$

Note that Q is a product of three diagonal matrices with strictly positive elements, and is therefore positive definite.

Next, note that if

$$f_2 + g_2(z_3) = -(e_2 - k_1 e_1), \quad (3.83)$$

then $\dot{V}_2 < 0$. However, $g_2(z_3)$ can not be chosen arbitrarily. Instead, we continue the design process as shown below to formulate a control law that ensures the desired $g_2(z_3)$ response.

Define

$$e_3 \triangleq Q(f_2 + g_2(z_3)) + k_2 e_2, \quad (3.84)$$

where $k_2 > 0$. Differentiating (3.80) and letting $k_2 = k_1^{-1}$ yields

$$\dot{V}_2 = -(\sqrt{k_1}e_1 - \sqrt{k_2}e_2)^T(\sqrt{k_1}e_1 - \sqrt{k_2}e_2) + e_3^T(e_2 - k_1 e_1). \quad (3.85)$$

Remark 3.9.1 *The choice $k_2 = k_1^{-1}$ allows the cancellation of several terms and the construction of the positive semidefinite first term in $\dot{V}_2$ in (3.85).*

3.9.1.3 e_3 stabilization

Consider the function

$$V_3 \triangleq V_2 + \frac{1}{2}e_3^T e_3. \quad (3.86)$$

Differentiating (3.86) and using (3.85) yields

$$\begin{aligned}\dot{V}_3 = \dot{V}_2 + e_3^T \dot{e}_3 = & -(\sqrt{k_1}e_1 - \sqrt{k_2}e_2)^T(\sqrt{k_1}e_1 - \sqrt{k_2}e_2) \\ & + e_3^T \left(e_2 - k_1 e_1 + \dot{Q}(f_2 + g_2(z_3)) + QNz_4 + k_2 \dot{e}_2 \right),\end{aligned}\quad (3.87)$$

where

$$N(z_3) \triangleq \partial_{z_3} g_2(z_3) \in \mathbb{R}^{2 \times 2}. \quad (3.88)$$

Note that if

$$e_2 - k_1 e_1 + \dot{Q}(f_2 + g_2(z_3)) + QNz_4 + k_2 \dot{e}_2 = -k_3 e_3, \quad (3.89)$$

then

$$\dot{V}_3 = -(\sqrt{k_1}e_1 - \sqrt{k_2}e_2)^T(\sqrt{k_1}e_1 - \sqrt{k_2}e_2) - k_3 e_3^T e_3. \quad (3.90)$$

However, z_4 can not be chosen arbitrarily. Instead, we continue the design process as shown below to formulate a control law that ensures the desired z_4 response.

Define

$$e_4 \triangleq e_2 - k_1 e_1 + \dot{Q}(f_2 + g_2(z_3)) + QNz_4 + k_2 \dot{e}_2 + k_3 e_3, \quad (3.91)$$

where $k_3 > 0$. Then, differentiating (3.86) yields

$$\dot{V}_3 = -(\sqrt{k_1}e_1 - \sqrt{k_2}e_2)^T(\sqrt{k_1}e_1 - \sqrt{k_2}e_2) - k_3 e_3^T e_3 + e_4^T e_3. \quad (3.92)$$

3.9.1.4 e_4 stabilization

Consider the function

$$V \triangleq V_3 + \frac{1}{2} e_4^T e_4. \quad (3.93)$$

Differentiating (3.93) yields

$$\begin{aligned}\dot{V} &= \dot{V}_3 + e_4^T \dot{e}_4 \\ &= -(\sqrt{k_1}e_1 - \sqrt{k_2}e_2)^T(\sqrt{k_1}e_1 - \sqrt{k_2}e_2) - k_3 e_3^T e_3 + e_4^T (\Sigma + \Upsilon u),\end{aligned}\quad (3.94)$$

where

$$\Sigma \triangleq e_3 + \dot{e}_2 - k_1 \dot{z}_1 + \ddot{Q}(f_2 + g_2(z)_3) + k_2 \ddot{e}_2 + k_3 \dot{e}_3 + 2\dot{Q}Nz_4 + \dot{N}z_4, \quad (3.95)$$

$$\Upsilon \triangleq QNg_4. \quad (3.96)$$

Finally, consider the control law

$$u = -\Upsilon^{-1}(\Sigma + k_4 e_4), \quad (3.97)$$

where $k_4 > 0$. Substituting the controller (3.97) in (3.44) yields

$$\dot{V} = -(\sqrt{k_1}e_1 - \sqrt{k_2}e_2)^T(\sqrt{k_1}e_1 - \sqrt{k_2}e_2) - k_3 e_3^T e_3 - k_4 e_4^T e_4. \quad (3.98)$$

Remark 3.9.2 Note that the control law in (3.97) requires Υ given by (3.96) to be nonsingular. Since Q given by (3.82) and g_4 given by (2.41) are positive definite, Υ is nonsingular if and only if $N(z_3)$ is nonsingular. As shown in (B.3.10), $N(z_3)$ is nonsingular if and only if $z_{32} = x_{32} = F \neq 0$.

Remark 3.9.3 The control law (3.97) requires Υ and Σ , which require $\dot{e}_2, \dot{z}_1, \dot{Q}, \ddot{Q}$, and $\dot{N}$. These derivatives are not obtained through differentiation, but are explicitly computed as functions of the measured state, as shown in Appendix B.3.

Remark 3.9.4 Although the control (3.97) does not explicitly guarantee that $F \neq 0$ for all $t \geq 0$, the net force required near any desired equilibrium is not-zero, and thus

the net force is not likely to converge to zero in well-posed problems. In practice, to deal with zero-crossings of the net force, projecting the net force F to $-\epsilon$ and ϵ , where $\epsilon > 0$ is a small number, if $F \in (-\epsilon, \epsilon)$, ensures well-posed computation of (3.97).

The control architecture is shown in Figure 3.8

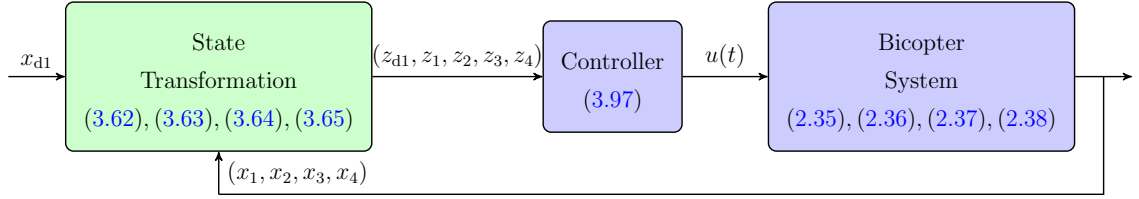


Figure 3.8: Block diagram illustrating the Constraint Lifting controller implementation in the Bicopter Problem.

3.9.2 Stability Analysis

The stability analysis of the closed-loop dynamics with the proposed control law uses the Barbashin-Krasovskii-LaSalle invariance principle, which originally appeared as Theorems 1 and 2 in [44]. In particular, the Barbashin-Krasovskii-LaSalle's invariance principle is used to prove the asymptotic stability of the equilibrium point

In the following, Propositions 3.9.1, 3.9.2, 3.9.3, and 3.9.4 define sets used in the stability proof and establish their properties.

Proposition 3.9.1 *Consider the system (3.70)–(3.73), The solution*

$$(z_1, z_2, z_3, z_4, u) = \left(z_{d1}, 0, \begin{bmatrix} 0 \\ mg \end{bmatrix}, 0, 0 \right) \quad (3.99)$$

is an equilibrium point.

Proof: Substituting the z_1, z_2, z_3, z_4 , and u given by (3.99) in the RHS of (3.70)–(3.73) trivially shows that $\dot{z}_1 = \dot{z}_2 = \dot{z}_3 = \dot{z}_4 = 0$. ■

Proposition 3.9.2 *Let $\ell > 0$. Define*

$$\Omega \triangleq \{z_1, z_2, z_3, z_4, u \in \mathbb{R}^2, : V \leq \ell\}. \quad (3.100)$$

Then, Ω is bounded and a positively invariant set.

Proof: Since V is radially unbounded and Ω is defined by $V \leq \ell$, it follows from Theorem 2 in [44] that Ω is bounded. Since $\dot{V} \leq 0$ and $V > 0$, Ω is a positively invariant set. ■

Proposition 3.9.3 *Define $E \subset \Omega$ such that*

$$\begin{aligned} z_1 &= z_{d1} - Q(f_2 + g_2(z_3)), \\ z_2 &= 0, \\ z_4 &= -k_2 D(\bar{x}_2) D(\text{Ch}_{\zeta_1})^4 (f_2 + g_2(z_3)). \end{aligned}$$

Then, E is the set of all points in Ω where $\dot{V} = 0$.

Proof: It follows from (3.98) that $\dot{V} = 0$ if and only if $k_1 e_1 = e_2$, and $e_3 = e_4 = 0$.

First, $k_1 e_1 = e_2$, in (3.78), necessarily requires that $\tan(\zeta_2) = 0$, which implies that $z_2 = 0$, $\zeta_2 = 0$, $D(\text{Ch}_{\zeta_2}) = I_2$, where I_2 is the 2 by 2 identity matrix, and $D(\text{Sh}_{\zeta_2}) = 0$, where, for $q \in \mathbb{R}^n$,

$$\text{Sh}_q \triangleq \sinh \left(\begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}^T \right) \in \mathbb{R}^n \quad (3.101)$$

Next, (3.82) with $\zeta_2 = 0$ yields $Q = D(\text{Ch}_{\zeta_1})^2 D_I(\bar{x}_2)^2$, and (B.3.5) with $\zeta_2 = 0$ yields $\dot{e}_2 = D(\text{Ch}_{\zeta_1})^2 (f_2 + g_2(z_3))$. Furthermore, it follows from (B.3.4) that $\dot{\zeta}_2 = f_2 + g_2(z_3)$

Therefore, (3.91) yields

$$e_4 = D_I(\text{Ch}_{\zeta_1})^2 D_I(\bar{x}_2) z_4 + k_2 D(\text{Ch}_{\zeta_1})^2 (f_2 + g_2(z_3)). \quad (3.102)$$

From (3.102), $e_4 = 0$ if and only if

$$z_4 = -k_2 D(\bar{x}_2) D(\text{Ch}_{\zeta_1})^4 (f_2 + g_2(z_3)). \quad (3.103)$$

Next, it follows from (3.84) and $e_2 = k_1 e_1$ that $e_3 = 0$ if and only if

$$z_1 = z_{d1} - Q(f_2 + g_2(z_3)), \quad (3.104)$$

which completes the proof. ■

Proposition 3.9.4 *Let $\mathcal{M} \subset E$ such that $z_1 = z_{d1}$, $z_3 = \begin{bmatrix} 0 & mg \end{bmatrix}^T$, and $z_4 = u = 0$. Then, $\mathcal{M}$ is the largest positively invariant set in E .*

Proof: It follows from Proposition 3.9.1 that $z_1 = z_{d1} = D(\bar{x}_1) \text{atanh}(\chi_{d1})$, $z_2 = z_4 = u = 0$, $z_3 = \begin{bmatrix} 0 & mg \end{bmatrix}^T$, is an equilibrium point. Furthermore, $z_1 = z_{d1}$, $z_2 = z_4 = 0$, and $z_3 = \begin{bmatrix} 0 & mg \end{bmatrix}^T$, fulfills equations (3.103) and (3.104), which implies that the equilibrium point $z_1 = z_{d1} = D(\bar{x}_1) \text{atanh}(\chi_{d1})$, $z_2 = z_4 = u = 0$, $z_3 = \begin{bmatrix} 0 & mg \end{bmatrix}^T$, is a subset of E . Thus, $\mathcal{M}$ is the largest positively invariant set in E . ■

Theorem 3.9.1 *Consider the system (3.70)–(3.73). Let $(x_1(0), x_2(0)) \in \mathcal{S}$ and let $|x_{d1}| < \bar{x}_1$. Consider the control law (3.97), where $k_1, k_3, k_4 > 0$, and $k_2 = k_1^{-1}$.*

Then,

$$\lim_{t \rightarrow \infty} z_1(t) = z_{d1} = D(\bar{x}_1) \operatorname{atanh}(\chi_{d1}), \quad (3.105)$$

$$\lim_{t \rightarrow \infty} z_2(t) = 0, \quad (3.106)$$

$$\lim_{t \rightarrow \infty} z_3(t) = \begin{bmatrix} 0 & mg \end{bmatrix}^T, \quad (3.107)$$

$$\lim_{t \rightarrow \infty} z_4(t) = 0, \quad (3.108)$$

$$\lim_{t \rightarrow \infty} u(t) = 0, \quad (3.109)$$

$$\lim_{t \rightarrow \infty} x_1(t) = \begin{bmatrix} r_{d1} & r_{d2} \end{bmatrix}^T, \quad (3.110)$$

Proof: Consider the set Ω defined in Proposition 3.9.2 and the set E defined in Proposition 3.9.3. It follows from Proposition 3.9.4 that $\mathcal{M}$ is the largest positively invariant set in E . It follows from Theorem 2 in [44] that any solution starting in Ω converge to $\mathcal{M}$ as $t \rightarrow \infty$, which implies (3.105)–(3.109).

Finally, since $\lim_{t \rightarrow \infty} z_1(t) = z_{d1}$, (3.62) implies (3.110). ■

3.9.3 Numerical Simulation

To simulate the bicopter, we assume that the mass of the bicopter m is 1 kg, its moment of inertia J is 0.2 kgm², and the length of the bicopter arm l is 0.2 m. In the controller, we set $k_1 = 1$, $k_3 = 1$, and $k_4 = 1$.

The safety set $\mathcal{S}$ is specified by

$$\bar{x}_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \quad \bar{x}_2 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \quad (3.111)$$

which implies that the bicopter should stay within a rectangle of width and height 7 m and 5 m, respectively, and its horizontal and vertical velocity should not exceed

0.5 m/s.

The bicopter is commanded to follow a set of waypoints defined by the vertices of an polygon, which is chosen as the largest polygon within the safe set.

The trajectory between the waypoints is constructed using the algorithm described in Appendix A of [70] with a maximum velocity $v_{\max} = 1$ m/s and a maximum acceleration $a_{\max} = 1$ m/s². The bicopter is commanded to approach the polygon, traverse around the polygon once, and return to the origin.

Figure 3.9 shows the trajectory-tracking response of the bicopter, where the desired trajectory is shown in black dashes, the position response is shown in blue, and the safe set is shown in yellow. Figure 3.10 shows the velocity response of the bicopter, where the velocity response is shown in solid blue, and the safe set is shown in pink. Figure 3.11 shows the closed-loop response of the bicopter with respect to time. Note that the position and velocity responses are always contained within the desired safe set $\mathcal{S}$. Finally, Figure 3.12 shows the pitch response and the force and the moment applied to the bicopter.

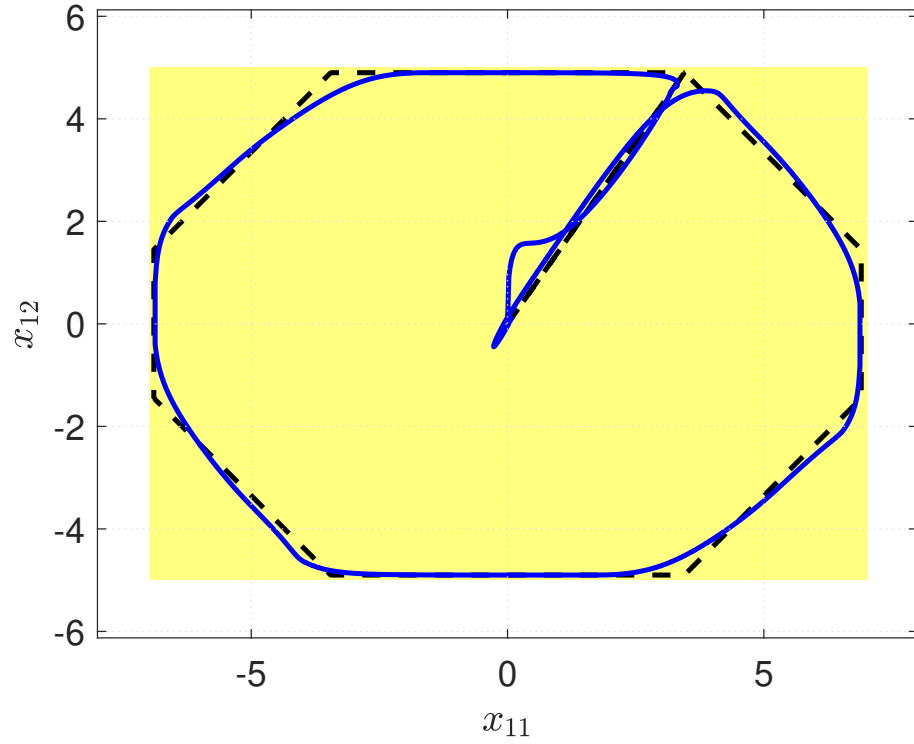


Figure 3.9: **Constrained trajectory tracking response.** Closed-loop position tracking response of the bicopter. The desired position is shown in dashed black, the position response is shown in solid blue, and the safe set is shown in yellow. Note that the position response is always contained within the desired safe set for positions $\mathcal{S}$.

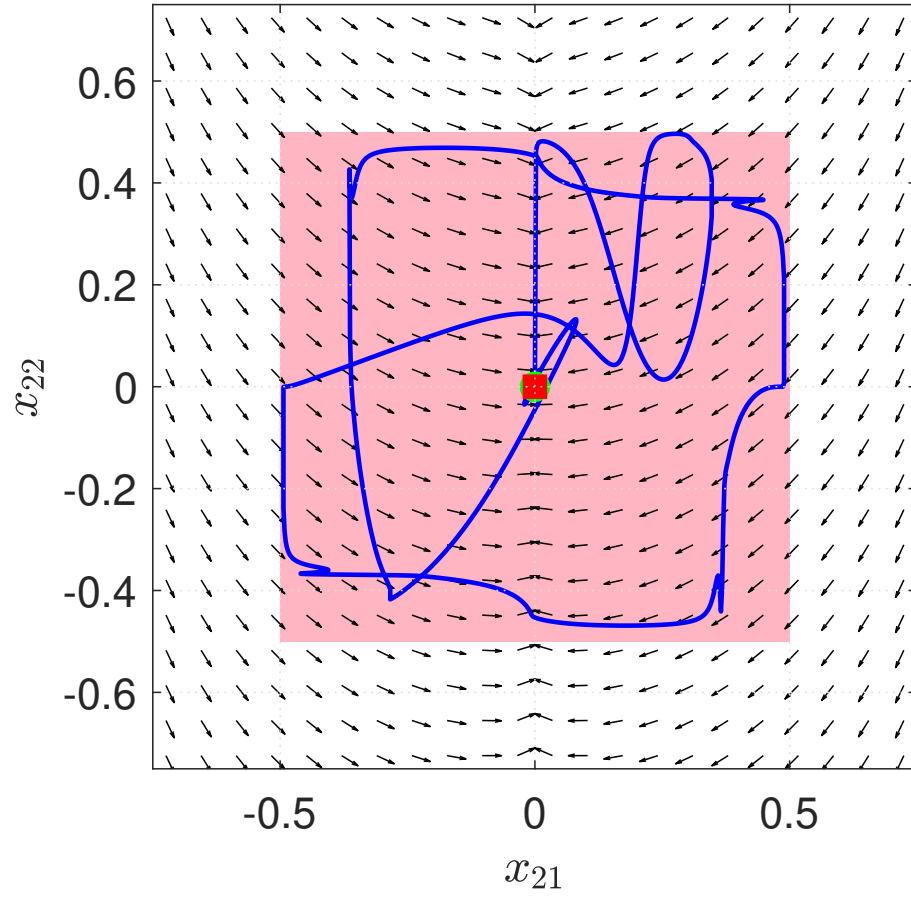


Figure 3.10: **Constrained trajectory tracking response.** Closed-loop velocity response of the bicopter. The velocity response is shown in solid blue, and the safe set is shown in pink. Note that the velocity response is always contained within the desired safe set for velocities $\mathcal{S}$.

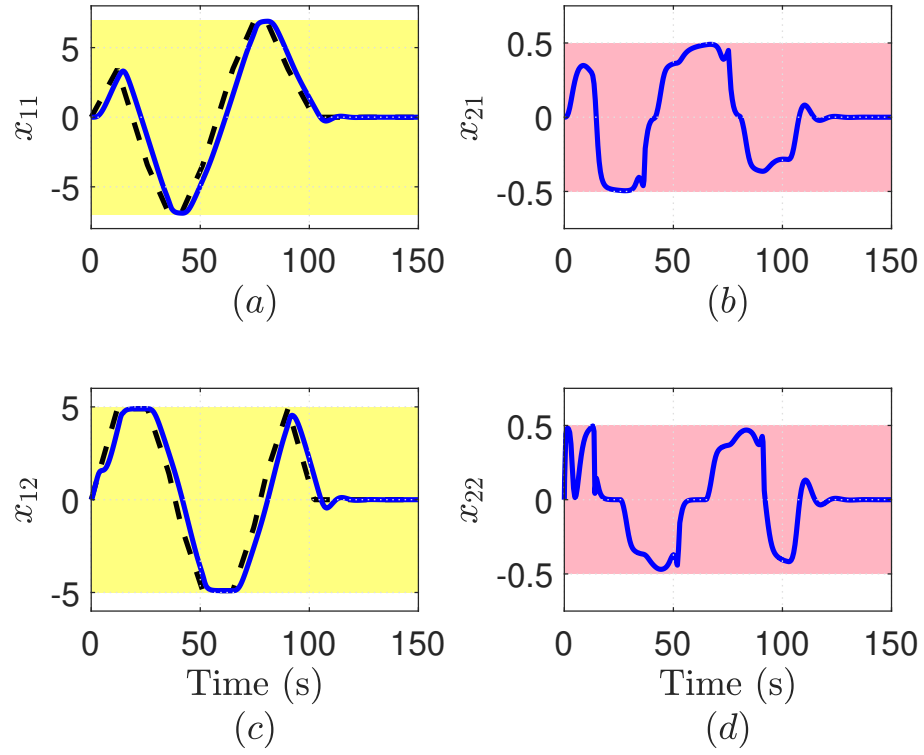


Figure 3.11: **Constrained trajectory tracking response.** Closed-loop position and velocity tracking response with respect to time of the bicopter. Note that the position and velocity responses are always contained within the desired safe set $\mathcal{S}$.

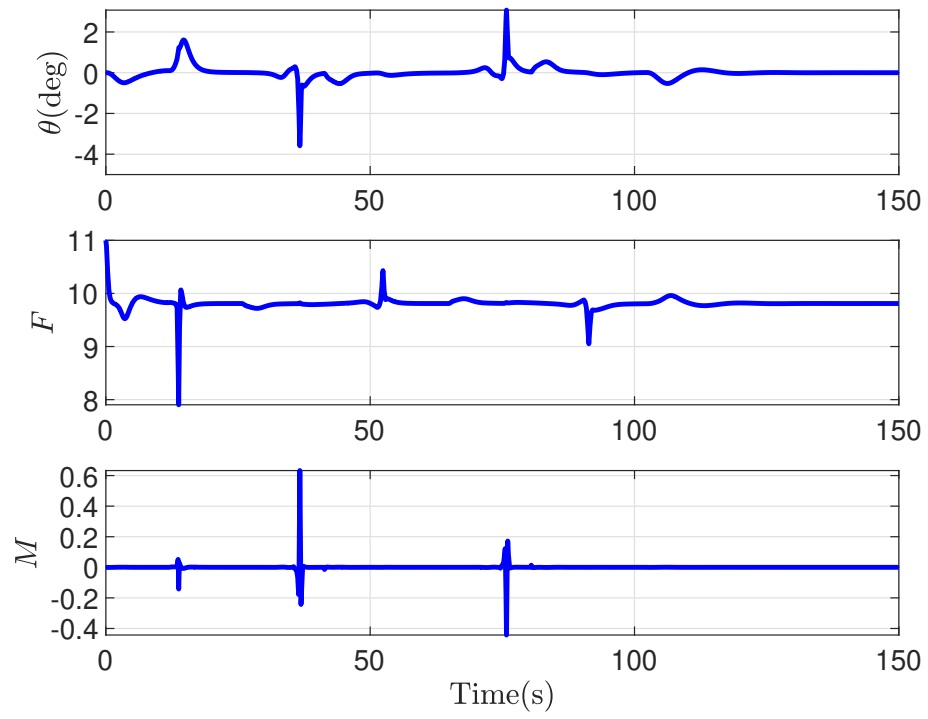


Figure 3.12: **Polygon Constrained Trajectory:** Pitch angle θ , and inputs F , and M obtained with the control law (3.97). Note that the values for the three of them exhibit reasonable values of operation.

3.10 Runtime Comparison of MPC and Constraint-Lifting Controller for the Tracking Problem

This section presents a comparative analysis of the computational complexity of the model predictive controller (MPC) and the proposed constraint-lifting controller for stabilizing a double integrator while enforcing state constraints.

Consider the double integrator system

$$\dot{x}_1 = x_2, \tag{3.112}$$

$$\dot{x}_2 = u. \tag{3.113}$$

The objective is to track x_{1d} while enforcing constraints on both position x_1 and velocity x_2 . In this application, the safe set is defined by $\bar{x} \triangleq \begin{bmatrix} 3 & 3 \end{bmatrix}^T$. These constraints imply that both position and velocity should remain smaller than 3 m and 3 m/s.

To benchmark the computational complexity of the proposed constraint-lifting controller, its performance is compared with that of a standard model predictive controller (MPC). For this purpose, an MPC scheme is implemented for a double-integrator system using the MATLAB toolbox [90].

The initial conditions and command signals for both positions and velocities are randomly set to values within the safe set. The sampling time for both controllers is chosen as $T_s = 0.1$. We performed five different runs for each set of randomly generated initial conditions and command references.

The Matlab files corresponding to the simulation performed can be found

Run Number	MPC (s)	Constraint-Lifting (s)
Run 1	0.030	0.005
Run 2	0.042	0.008
Run 3	0.035	0.004
Run 4	0.030	0.004
Run 5	0.028	0.004
Average	0.033	0.005

Table 3.3: Random Set 1: Comparison of computation times for a constrained double integrator controlled using MPC and the proposed constraint-lifting controller.

Run Number	MPC (s)	Constraint-Lifting (s)
Run 1	0.030	0.005
Run 2	0.037	0.005
Run 3	0.025	0.006
Run 4	0.025	0.004
Run 5	0.026	0.004
Average	0.0236	0.0048

Table 3.4: Random Set 2: Comparison of computation times for a constrained double integrator controlled using MPC and the proposed constraint-lifting controller.

in the accompanying repository.¹ These numerical experiments suggest that the technique presented in this thesis is approximately five times faster than MPC.

¹https://github.com/estimation-control-learning-laboratory/adaptive_state_constraint_ctrl

Run Number	MPC (s)	Constraint-Lifting (s)
Run 1	0.025	0.005
Run 2	0.032	0.007
Run 3	0.027	0.005
Run 4	0.027	0.005
Run 5	0.025	0.005
Average	0.0272	0.0054

Table 3.5: Random Set 3: Comparison of computation times for a constrained double integrator controlled using MPC and the proposed constraint-lifting controller.

Run Number	MPC (s)	Constraint-Lifting (s)
Run 1	0.024	0.004
Run 2	0.025	0.004
Run 3	0.024	0.004
Run 4	0.027	0.004
Run 5	0.029	0.005
Average	0.0258	0.0042

Table 3.6: Random Set 4: Comparison of computation times for a constrained double integrator controlled using MPC and the proposed constraint-lifting controller.

Furthermore, the proposed technique uses algebraic functions to compute the control signal, whereas MPC relies on additional optimization tools.

Run Number	MPC (s)	Constraint-Lifting (s)
Run 1	0.024	0.004
Run 2	0.027	0.007
Run 3	0.028	0.005
Run 4	0.026	0.004
Run 5	0.026	0.006
Average	0.0262	0.0052

Table 3.7: Random Set 5: Comparison of computation times for a constrained double integrator controlled using MPC and the proposed constraint-lifting controller.

3.11 Conclusions

This chapter presented a constraint-enforcing control framework that guarantees closed-loop stability for a class of nonlinear systems while ensuring the positive invariance of a prescribed safe set. The proposed approach transforms the constrained control problem into an equivalent unconstrained one through *constraint-lifting functions* constructed using strictly increasing sigmoid functions. A class of Lyapunov candidate functions, called *sigmoid integral functions*, was introduced. These functions behave quadratically near the origin and linearly far away from it, a key property that eliminates exponentially growing terms that lead to numerical instability in the classical Lyapunov-based controller design. The sigmoid integral functions are then employed in the controller synthesis to design a stabilizing feedback law for the transformed system. A rigorous theoretical analysis establishes the asymptotic stability of the resulting closed-loop system. Numerical simulations

demonstrate the effectiveness of the proposed technique for a constrained attitude control problem and for a Bicopter with constrained positions and velocities.

Chapter 4

Conclusions and Future Work

4.1 Conclusion

This thesis has presented two complementary methodologies for the control of pure-feedback nonlinear systems:

1. Inverse-free adaptive stabilization under multiplicative uncertainties.
2. Constraint-lifting for safe control under state constraints.

Although motivated by distinct challenges, both directions address fundamental limitations in existing nonlinear control techniques and provide unified advances for systems where classical approaches fall short.

4.1.1 Inverse-free Adaptive Control

The first part of this thesis tackled the problem of adaptive stabilization in pure-feedback nonlinear systems whose input channels contain multiplicative uncertainties. Such systems exhibit nonlinear dependence not only on the control input but also on intermediate states, making classical backstepping difficult to apply directly. In particular, standard backstepping techniques typically require the inversion of uncertain or state-dependent input coefficients, an operation that may be ill-defined or lead to singularities.

To overcome this difficulty, this work proposed an inverse-free adaptive backstepping algorithm that preserves the recursive design structure of classical backstepping while eliminating the need to invert uncertain parameters. The key idea is to construct stabilizing control laws and parameter update rules that depend only on measurable variables and estimated parameters, but never on inverse estimates that may approach zero.

A rigorous Lyapunov-based framework was developed to analyze the resulting closed-loop system. This framework guarantees

1. boundedness of all system states,
2. boundedness of parameter estimates,
3. stability of the overall closed-loop dynamics.

This methodology extends the applicability of adaptive backstepping to a broad class of systems with uncertain input mappings, commonly encountered in fields such as robotics, aerospace vehicles, mechanical systems with flexible links, and electromechanical actuators. By resolving the long-standing challenge of unstable or undefined inversions in adaptive designs, the proposed approach provides a robust and systematic solution for stabilization in uncertainty-dominated environments.

4.1.2 Constraint-lifting Control

The second research direction focused on guaranteeing state constraint satisfaction in pure-feedback nonlinear systems. In many real-world engineering appli-

cations, robotics, automotive systems, and safety-critical mechatronics, state constraints arise naturally from physical limits, safety requirements, or operational specifications. Classical backstepping and Lyapunov-based designs, however, do not directly accommodate such constraints, often requiring nontrivial modifications or leading to overly conservative solutions.

To address this challenge, this thesis introduced a constraint-lifting framework that transforms a constrained control problem into an unconstrained one, enabling systematic controller synthesis without violating safety limits. The key mechanism behind this transformation is the use of sigmoid functions, which smoothly and monotonically map unconstrained variables into constrained domains. This ensures that constraint satisfaction is embedded directly into the system’s coordinate representation.

Furthermore, to prevent the rapid growth or stiffness that often arises from differentiating sigmoid functions in pure-feedback structures, the methodology incorporates sigmoid integral functions into the backstepping design. These functions retain the desirable mapping properties of sigmoids but mitigate excessive amplification of terms in the dynamics. As a result,

1. constructed virtual controls remain smooth,
2. state trajectories evolve safely within the prescribed bounds,
3. stabilizing feedback laws can be systematically derived for the lifted system.

This framework not only enforces constraints but also reveals structural insights into pure-feedback systems, enabling more transparent and intuitive controller designs.

Its applicability extends to problems requiring safety-critical behavior, soft landings, collision avoidance, and bounded-state performance.

4.1.3 Overall Problems Addressed

Together, the methodologies developed in this thesis address two complementary challenges in nonlinear control,

1. inverse-free adaptive control provides robustness to multiplicative uncertainties in the input mapping.
2. constraint-lifting control ensures safe evolution of the system states under explicit constraints.

While these problems are distinct, their solutions naturally reinforce one another: inverse-free adaptive control addresses uncertainty-driven instability, whereas constraint-lifting ensures that stability is achieved within feasible and safe operating regions.

4.2 Future Directions

This thesis opens several promising avenues for continued research in both inverse-free adaptive control and constraint-lifting control for pure-feedback nonlinear systems. The following subsections outline extensions that build directly on the theoretical foundations developed here, as well as broader opportunities to integrate these contributions into emerging control frameworks and practical applications.

4.2.1 Inverse-free Adaptive Control

The inverse-free adaptive backstepping approach developed in this thesis provides a foundation for addressing uncertainty in systems with nonlinear and multiplicative input mappings. Several directions remain open for expanding its theoretical scope and practical applicability:

1. *Multiple unknown parameters per state equation.* A natural extension is to generalize the adaptive laws to systems where each state equation contains several unknown parameters. This requires designing multi-dimensional update laws that remain stable and avoid estimation singularities, while preserving the inverse-free property. This research direction is illustrated by applications in advanced mechanical and aerospace systems, where multiple interacting uncertainties are present. For example, flexible mechanical structures such as robotic arms or lightweight aerospace components often exhibit distributed uncertainties in stiffness, damping, or payload variations. In robotic manipulators, multiple joints and actuators can introduce coupled unknown dynamics and friction parameters that must be simultaneously estimated for precise trajectory tracking. Similarly, in fluid-structure interaction problems, such as underwater vehicles or flexible wings in turbulent flows, the system dynamics involve multiple unknown parameters arising from the fluid forces and structural deformations. Extending adaptive laws to handle these multi-parameter scenarios enables robust performance and stability in complex, real-world systems where conventional single-parameter adaptive approaches would fail.

2. *Extension to time-varying and nonlinearly parametrized uncertainties.* While this thesis considers multiplicative uncertainties that are linearly parameterized, many applications involve coefficients that vary with time or appear in nonlinear parameterizations. Extending the methodology to these cases would require new Lyapunov function constructions and possibly hybrid update mechanisms. Such scenarios commonly arise in aerospace vehicles with time-varying aerodynamic coefficients, robotic manipulators with joint friction that changes during operation, and chemical reactors whose reaction rates evolve nonlinearly with temperature and concentration.
3. *Integration with learning-based estimators.* A forward-looking direction is the combination of inverse-free adaptive control with learning components such as
- Gaussian processes,
 - neural networks,
 - Koopman-based estimators.

This combination could allow uncertainty estimation in high-dimensional systems while ensuring that the learned model does not introduce inversions of uncertain quantities. Figure 4.1 shows an illustration of this research direction.

4. *Evaluation of the proposed adaptive control for applications with measurement noise.* Real-world systems operate under noise, quantization, and sensor imperfections. Testing the applicability of the proposed adaptive control in applications where sensor noise is present is an interesting future research op-

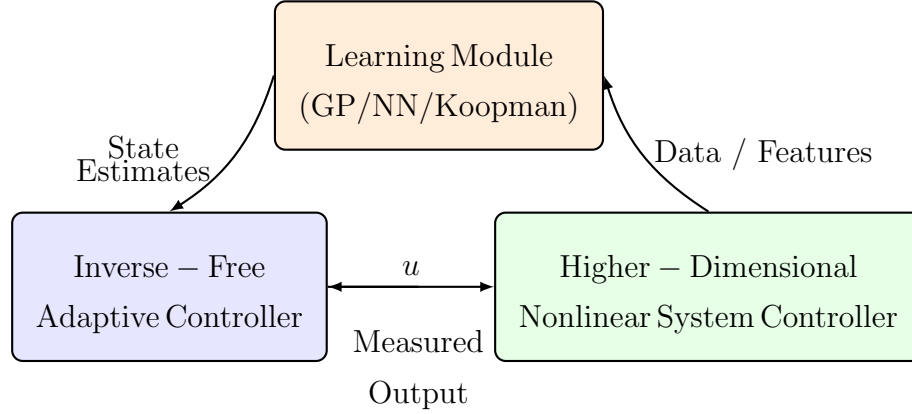


Figure 4.1: Integration with Learning-Based Estimators

portunity.

5. *Experimental validation on complex dynamics systems.* Ultimately, applying inverse-free adaptive control to complex platforms such as:

- quadrotors,
- robotic arms with flexible joints,
- hypersonic vehicles,
- underactuated balancing systems,

would provide strong evidence of its practical impact and robustness under uncertainty.

4.2.2 Constraint-lifting Control

The second part of this thesis proposes a systematic methodology for handling state constraints through smooth coordinate transformations and sigmoid-based lifting. Several rich avenues exist for extending this framework:

1. *Asymmetric and time-varying bounds.* Many systems operate under constraints that are not symmetric or that change over time. Extending the lifting mechanism to handle

- different upper and lower bounds,
- drifting limits due to environmental changes,
- time-varying envelopes for safety-critical missions,

would broaden its applicability.

2. *Recovery controllers and invariant set re-entry:* Although the lifting ensures trajectories remain within bounds, real systems may violate constraints due to disturbances or modeling errors. A major extension is to design recovery controllers capable of steering the state back into the constrained region while ensuring stability and avoiding aggressive corrective actions.

3. *Integration with internal model and tracking controllers.* For missions requiring trajectory tracking (e.g., aircraft guidance, robotic manipulation), embedding internal model principles into the constraint-lifting framework could enable reference tracking while guaranteeing constraint satisfaction.

4. *Coupling constraint-lifting with adaptive estimation.* In uncertain systems, the constraint map could be combined with adaptive estimators to create controllers that are both safe (respect constraints) and robust (handle multi-parametric uncertainties). This represents a promising unification of the two main contributions of this thesis.

5. *Coupled-state constraints.* Future work may consider constraints that involve

- coupled states (e.g., collision avoidance between moving bodies). Idea represented in Figure 4.2,
- norms or geometric regions (e.g., safe zones, funnels, tubes),
- safety constraints described by control barrier functions.

Extending sigmoid-based lifting to such sets would require new multidimensional transformation tools.

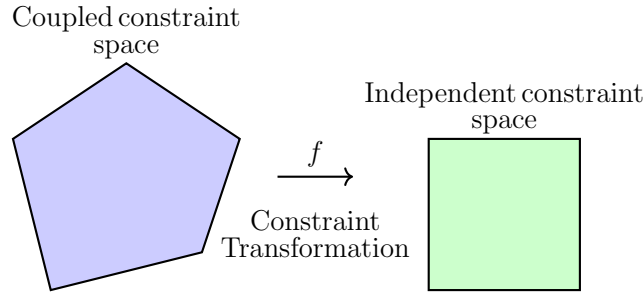


Figure 4.2: Transformation of Coupled Constraints into an Independent Constraint Space

6. *Fast online feasibility monitoring and switching strategies.* Safety-critical systems may benefit from real-time feasibility checks to determine when constraints could be violated. Future research could develop

- online constraint monitors,
- switching control laws,
- supervisory safety filters,

to complement the smooth lifting strategy.

7. *Application to full-dynamics platforms (quadcopters, robotic manipulators, UAVs).*

A direct and impactful extension is the application of the constraint-lifting framework to full nonlinear dynamics such as:

- quadcopters,
- aerial manipulators,
- mobile robots and ground vehicles,
- high-speed marine vehicles.

This would demonstrate its capacity to handle real-world constraints such as angle limits, thrust bounds, collision tubes, and actuator saturation.

8. *Adaptive-safe control for multi-agent collision avoidance.* Develop a framework where each agent uses inverse-free adaptive backstepping to handle uncertain dynamics while simultaneously ensuring safety constraints (e.g., minimum separation distances) using constraint lifting. This could provide provably safe cooperative control of UAV swarms or autonomous vehicles with uncertain input gains. Such scenarios arise in swarms of drones or fleets of autonomous vehicles, where each agent must stabilize its uncertain dynamics while avoiding collisions with others in real time.

9. *Hybrid adaptive-safe control for systems with mode switching.* Consider systems that switch between different dynamics (e.g., hybrid or piecewise systems). Develop adaptive controllers that estimate uncertain parameters in each mode while enforcing constraints via lifting, ensuring safety across all

modes and during transitions. Such scenarios arise in vehicles or robotic systems that switch between different operational modes, such as hybrid-electric vehicles transitioning between electric and combustion modes, or legged robots switching gait patterns, while maintaining safety and stability across all modes.

Appendix A

A.1 Details of Example 2.5.1

It follows from (2.6) and (2.7) that

$$e_2 = x_1^2 + x_2 + k_1 x_1, \quad (\text{A.1.1})$$

$$e_3 = u + \hat{p} \left(x_1 + (2x_1 + k_1)(x_1^2 + x_2) + x_1 x_2 \right) + k_2 \text{sgn}(\theta) e_2, \quad (\text{A.1.2})$$

and thus

$$\frac{\partial e_3}{\partial x_1} = \hat{p} \left(1 + 2x_2 + 6x_1^2 + 2k_1 x_1 + x_2 \right) + k_2 \text{sgn}(\theta) (2x_1 + k_1), \quad (\text{A.1.3})$$

$$\frac{\partial e_3}{\partial x_2} = \hat{p} \left(2x_1 + k_1 + x_1 \right) + k_2 \text{sgn}(\theta), \quad (\text{A.1.4})$$

$$\frac{\partial e_3}{\partial \hat{p}} = x_1 + (2x_1 + k_1)(x_2 + x_1^2) + x_1 x_2. \quad (\text{A.1.5})$$

A.2 Details of Example 2.5.2

It follows from (2.6) and (2.7) that

$$e_2 = x_2 + \frac{x_2^3}{5} + \hat{p}_1 x_1 + k_1 \text{sgn}(\theta_1) x_1, \quad (\text{A.2.1})$$

$$e_3 = \left(1 + \frac{3x_2^2}{5} \right) \left(u + \frac{u^3}{7} \right) + \hat{p}_2 \left[(x_1 + \hat{p}_1 + k_1 \text{sgn}(\theta_1) \hat{\theta}_1 + (\hat{p}_1 + k_1 \text{sgn}(\theta_1)) x_1 + \right. \\ \left. \left(1 + \frac{3x_2^2}{5} \right) x_1 x_2 + x_1 \dot{\hat{p}}_1 \right] + k_2 \text{sgn}(\theta_2) e_2, \quad (\text{A.2.2})$$

and thus

$$\begin{aligned} \frac{\partial e_3}{\partial x_1} &= \hat{p}_2 \left[\hat{\theta}_1 + \hat{p}_1 + k_1 \text{sgn}(\theta_1) + x_2 + \frac{3x_2^2}{5} + 3\gamma_1 \text{sgn}(\theta_1) x_1^2 \right] \\ &\quad + k_2 \text{sgn}(\theta_2) (\hat{p}_1 + k_1 \text{sgn}(\theta_1)), \end{aligned} \quad (\text{A.2.3})$$

$$\frac{\partial e_3}{\partial x_2} = \frac{6x_2}{5} \left(u + \frac{u^3}{7} \right) + \hat{p}_2 \left[x_1 + \frac{9x_2^2 x_1}{5} \right] + k_2 \text{sgn}(\theta_2) \left(1 + \frac{3x_2^2}{5} \right), \quad (\text{A.2.4})$$

$$\frac{\partial e_3}{\partial \hat{\theta}_1} = \hat{p}_2 [x_1 + \hat{p}_1 + k_1 \text{sgn}(\theta_1)], \quad (\text{A.2.5})$$

$$\frac{\partial e_3}{\partial \hat{p}_1} = \hat{p}_2 [\hat{\theta}_1 + x_1] + k_2 \text{sgn}(\theta_2) x_1, \quad (\text{A.2.6})$$

$$\begin{aligned} \frac{\partial e_3}{\partial \hat{p}_2} &= (x_1 + \hat{p}_1 + k_1 \text{sgn}(\theta_1)) + (\hat{p}_1 + k_1 \text{sgn}(\theta_1)) x_1 \\ &\quad + \left(1 + \frac{3x_2^2}{5} \right) x_1 x_2 + x_1 \dot{\hat{p}}_1, \end{aligned} \quad (\text{A.2.7})$$

$$\frac{\partial e_3}{\partial u} = \left(1 + \frac{3u^2}{7} \right) \left(1 + \frac{3x_2^2}{5} \right). \quad (\text{A.2.8})$$

A.3 Nussbaum-based control

This appendix briefly reviews the procedure to synthesize a Nussbaum-function-based control law as shown in [34]. Consider the strict-feedback system

$$\dot{x}_1 = f_1(x_1)x_2, \quad (\text{A.3.1})$$

$$\dot{x}_2 = f_2(x_1, x_2) + g_2(x_1, x_2)\theta u, \quad (\text{A.3.2})$$

where $x_1, x_2 \in \mathbb{R}$ are the states and $u \in \mathbb{R}$ is the control input. The map g_2 is assumed to be non-singular for all $x_1, x_2 \in \mathbb{R}$. Finally, $\theta \in \mathbb{R}$ is the unknown parameter. The objective is to design a globally asymptotically stable control law without the knowledge of θ .

Define $e_1 \triangleq x_1 - x_{1d}$, where x_{1d} is the desired trajectory for x_1 . Consider the

function $V_1 \triangleq \frac{1}{2}e_1^2$. Then, $\dot{V}_1 = e_1(f_1(x_1)x_2)$. Defining $e_2 \triangleq f_1(x_1)x_2 + k_1e_1$ yields $\dot{V}_1 = -k_1e_1^2 + e_2e_1$. Furthermore,

$$\dot{e}_2 = \frac{\partial f_1(x_1)}{\partial x_1}f_1(x_1)x_2^2 + f_1(x_1)\left(f_2(x_1, x_2) + g_2(x_1, x_2)\theta u\right) + k_1f_1(x_1)x_2 \quad (\text{A.3.3})$$

Next, define $V_2 \triangleq V_1 + \frac{1}{2}e_2^2$, which implies that $\dot{V}_2 = -k_1e_1^2 + e_2(\Sigma + \theta g_2(x_1, x_2)u)$, where

$$\Sigma \triangleq e_1 + \frac{\partial f_1(x_1)}{\partial x_1}f_1(x_1)x_2^2 + f_1(x_1)\left(f_2(x_1, x_2)\right) + k_1f_1(x_1)x_2. \quad (\text{A.3.4})$$

Finally, as shown in [34], define the function $h(s) \triangleq s^2 \cos(s)$, where s satisfies $\dot{s} \triangleq k_2e_2^2 + e_2\Sigma$, and define the *Nussbaum* control law $u = g_2(x_1, x_2)^{-1}h(s)(k_2e_2 + \Sigma)$.

Then,

$$\dot{V}_2 = -k_1e_1^2 - k_2e_2^2 + (\theta h(s) + 1)\dot{s}, \quad (\text{A.3.5})$$

which, as shown in [34], assures that the origin of system (A.3.1), (A.3.2) is asymptotically stable.

Appendix B

B.1 Dynamics in z Coordinate

It follows from (3.20) that

$$\begin{aligned}
\dot{z}_1 &= D(\bar{x}_1)\partial_{\chi_1}\phi(\chi_1)\dot{\chi}_1 \\
&= D(\bar{x}_1)\partial_{\chi_1}\phi(\chi_1)D_I(\bar{x}_1)\dot{x}_1 \\
&= \partial_{\chi_1}\phi(\chi_1)g_1(x_1)x_2 \\
&= \partial_{\chi_1}\phi(\psi_1(\zeta_1))g_1(D(\bar{x}_1)\psi_1(\zeta_1))D(\bar{x}_2)\psi_2(\zeta_2) \\
&= \Phi(\zeta_1)\psi_2(\zeta_2) \\
&= \mathcal{G}_1(z_1, z_2).
\end{aligned} \tag{B.1.1}$$

Similarly, it follows from (3.21) that

$$\begin{aligned}
\dot{z}_2 &= D(\bar{x}_2)\partial_{\chi_2}\phi(\chi_2)\dot{\chi}_2 \\
&= \partial_{\chi_2}\phi(\chi_2)D(\bar{x}_2)D_I(\bar{x}_2)\dot{x}_2 \\
&= \partial_{\chi_2}\phi(\chi_2)(f_2(x_1, x_2) + g_2(x_1, x_2)u) \\
&= \partial_{\chi_2}\phi(\psi_2(\zeta_2))\left(f_2(D(\bar{x}_1)\psi_1(\zeta_1), D(\bar{x}_2)\psi_2(\zeta_2)) + g_2(D(\bar{x}_2)\psi_1(\zeta_1), D(\bar{x}_2)\psi_2(\zeta_2))u\right) \\
&= \mathcal{F}_2(z_1, z_2) + \mathcal{G}_2(z_1, z_2)u.
\end{aligned} \tag{B.1.2}$$

B.2 Time-derivative of Sigmoid Integral-based Lyapunov Function

Note that

$$d_t \sum_{i=1}^n \mathcal{V}(\zeta_{2i}) = \sum_{i=1}^n d_t \int_0^{\zeta_{2i}} \sigma(s)ds \tag{B.2.1}$$

For each $i = 1, \dots, n$, it follows from the Leibniz's formula that

$$d_t \int_0^{\zeta_{2i}} \sigma(s) ds = \sigma(\zeta_{2i}) \dot{\zeta}_{2i}, \quad (\text{B.2.2})$$

which, using (3.23), implies that

$$d_t \sum_{i=1}^n \mathcal{V}(\zeta_{2i}) = \sum_{i=1}^n \sigma(\zeta_{2i}) \dot{\zeta}_{2i} = \psi(\zeta)^T \dot{\zeta}_2 = \psi(\zeta_2)^T D_I(\bar{x}_2) \dot{z}_2. \quad (\text{B.2.3})$$

B.3 Math expressions used in the synthesis of the Bicopter Example with position and velocity constraints

It follows from (3.70), and (3.71) that

$$\dot{\zeta}_1 = D_I(\bar{x}_1) \mathcal{F}_1(z_1, z_2), \quad (\text{B.3.1})$$

$$\dot{\zeta}_2 = D_I(\bar{x}_2) D(\text{Ch}_{\zeta_2})^2 (f_2 + g_2(z_3)), \quad (\text{B.3.2})$$

and thus

$$\ddot{\zeta}_1 = D_I(\bar{x}_1) \left(D(\text{Sh}_{2\zeta_2}) D(\dot{\zeta}_1) D(\bar{x}_2) \tanh(\zeta_2) + D(\text{C}_{\zeta_1})^2 D(\bar{x}_2) D_I(\text{Ch}_{\zeta_2})^2 \dot{\zeta}_2 \right), \quad (\text{B.3.3})$$

$$\ddot{\zeta}_2 = D_I(\bar{x}_2) \left(D(\text{Sh}_{2\zeta_2}) D(\dot{\zeta}_2) (f_2 + g_2(z_3)) + D(\text{Ch}_{\zeta_2})^2 N(z_3) z_4 \right). \quad (\text{B.3.4})$$

It follows from (3.78) that

$$\dot{e}_2 = D(\text{Sh}_{2\zeta_2}) D(\dot{\zeta}_1) D(\bar{x}_2) \tanh(\zeta_2) + D(\text{Ch}_{\zeta_1})^2 D(\bar{x}_2) D_I(\text{Ch}_{\zeta_2})^2 \dot{\zeta}_2, \quad (\text{B.3.5})$$

and thus

$$\begin{aligned} \ddot{e}_2 = & 2 D(\cosh(2\zeta_1)) D(\dot{\zeta}_1)^2 D(\bar{x}_2) \tanh(\zeta_2) + D(\text{Sh}_{2\zeta_2}) D(\ddot{\zeta}_1) D(\bar{x}_2) \tanh(\zeta_2) \\ & + D(\text{Sh}_{2\zeta_2}) D(\dot{\zeta}_1) D(\bar{x}_2) D_I(\text{Ch}_{\zeta_2})^2 \dot{\zeta}_2 + D(\text{Sh}_{2\zeta_2}) D(\bar{x}_2) D_I(\text{Ch}_{\zeta_2})^2 \dot{\zeta}_2 \\ & - 2 D(\text{Ch}_{\zeta_1})^2 D(\bar{x}_2) D_I(\text{Ch}_{\zeta_2})^2 D(\tanh(\zeta_2)) \dot{\zeta}_2 + D(\text{Ch}_{\zeta_1})^2 D(\bar{x}_2) D_I(\text{Ch}_{\zeta_2})^2 \ddot{\zeta}_2. \end{aligned} \quad (\text{B.3.6})$$

It follows from (3.84) that

$$\dot{e}_3 = \dot{Q}(f_2 + g_2(z_3)) + QN(z_3)z_4 + k_2\dot{e}_2. \quad (\text{B.3.7})$$

It follows from (3.82) that

$$\begin{aligned} \dot{Q} &= 2D(\text{Ch}_{\zeta_2})D(\text{Sh}_{\zeta_2})D(\dot{\zeta}_2)D_I(\text{Ch}_{\zeta_1})^2D_I(\bar{x}_2)^2 \\ &\quad - 2D_I(\text{Ch}_{\zeta_1})^3D(\text{Sh}_{\zeta_1})D(\dot{\zeta}_1)D(\text{Ch}_{\zeta_2})^2D_I(\bar{x}_2)^2, \end{aligned} \quad (\text{B.3.8})$$

and thus

$$\begin{aligned} \ddot{Q} &= 2D(\text{Sh}_{\zeta_2})^2D(\dot{\zeta}_2)^2D_I(\text{Ch}_{\zeta_1})^2D_I(\bar{x}_2)^2 + 2D(\text{Ch}_{\zeta_2})^2D(\dot{\zeta}_2)^2D_I(\text{Ch}_{\zeta_1})^2D_I(\bar{x}_2)^2 \\ &\quad + D(\text{Sh}_{2\zeta_2})D(\ddot{\zeta}_2)D_I(\text{Ch}_{\zeta_1})^2D_I(\bar{x}_2)^2 - 2D(\text{Sh}_{2\zeta_2})D(\dot{\zeta}_2)D(\tanh(\zeta_1))D_I(\bar{x}_2\text{Ch}_{\zeta_1})^2D(\dot{\zeta}_1) \\ &\quad + 6D(\tanh(\zeta_1)\text{Sh}_{\zeta_1})D_I(\text{Ch}_{\zeta_1})^3D(\text{Ch}_{\zeta_2}\dot{\zeta}_1)^2D_I(\bar{x}_2)^2 - 2D_I(\text{Ch}_{\zeta_1})^2D(\dot{\zeta}_1)^2D(\text{Ch}_{\zeta_2})^2D_I(\bar{x}_2)^2 \\ &\quad - 2D_I(\text{Ch}_{\zeta_1})^3D(\text{Sh}_{\zeta_1})D(\ddot{\zeta}_1)D(\text{Ch}_{\zeta_2})^2D_I(\bar{x}_2)^2 \\ &\quad - 2D_I(\text{Ch}_{\zeta_1})^3D(\text{Sh}_{\zeta_1})D(\dot{\zeta}_1)D(\text{Sh}_{2\zeta_2})D(\dot{\zeta}_2)D_I(\bar{x}_2)^2, \end{aligned} \quad (\text{B.3.9})$$

It follows from (3.88) that

$$N = m^{-1} \begin{bmatrix} -C_{z_{31}}z_{32} & -S_{z_{31}} \\ -S_{z_{31}}z_{32} & C_{z_{31}} \end{bmatrix}, \quad (\text{B.3.10})$$

$$\dot{N} = m^{-1} \begin{bmatrix} S_{z_{31}}z_{41}z_{32} - C_{z_{31}}z_{42} & -C_{z_{31}}z_{41} \\ -C_{z_{31}}z_{41}z_{32} - S_{z_{31}}z_{42} & -S_{z_{31}}z_{41} \end{bmatrix}. \quad (\text{B.3.11})$$

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